

SingleRAN

CPRI MUX Feature Parameter Description

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 SRAN22.1 Draft A (2025-12-31)

This issue does not introduce any changes to SRAN21.1 02 (2025-04-27).

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to GSM, UMTS, LTE FDD, LTE TDD, LTE NB-IoT, and NR.

Unless otherwise specified, in this document, LTE and eNodeB always include FDD, TDD, and NB-IoT. In scenarios where they need to be distinguished, LTE FDD, LTE TDD, and LTE NB-IoT are used. The same rules apply to eNodeB.

Unless otherwise specified, in this document, NR and gNodeB always include FDD and TDD. In scenarios where they need to be distinguished, NR FDD and NR TDD are used. The same rules apply to gNodeB.

The "G", "C", "U", "L", "T", "M", "N", "N (TDD)", and "N (FDD)" in RAT acronyms refer to GSM, CDMA, UMTS, LTE FDD, LTE TDD, LTE NB-IoT, NR, NR TDD, and NR FDD, respectively.

Unless otherwise specified, the application scenarios for CPRI MUX involving LTE NB-IoT are the same as those for CPRI MUX involving LTE FDD. This document uses LTE FDD as an example. For example, to learn the applications of CPRI MUX in GM/GLM, UM/ULM, MN/MLN, GUM/GULM, GMN/GLMN, UMN/ULMN, and GUMN/GULMN multimode base stations, see CPRI MUX in GL, UL, LN, GUL, GLN, ULN, and GULN multimode base stations.

2.3 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
-	CPRI MUX	9 Operation and Maintenance

2.4 Differences

Differences Between NR FDD and NR TDD

Function Name	Difference
CPRI MUX	None

Differences Between NSA and SA

Function Name	Difference
CPRI MUX	None

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference
CPRI MUX	Supported only in low frequency bands

3 Overview

CPRI Multiplex (CPRI MUX) enables multiple network modes in a multimode base station to use the same CPRI port for data transmission, reducing costs of new deployment or refarming.

- New deployment: In CPRI MUX, multiple network modes can use the same CPRI port to transmit data, decreasing the number of required CPRI optical fibers for newly deployed multimode base stations. This greatly reduces optical module and fiber purchase costs, and optical fiber installation and maintenance costs.
- Refarming: Assume that a multimode RF module uses a single CPRI cable during initial site deployment. CPRI MUX enables new network modes to share the existing RF module and CPRI cable without adding CPRI cables. This reduces both engineering costs and service interruption duration.

NOTE

- This document applies to 3900 and 5900 series base stations, excluding DBS3900 and DBS5900 LampSite base stations. For CPRI MUX in DBS3900 and DBS5900 LampSite base stations, see *CPRI MUX (LampSite)*.
- This document does not involve any basic CPRI topologies, such as chain, star, ring, dual-star, and load-sharing topologies. For details, see *RF Unit and Topology Management*.
- This document does not involve the CPRI trunk chain and description of the trunk chain topology with CPRI MUX. For details, see *RMU* in *3900 & 5900 Series Base Station Product Documentation*.
- If LTE and NR uplink spectrum sharing is enabled, data convergence is supported between LTE and NR. For details, see *LTE FDD and NR SUL Dynamic Spectrum Sharing* in *eRAN Feature Documentation* or *5G RAN Feature Documentation*.
- This document applies only to CPRI, and does not apply to enhanced common public radio interfaces (eCPRI). For details about eCPRI MUX for LTE TDD and NR TDD cells, see *eCPRI* in *eRAN Feature Documentation* and *eCPRI* in *5G RAN Feature Documentation*, respectively.
- For details about slot assignment principles for different BBUs, see *BBU3900 and BBU3910 Hardware Description*, *BBU5900 Hardware Description*, *BBU5910 Hardware Description*, and *BBU5900A Hardware Description* in *3900 & 5900 Series Base Station Product Documentation*.

3.1 Principles

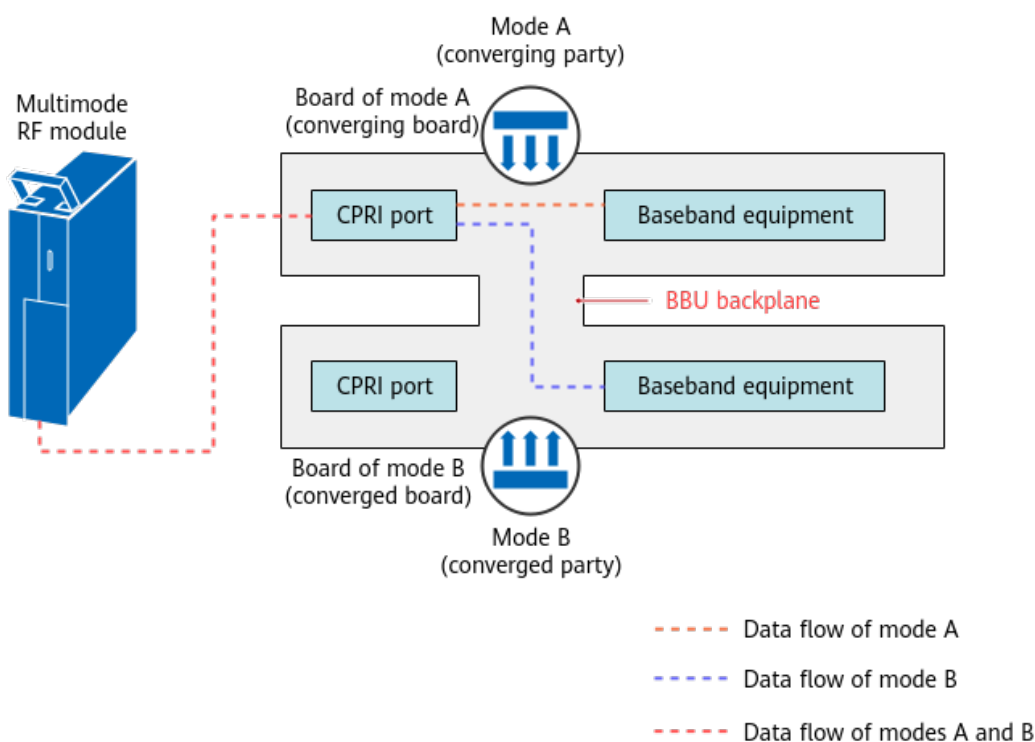
BBUs that support CPRI MUX include BBU3900/BBU3910/BBU5900/BBU5910/BBU5900A/BBU3910A/BBU3910C/BBU5900C/BookBBU5901/BookBBU5901a.

CPRI MUX Used with a BBU3900/BBU3910/BBU5900/BBU5910/BBU5900A

The BBU3900/BBU3910/BBU5900/BBU5910/BBU5900A supports CPRI MUX in both co-MPT and separate-MPT scenarios. For example, a BBU5900 is configured with two baseband processing units (BBPs) in a multimode base station. One of the BBPs works in network mode A while the other works in network mode B. The two BBPs are connected through the backplane. This way, data of network mode B is converged on the BBP of network mode A. Then, a CPRI optical fiber connecting the BBP of network mode A and a multimode RF module can carry data of both network modes A and B. **Figure 3-1** shows working principles of data convergence.

In CPRI MUX, the board of network mode A serves as the converging board and network mode A is the converging party. The board of network mode B serves as the converged board and network mode B is the converged party.

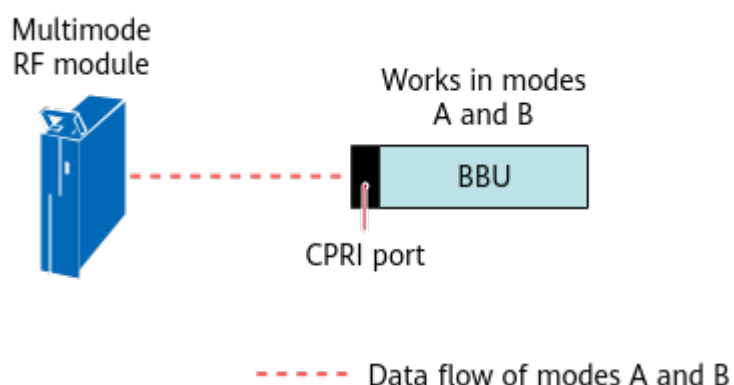
Figure 3-1 CPRI MUX principles when data of multiple network modes is converged (BBU3900/BBU3910/BBU5900/BBU5910/BBU5900A)



CPRI MUX Used with a BBU3910A/BBU3910C/BBU5900C/BookBBU5901/BookBBU5901a

The BBU3910A/BBU3910C/BBU5900C/BookBBU5901/BookBBU5901a supports CPRI MUX only in co-MPT scenarios. For example, for a BBU3910A in an A+B co-MPT base station, data of network modes A and B is transmitted to a multimode RF module through a CPRI port. [3.1 Principles](#) shows working principles of data convergence.

Figure 3-2 CPRI MUX principles when data of multiple network modes is converged (BBU3910A/BBU3910C/BBU5900C/BookBBU5901/BookBBU5901a)



3.2 Application Scenarios

In multimode base stations, CPRI MUX can work with a variety of CPRI topologies to meet the requirements of different application scenarios. The following table lists typical application scenarios.

Typical Application Scenario	Topology	Related Section/ Document
Most areas, especially densely populated areas	Star topology with CPRI MUX (recommended for co-MPT scenarios)	5.1 Star Topology with CPRI MUX
Long and narrow areas, such as highways and railways	Chain topology with CPRI MUX	5.2 Chain Topology with CPRI MUX
- Insufficient CPRI bandwidth for co-MPT base stations - Reconstruction from separate-MPT base stations using the dual-star topology to co-MPT base stations using the dual-star topology	Multimode load sharing topology with CPRI MUX	5.3 Load Sharing Topology with CPRI MUX

Typical Application Scenario	Topology	Related Section/ Document
Separate-MPT triple-mode/ quadruple-mode base stations	Dual-star topology with CPRI MUX	5.4 Dual-Star Topology with CPRI MUX
Insufficient CPRI links for long- distance transmission or insufficient CPRI bandwidth	Trunk chain topology with CPRI MUX	For details about this topology, see <i>RMU Feature Parameter Description in 3900 & 5900 Series Base Station Product Documentation</i> .

4 Boards That Support CPRI MUX and Constraints

Boards that support CPRI MUX and constraints vary as backplane capabilities and board configurations vary with BBUs.

4.1 BBU3900 Boards Supporting CPRI MUX and Constraints

The BBU3900 supports CPRI MUX in co-MPT scenarios for data convergence among two or more network modes (including GSM, UMTS, and LTE), and supports CPRI MUX in separate-MPT scenarios for data convergence between GSM and UMTS, GSM and LTE, and UMTS and LTE.

4.1.1 Co-MPT Scenarios

Topology Constraints

The constraints on CPRI MUX used with a BBU3900 in co-MPT scenarios are listed as follows:

- Converging and converged parties must be deployed in the same BBU.
- If the converged party is LTE, the converged board cannot be an LBBPc.
- If the converging or converged party is UMTS, a UBBP_Ux⁽¹⁾ must be installed in slot 3 or 2.
- In non-baseband extension mode⁽²⁾, GSM can serve as the converged party, and GSM RF modules are connected to the converging board. This eliminates the need for adding a GSM BBP or baseband radio interface unit (BRI).
In GSM baseband extension mode⁽³⁾, a UBBP_Gx⁽¹⁾ must be installed in slot 1, 2, or 3.
- If the converging board is a UBR1b, GSM supports the non-baseband extension mode only.

In UL or GUL CPRI MUX, if the converging board is a UBR1b, and UMTS and LTE serve as converged parties, then the cells of the converged parties must

be served by the specified baseband equipment. The baseband equipment can only include single-mode BBPs.

- If a BBU is configured with a co-BBP board, and network modes involved in CPRI MUX are the same as or included in the network modes that work in co-BBP mode, connect this co-BBP board to RF modules. For example, if a BBU is configured with a UBBP_GL, the GL CPRI port connecting to RF modules is provided by this UBBP_GL instead of a GO or LO BBP/BRI.

 NOTE

- ⁽¹⁾: UBBP_Ux indicates that the working mode of this UBBP includes UMTS. UBBP_Gx indicates that the working mode of this UBBP includes GSM.
- ⁽²⁾: In non-baseband extension mode, GSM baseband signals are processed by RRUs instead of a UBBP_Gx in the BBU.
- ⁽³⁾: In GSM baseband extension mode, GSM baseband signals are processed by a UBBP_Gx. In this case, a UBBP_Gx must be configured in the BBU.

Boards and Their Slot Assignment

Table 4-1 describes BBU3900 boards that can serve as the converging board of CPRI MUX in co-MPT scenarios and their slot assignment. See [Topology Constraints](#) for the converged board and slot assignment.

Table 4-1 BBU3900 boards that can serve as the converging board of CPRI MUX in co-MPT scenarios and their slot assignment

Network Mode Served by the CPRI Port	Converging Board	Slot of the Converging Board
UL	UBRlb ⁽¹⁾	Slot 1
UL	LBBPd, or UBBP_U/L/GU/GL/UL/GUL	Slot 2 or 3
GU	UBRlb ⁽¹⁾	Slot 0 or 1
	UBBP_G/U/GU/GL/UL/GUL	Slot 2 or 3
GL	UBRlb ⁽¹⁾	Slot 1
	UBBP_G/L	Slot 1, 2, or 3
	UBBP_GU/GL/UL/GUL	Slot 2 or 3
GUL	UBRlb ⁽¹⁾	Slot 1
	UBBP_G/U/L/GU/GL/UL/GUL	Slot 2 or 3

Network Mode Served by the CPRI Port	Converging Board	Slot of the Converging Board
Note: In this table, UBBP_G/U/L/GU/GL/UL/GUL indicates that this UBBP works in GO, UO, LO, GU, GL, UL, or GUL mode. For example, UBBP_UL indicates that this UBBP works in UL mode, and there is a possibility that it supports GUL. (1): If CPRI ports on a UBR1b work at a data rate higher than 4.9 Gbit/s, each port can carry services of two network modes. If the CPRI ports work at a data rate lower than or equal to 4.9 Gbit/s, each port can carry GUL services. If GLM/ULM concurrency is applied, GLM/ULM is equivalent to two network modes when CPRI compression is not used for LTE and equivalent to three network modes when CPRI compression is used for LTE.		

4.1.2 Separate-MPT Scenarios

Topology Constraints

The constraints on CPRI MUX used with a BBU3900 in separate-MPT scenarios are listed as follows:

- CPRI MUX is only supported in one BBU. Inter-BBU CPRI MUX is not supported.
- Converging and converged parties must share one BBU. One mode can only converge data or have its data converged. Data of a network mode can only be converged to one board.
- Only UMTS and LTE can serve as the converging party. If CPRI MUX involves GSM, GSM supports the non-baseband extension mode only.
- Each BBU provides a maximum of six ports for data convergence. These ports must be located on the same LBBPd, UBBP_U, or UBBP_L.
- For a combined-RRU cell, a distributed cell, a cell in 0.5/0.5 configuration mode, and a cell for which signals from multiple RRUs are independently demodulated, the CPRI ports on RRUs serving such a cell must connect to the BBPs of the same mode.
- When a UBBPg1a serves as the converging board in GU/GL CPRI MUX, the converged party GSM must be upgraded to SRAN16.1 SPC200 or later.

NOTE

In multi-level cascading scenarios, before removing an entire chain/ring from the converged party, the nodes on the chain/ring must be removed in sequence, starting from the last-level node.

Boards and Their Slot Assignment

Table 4-2 describes BBU3900 boards that support CPRI MUX in separate-MPT scenarios and their slot assignment.

Table 4-2 BBU3900 boards that support CPRI MUX in separate-MPT scenarios and their slot assignment

Network Mode Served by the CPRI Port	Converging Party	Converging Board	Slot of the Converging Board	Converged Board	Slot of the Converged Board
UL	UMTS	UBBP_U	Slot 2 or 3	LBBPd UBBP_L	An LBBPd or UBBP_L must be installed in slot 3 or 2.
	LTE	LBBPd UBBP_L	Slot 2 or 3	UBBP_U	A UBBP_U must be installed in slot 3 or 2.
GU	UMTS	UBBP_U	Slot 2 or 3	GTMUc	Slot6 ⁽¹⁾
GL	LTE	LBBPd UBBP_L	Slot 2 or 3	GTMUc	Slot6 ⁽¹⁾
Note: In this table, UBBP_G/U/L/GU/GL/UL/GUL indicates that this UBBP works in GO, UO, LO, GU, GL, UL, or GUL mode. For example, UBBP_UL indicates that this UBBP works in UL mode, and there is a possibility that it supports GSM, UMTS, and LTE. ⁽¹⁾: In this scenario, data is converged from slot 6 to slot 2 or 3 instead of both of them.					

4.2 BBU3910 Boards Supporting CPRI MUX and Constraints

The BBU3910 supports CPRI MUX in co-MPT scenarios for data convergence among two or more network modes (including GSM, UMTS, LTE, and NR), and supports CPRI MUX in separate-MPT scenarios for data convergence between GSM and UMTS, GSM and LTE, UMTS and LTE, UMTS and NR, and LTE and NR.

4.2.1 Co-MPT Scenarios

Topology Constraints

The constraints on CPRI MUX used with a BBU3910 in co-MPT scenarios are listed as follows:

- Converging and converged parties must be deployed in the same BBU. Inter-BBU CPRI MUX is not supported.

- In non-baseband extension mode, GSM can serve as the converged party, and GSM RF modules are connected to the converging board. This eliminates the need for adding a GSM BBP or BRI.
- If the converging board is a UBR1b, GSM supports the non-baseband extension mode only.
In UL or GUL CPRI MUX, if the converging board is a UBR1b, and UMTS and LTE serve as converged parties, then the cells of the converged parties must be served by the specified baseband equipment. The baseband equipment can only include single-mode BBPs.
- If a BBU is configured with a multimode co-BBP board, and network modes involved in CPRI MUX are the same as or included in the network modes that work in co-BBP mode, it is recommended that the CPRI port connecting to RF modules be provided by this multimode co-BBP board. For example, if a BBU is configured with a UBBP_GL, it is recommended that the GL CPRI port connecting to RF modules be provided by this UBBP_GL instead of a GO or LO BBP/BRI.

Boards and Their Slot Assignment

Table 4-3 describes BBU3910 boards that can serve as the converging board of CPRI MUX in co-MPT scenarios and their slot assignment. See [Topology Constraints](#) for BBU3910 boards that can serve as the converged board of CPRI MUX in co-MPT scenarios and their slot assignment.

Table 4-3 BBU3910 boards that can serve as the converging board of CPRI MUX in co-MPT scenarios and their slot assignment

Network Mode Served by the CPRI Port	Converging Board	Slot of the Converging Board
GU	UBR1b ⁽¹⁾ UBBP_G/U/L/GU/GL/UL/GUL	Slot 0, 1, 2, 3, 4, or 5
GL		
UL		
GUL		
UN	UBBP_G/U/L/N/GU/GL/UL/GUL/LN	Slot 0, 1, 2, 3, 4, or 5
LN		
GLN		
ULN		
GULN		

Network Mode Served by the CPRI Port	Converging Board	Slot of the Converging Board
Note: In this table, UBBP_G/U/L/GU/GL/UL/GUL/N/LN indicates the working mode of the UBBP instead of the mode supported by the UBBP. For example, UBBP_UL indicates that this UBBP works in UL mode, and there is a possibility that it supports GSM, UMTS, and LTE. (1): If CPRI ports on a UBR1b work at a data rate higher than 4.9 Gbit/s, each port can carry services of two network modes. If the CPRI ports work at a data rate lower than or equal to 4.9 Gbit/s, each port can carry GUL services. If GLM/ULM concurrency is applied, GLM/ULM is equivalent to two network modes when CPRI compression is not used for LTE and equivalent to three network modes when CPRI compression is used for LTE.		

4.2.2 Separate-MPT Scenarios

Topology Constraints

The constraints on CPRI MUX used with a BBU3910 in separate-MPT scenarios are listed as follows:

- CPRI MUX is only supported in one BBU. Inter-BBU CPRI MUX is not supported.
- Converging and converged parties must share one BBU. One mode can only converge data or have its data converged and cannot serve as both the converging and converged parties.
- CPRI MUX and eCPRI MUX cannot be supported simultaneously by a group of converging and converged boards.
- Only UMTS, LTE, and NR can serve as the converging party. If CPRI MUX involves GSM, GSM supports the non-baseband extension mode only.
- For GU or GL CPRI MUX in separate-MPT scenarios, a single BBU provides a maximum of six ports for data convergence. These ports must be located on the same UBBP_U or UBBP_L.
- For UL or LN CPRI MUX in separate-MPT scenarios, a single BBU provides a maximum of 24 ports for data convergence. One converged board corresponds to one converging board (one-to-one convergence) or two converging boards (one-to-two convergence). A maximum of three one-to-one convergence relationships or two one-to-two convergence relationships are supported. The two-to-one convergence relationship is not supported. After the converged party supports one-to-two convergence relationships, CPRI chains/rings with **Access Type** set to **LOCALPORT** cannot be configured on the converged party.
- For a combined-RRU cell, a distributed cell, a cell in 0.5/0.5 configuration mode, and a cell for which signals from multiple RRUs are independently demodulated, the CPRI ports on RRUs serving such a cell must connect to the BBPs of the same mode.

- When the converging party in UL/GU/GL CPRI MUX uses a data rate of 24.3 Gbit/s, the versions of the converging and converged parties must be upgraded to SRAN16.1 SPC180 or later.

 NOTE

In multi-level cascading scenarios, before removing an entire chain/ring from the converged party, the nodes on the chain/ring must be removed in sequence, starting from the last-level node.

Boards and Their Slot Assignment

Table 4-4 describes BBU3910 boards that support CPRI MUX in separate-MPT scenarios and their slot assignment.

Table 4-4 BBU3910 boards that support CPRI MUX in separate-MPT scenarios and their slot assignment

Network Mode Served by the CPRI Port	Converging Board ⁽¹⁾	Slot of the Converging Board	Converged Board ⁽¹⁾⁽²⁾	Slot of the Converged Board
UL	UBBP_U	Slot 0, 1, 2, 3, 4, or 5	UBBP_L	Slot 0, 1, 2, 3, 4, or 5
	UBBP_L	Slot 0, 1, 2, 3, 4, or 5	UBBP_U	Slot 0, 1, 2, 3, 4, or 5
UL	UBBPei_U	Slot 0, 1, 2, 3, 4, or 5	UBBPd/UBBPe_L	Slot 0, 1, 2, 3, 4, or 5
	UBBPei_L	Slot 0, 1, 2, 3, 4, or 5	UBBPd/UBBPe_U	Slot 0, 1, 2, 3, 4, or 5
GU	UBBP_U	Slot 0, 1, 2, or 3	GTMUc	Slot 6 ⁽³⁾
GL	UBBP_L	Slot 0, 1, 2, or 3	GTMUc	Slot 6 ⁽³⁾
LN	UBBP_L	Slot 0, 1, 2, 3, 4, or 5	UBBP_N	Slot 0, 1, 2, 3, 4, or 5
	UBBP_N	Slot 0, 1, 2, 3, 4, or 5	UBBP_L	Slot 0, 1, 2, 3, 4, or 5

Network Mode Served by the CPRI Port	Converging Board ⁽¹⁾	Slot of the Converging Board	Converged Board ⁽¹⁾⁽²⁾	Slot of the Converged Board
Note: In this table, UBBP_G/U/L/N/LN/GU/GL/UL/GUL indicates the working mode of the UBBP. For example, a UBBP supporting GUL but working in UL mode is expressed as UBBP_UL. The UBBPg series boards using DSFP optical modules cannot serve as the converged board. If they serve as the converging board, the two LC port channels of each DSFP optical module support CPRI MUX only in LN CPRI MUX scenarios. In other scenarios, only the first LC port channel of each DSFP optical module supports CPRI MUX. (1): A UBBPem or UBBPf1 cannot serve as the converging or converged board. A UBBPex2 cannot serve as the converging board. (2): A UBBPei cannot serve as the converged board in separate-MPT scenarios. (3): Data is converged from slot 6 to one slot (slot 0 to 3) instead of two or more slots (for example, from slot 6 to slot 0 or 1).				

4.2.3 Hybrid-MPT Scenarios

In hybrid-MPT scenarios, there are both co-MPT and separate-MPT in a BBU.

NOTE

The ampersand (&) indicates separate-MPT and the asterisk (*) indicates co-MPT. For example, U&[L*N] indicates a hybrid-MPT scenario where UMTS, LTE, and NR modes are deployed in one BBU. The BBU is configured with two main control boards, one working in UMTS mode and the other working in LTE and NR modes at the same time. In this case, LTE and NR work in co-MPT mode. UMTS works with LTE and NR in separate-MPT mode. The three modes work together in U&[L*N] hybrid-MPT mode. Data convergence from UMTS to NR and that from LTE to NR apply to CPRI MUX in separate-MPT scenarios and CPRI MUX in co-MPT scenarios, respectively.

Topology Constraints

For the BBU3910, CPRI MUX can be used in U&[L*N] or [G*U]&[L*N] hybrid-MPT scenarios only when the following constraints are met:

- Converging and converged parties must be deployed in the same BBU. Inter-BBU CPRI MUX is not supported.
- Converging and converged parties must share one BBU. One mode can only converge data or have its data converged and cannot serve as both the converging and converged parties.
- CPRI MUX and eCPRI MUX cannot be supported simultaneously by a converging board.
- LTE and NR can serve as the converging party.

- For CPRI MUX in separate-MPT scenarios, a single BBU provides a maximum of 24 ports for data convergence. One converged board corresponds to one converging board (one-to-one convergence) or two converging boards (one-to-two convergence). A maximum of three one-to-one convergence relationships or two one-to-two convergence relationships are supported. The two-to-one convergence relationship is not supported. After the converged party supports one-to-two convergence relationships, CPRI chains/rings with **Access Type** set to **LOCALPORT** cannot be configured on the converged party.
- For a combined-RRU cell, a distributed cell, a cell in 0.5/0.5 configuration mode, and a cell for which signals from multiple RRUs are independently demodulated, the CPRI ports on RRUs serving such a cell must connect to the BBPs of the same mode.
- For CPRI MUX in U&[L*N] and [G*U]&[L*N] hybrid-MPT scenarios with a data rate of 24.3 Gbit/s, the optical fiber connections can be provided by LTE/NR BBP. The LTE/NR and UMTS sides must be upgraded to SRAN16.1 SPC180 or later. In versions earlier than SRAN16.1 SPC180, ULN/GULN CPRI MUX with a data rate of 24.3 Gbit/s is not supported. When the RF module with a data rate of 24.3 Gbit/s is used, the CPRI rate on the LTE/NR side must be fixed to 9.8 Gbit/s.
- When the CPRI MUX extension switch (specified by **EQUIPMENT.CPRIMUXEXTSW (LTE eNodeB, 5G gNodeB)**) is turned off, the LTE and NR co-carrier co-CPRI data function is not supported in the ULN/GULN CPRI MUX topology.
- When the CPRI MUX extension switch (specified by **EQUIPMENT.CPRIMUXEXTSW (LTE eNodeB, 5G gNodeB)**) is turned on, the LTE and NR co-carrier co-CPRI data function is supported in the ULN/GULN CPRI MUX topology. Cells in GSM baseband extension mode are not supported. For cells in GSM non-baseband extension mode, baseband frequency hopping between boards of different modes or cross-RF module frequency hopping is not supported. UMTS RF interconnection cells do not support switching of the two receive channels and two 1T UMTS cells cannot be combined into one virtual antenna mapping (VAM) cell.

Boards and Their Slot Assignment

Table 4-5 describes BBU3910 boards that support CPRI MUX in U&[L*N]/[G*U]&[L*N] hybrid-MPT scenarios and their slot assignment. When CPRI ports of different main control boards serve two network modes, see **Table 4-4** for boards that support CPRI MUX and their slot assignments.

Table 4-5 BBU3910 boards that support CPRI MUX in U&[L*N]/[G*U]&[L*N] hybrid-MPT scenarios and their slot assignment

Network Mode Served by the CPRI Port	Converging Board ⁽¹⁾	Slot of the Converging Board	Converged Board ⁽¹⁾	Slot of the Converged Board
ULN	UBBP_L/N/LN	Slot 0, 1, 2, 3, 4, or 5	UBBP_U	Slot 0, 1, 2, 3, 4, or 5

Network Mode Served by the CPRI Port	Converging Board ⁽¹⁾	Slot of the Converging Board	Converged Board ⁽¹⁾	Slot of the Converged Board
GULN	UBBP_L/N/LN	Slot 0, 1, 2, 3, 4, or 5	UBBP_U	Slot 0, 1, 2, 3, 4, or 5
GUL	UBBP_L	Slot 0, 1, 2, 3, 4, or 5	UBBP_U	Slot 0, 1, 2, 3, 4, or 5
GUN	UBBP_N	Slot 0, 1, 2, 3, 4, or 5	UBBP_U	Slot 0, 1, 2, 3, 4, or 5

Note: In this table, UBBP_U/LN indicates the working mode of the UBBP. For example, a UBBP supporting LTE and NR but working in LTE mode is expressed as UBBP_L.

The UBBP_g series boards using DSFP optical modules cannot serve as the converged board. If they serve as the converging board, the two LC port channels of each DSFP optical module support CPRI MUX only in LN CPRI MUX scenarios. In other scenarios, only the first LC port channel of each DSFP optical module supports CPRI MUX.

⁽¹⁾: A UBBP_{em}, UBBP_{ei}, or UBBP_{f1} cannot serve as the converging or converged board. A UBBP_{ex2} cannot serve as the converging board.

4.3 BBU5900/BBU5910 Boards Supporting CPRI MUX and Constraints

The BBU5900/BBU5910 supports CPRI MUX in co-MPT scenarios for data convergence among two or more network modes (including GSM, UMTS, LTE, and NR). In separate-MPT scenarios, data convergence between UMTS and LTE and between LTE and NR is supported.

NOTE

The BBU5900 and BBU5910 share the boards supporting CPRI MUX and constraints, except that the BBU5910 cannot be configured with the UBBP_{ei} or UBBP_{fw}.

4.3.1 Co-MPT Scenarios

Topology Constraints

The constraints on CPRI MUX used with a BBU5900/BBU5910 in co-MPT scenarios are listed as follows:

- Converging and converged parties must be deployed in the same BBU. Inter-BBU CPRI MUX is not supported.
- In non-baseband extension mode, GSM can serve as the converged party, and GSM RF modules are connected to the UBBP_U/L/UL/LN. This eliminates the need for adding a GSM baseband processing unit.

- If a BBU is configured with a multimode co-BBP board, and network modes involved in CPRI MUX are the same as or included in the network modes that work in co-BBP mode, it is recommended that the CPRI port connecting to RF modules be provided by this multimode co-BBP board. For example, if a BBU is configured with a UBBP_GL, it is recommended that the GL CPRI port connecting to RF modules be provided by this UBBP_GL instead of a GO or LO BBP/BRI.

Boards and Their Slot Assignment

Table 4-6 describes BBU5900/BBU5910 boards that can serve as the converging board of CPRI MUX in co-MPT scenarios and their slot assignment.

See **Topology Constraints** for BBU5900 boards that can serve as the converged board of CPRI MUX in co-MPT scenarios and their slot assignment.

Table 4-6 BBU5900/BBU5910 boards that can serve as the converging board of CPRI MUX in co-MPT scenarios and their slot assignment

Network Mode Served by the CPRI Port ⁽¹⁾	Converging Board	Slot of the Converging Board
GU	UBBP_G/U/L/GU/GL/UL/GUL	Slot 0, 1, 2, 3, 4, or 5
GL		
UL		
GUL		
UN	UBBP_G/U/L/N/GU/GL/UL/GUL/LN	Slot 0, 1, 2, 3, 4, or 5
LN		
GLN		
ULN		
GULN		

Note: In this table, UBBP_G/U/L/N/GU/GL/UL/GUL/LN indicates the working mode of the UBBP. For example, UBBP_UL indicates that this UBBP works in UL mode, and there is a possibility that it supports GUL.

⁽¹⁾: When a UBBPh series board serves as the converging board and is used with DSFP optical modules, GSM is supported only over ports 0 to 5.

4.3.2 Separate-MPT Scenarios

Topology Constraints

The constraints on CPRI MUX used with a BBU5900/BBU5910 in separate-MPT scenarios are listed as follows:

- CPRI MUX is only supported in one BBU. Inter-BBU CPRI MUX is not supported.
- Converging and converged parties must share one BBU. One mode can only converge data or have its data converged and cannot serve as both the converging and converged parties.
- CPRI MUX and eCPRI MUX can be configured simultaneously only when the converging board is a UBBPh3/UBBPh3a/UBBPh3b and so is the converged board. In other cases, CPRI MUX and eCPRI MUX cannot be configured simultaneously for the converging and converged boards in a group.
- The converging party can be UMTS/LTE/NR.
- For UL or LN CPRI MUX in separate-MPT scenarios, a single BBU provides a maximum of 24 ports for data convergence. One converged board corresponds to one converging board (one-to-one convergence) or two converging boards (one-to-two convergence). A maximum of three one-to-one convergence relationships or two one-to-two convergence relationships are supported. The two-to-one convergence relationship is not supported. After the converged party supports one-to-two convergence relationships, CPRI chains/rings with **Access Type** set to **LOCALPORT** cannot be configured on the converged party.
- For a combined-RRU cell, a distributed cell, a cell in 0.5/0.5 configuration mode, and a cell for which signals from multiple RRUs are independently demodulated, the CPRI ports on RRUs serving such a cell must connect to the BBPs of the same mode.
- When the converging party in UL CPRI MUX uses a data rate of 24.3 Gbit/s, the versions of the converging and converged parties must be upgraded to SRAN16.1 SPC180 or later.
- In UL CPRI MUX scenarios, if a UBBPh series board serves as the converging board and a UBBP_U serves as the converged board, only ports 0 to 5 on the UBBPh series board can provide optical fiber connections. If a UBBPh1u serves as the converged board, only ports 0 to 5 can provide optical fiber connections.

NOTE

In multi-level cascading scenarios, before removing an entire chain/ring from the converged party, the nodes on the chain/ring must be removed in sequence, starting from the last-level node.

Boards and Their Slot Assignment

Table 4-7 describes BBU5900 boards that support CPRI MUX in separate-MPT scenarios and their slot assignment. **Table 4-8** describes BBU5910 boards that support CPRI MUX in separate-MPT scenarios and their slot assignment.

Table 4-7 BBU5900 boards that support CPRI MUX in separate-MPT scenarios and their slot assignment

Network Mode Served by the CPRI Port	Converging Board ⁽¹⁾	Slot of the Converging Board	Converged Board ⁽¹⁾⁽²⁾	Slot of the Converged Board
UL	UBBP_U	Slot 0, 1, 2, 3, 4, or 5	UBBP_L	Slot 0, 1, 2, 3, 4, or 5
	UBBP_L	Slot 0, 1, 2, 3, 4, or 5	UBBP_U	Slot 0, 1, 2, 3, 4, or 5
UL	UBBP _{ei} _U	Slot 0, 1, 2, 3, 4, or 5	UBBP _d /UBBP _e _L	Slot 0, 1, 2, 3, 4, or 5
	UBBP _{ei} _L	Slot 0, 1, 2, 3, 4, or 5	UBBP _d /UBBP _e _U	Slot 0, 1, 2, 3, 4, or 5
LN	UBBP_L	Slot 0, 1, 2, 3, 4, or 5	UBBP _{fw} _N	Slot 0, 2, or 4
	UBBP_L	Slot 0, 1, 2, 3, 4, or 5	UBBP_N	Slot 0, 1, 2, 3, 4, or 5
	UBBP_N	Slot 0, 1, 2, 3, 4, or 5	UBBP_L	Slot 0, 1, 2, 3, 4, or 5
Note: In this table, UBBP_G/U/L/N/GU/GL/UL/GUL/LN indicates the working mode of the UBBP. For example, UBBP_UL indicates that this UBBP works in UL mode, and there is a possibility that it supports GSM, UMTS, and LTE. The UBBPg series boards using DSFP optical modules cannot serve as the converged board. If they serve as the converging board, the two LC port channels of each DSFP optical module support CPRI MUX only in LN CPRI MUX scenarios. In other scenarios, only the first LC port channel of each DSFP optical module supports CPRI MUX. ⁽¹⁾: A UBBPem, UBBPf1, or UBBPg4 cannot serve as the converging or converged board. A UBBPex2 cannot serve as the converging board. ⁽²⁾: A UBBPei cannot serve as the converged board in separate-MPT scenarios.				

Table 4-8 BBU5910 boards that support CPRI MUX in separate-MPT scenarios and their slot assignment

Network Mode Served by the CPRI Port	Converging Board ⁽¹⁾	Slot of the Converging Board	Converged Board ⁽¹⁾	Slot of the Converged Board
UL	UBBP_U	Slot 0, 1, 2, 3, 4, or 5	UBBP_L	Slot 0, 1, 2, 3, 4, or 5

Network Mode Served by the CPRI Port	Converging Board ⁽¹⁾	Slot of the Converging Board	Converged Board ⁽¹⁾	Slot of the Converged Board
	UBBP_L	Slot 0, 1, 2, 3, 4, or 5	UBBP_U	Slot 0, 1, 2, 3, 4, or 5
LN	UBBP_L	Slot 0, 1, 2, 3, 4, or 5	UBBP_N	Slot 0, 1, 2, 3, 4, or 5
	UBBP_N	Slot 0, 1, 2, 3, 4, or 5	UBBP_L	Slot 0, 1, 2, 3, 4, or 5
Note: In this table, UBBP_G/U/L/N/GU/GL/UL/GUL/LN indicates the working mode of the UBBP. For example, UBBP_UL indicates that this UBBP works in UL mode, and there is a possibility that it supports GSM, UMTS, and LTE. The UBBPg series boards using DSFP optical modules cannot serve as the converged board. If they serve as the converging board, the two LC port channels of each DSFP optical module support CPRI MUX only in LN CPRI MUX scenarios. In other scenarios, only the first LC port channel of each DSFP optical module supports CPRI MUX. ⁽¹⁾: A UBBPem, UBBPf1, or UBBPg4 cannot serve as the converging or converged board. A UBBPex2 cannot serve as the converging board.				

4.3.3 Hybrid-MPT Scenarios

In hybrid-MPT scenarios, there are both co-MPT and separate-MPT in a BBU.

NOTE

The ampersand (&) indicates separate-MPT and the asterisk (*) indicates co-MPT. For example, U&[L*N] indicates a hybrid-MPT scenario where UMTS, LTE, and NR modes are deployed in one BBU. The BBU is configured with two main control boards, one working in UMTS mode and the other working in LTE and NR modes at the same time. In this case, LTE and NR work in co-MPT mode. UMTS works with LTE and NR in separate-MPT mode. The three modes work together in U&[L*N] hybrid-MPT mode. Data convergence from UMTS to NR and that from LTE to NR apply to CPRI MUX in separate-MPT scenarios and CPRI MUX in co-MPT scenarios, respectively.

Topology Constraints

For the BBU5900/BBU5910, CPRI MUX can be used in U&[L*N] or [G*U]&[L*N] hybrid-MPT scenarios only when the following constraints are met:

- CPRI MUX is only supported in one BBU.
- CPRI MUX can only involve two network modes deployed in the same BBU. For CPRI MUX in separate-MPT scenarios, the converging party cannot serve as the converged party.
- CPRI MUX and eCPRI MUX can be configured simultaneously only when the converging board is a UBBPh3/UBBPh3a/UBBPh3b and so is the converged board. In other cases, CPRI MUX and eCPRI MUX cannot be configured simultaneously for the converging and converged boards in a group.

- LTE and NR can serve as the converging party.
- For CPRI MUX in separate-MPT scenarios, a single BBU provides a maximum of 24 ports for data convergence. One converged board corresponds to one converging board (one-to-one convergence) or two converging boards (one-to-two convergence). A maximum of three one-to-one convergence relationships or two one-to-two convergence relationships are supported. The two-to-one convergence relationship is not supported. After the converged party supports one-to-two convergence relationships, CPRI chains/rings with **Access Type** set to **LOCALPORT** cannot be configured on the converged party.
- For a combined-RRU cell, a distributed cell, a cell in 0.5/0.5 configuration mode, and a cell for which signals from multiple RRUs are independently demodulated, the CPRI ports on RRUs serving such a cell must connect to the BBPs of the same mode.
- For CPRI MUX in U&[L*N] and [G*U]&[L*N] hybrid-MPT scenarios with a data rate of 24.3 Gbit/s, the optical fiber connections can be provided by LTE/NR BBP. The LTE/NR and UMTS sides must be upgraded to SRAN16.1 SPC180 or later. In versions earlier than SRAN16.1 SPC180, ULN/GULN CPRI MUX with a data rate of 24.3 Gbit/s is not supported. When the RF module with a data rate of 24.3 Gbit/s is used, the CPRI rate on the LTE/NR side must be fixed to 9.8 Gbit/s.
- When the CPRI MUX extension switch (specified by **EQUIPMENT.CPRIMUXEXTSW (LTE eNodeB, 5G gNodeB)**) is turned off, the LTE and NR co-carrier co-CPRI data function is not supported in the ULN/GULN CPRI MUX topology.
- When the CPRI MUX extension switch (specified by **EQUIPMENT.CPRIMUXEXTSW (LTE eNodeB, 5G gNodeB)**) is turned on, the LTE and NR co-carrier co-CPRI data function is supported in the ULN/GULN CPRI MUX topology. Cells in GSM baseband extension mode are not supported. For cells in GSM non-baseband extension mode, baseband frequency hopping between boards of different modes or cross-RF module frequency hopping is not supported. UMTS RF interconnection cells do not support switching of the two receive channels and two 1T UMTS cells cannot be combined into one VAM cell.
- In U&[L*N]/[G*U]&[L*N] hybrid-MPT scenarios, if a UBBPh series board serves as the converging board and a UBBP_U serves as the converged board, only ports 0 to 5 on the UBBPh series board can provide optical fiber connections. If a UBBPh1u serves as the converged board, only ports 0 to 5 on the board can provide optical fiber connections.

Boards and Their Slot Assignment

Table 4-9 describes BBU5900/BBU5910 boards that support CPRI MUX in U&[L*N]/[G*U]&[L*N] hybrid-MPT scenarios and their slot assignment. When CPRI ports of different main control boards serve two network modes, see **Table 4-7** for boards that support CPRI MUX and their slot assignments.

Table 4-9 BBU5900/BBU5910 boards that support CPRI MUX in U&[L*N]/[G*U]&[L*N] hybrid-MPT scenarios and their slot assignment

Network Mode Served by the CPRI Port	Converging Board ⁽¹⁾	Slot of the Converging Board	Converged Board ⁽¹⁾⁽²⁾	Slot of the Converged Board
ULN	UBBP_L/N/LN	Slot 0, 1, 2, 3, 4, or 5	UBBP_U	Slot 0, 1, 2, 3, 4, or 5
GULN	UBBP_L/N/LN	Slot 0, 1, 2, 3, 4, or 5	UBBP_U	Slot 0, 1, 2, 3, 4, or 5
GUL	UBBP_L	Slot 0, 1, 2, 3, 4, or 5	UBBP_U	Slot 0, 1, 2, 3, 4, or 5
GUN	UBBP_N	Slot 0, 1, 2, 3, 4, or 5	UBBP_U	Slot 0, 1, 2, 3, 4, or 5

Note: In this table, UBBP_U/LN indicates the working mode of the UBBP. For example, a UBBP supporting LTE and NR but working in LTE mode is expressed as UBBP_L.

The UBBPg series boards using DSFP optical modules cannot serve as the converged board. If they serve as the converging board, the two LC port channels of each DSFP optical module support CPRI MUX only in LN CPRI MUX scenarios. In other scenarios, only the first LC port channel of each DSFP optical module supports CPRI MUX.

(1): A UBBPem, UBBPei, UBBPg4, or UBBPf1 cannot serve as the converging or converged board. A UBBPex2 cannot serve as the converging board.

(2): In scenarios where a UBBPh series board serves as the converged board, the network modes served by the CPRI port can include GSM only when the CPRI MUX extension switch (specified by **EQUIPMENT.CPRIMUXEXTSW (LTE eNodeB, 5G gNodeB)**) is turned on.

4.4 BBU3910A Boards Supporting CPRI MUX and Constraints

Topology Constraints

The constraints on CPRI MUX used with a BBU3910A are listed as follows:

- Only GU, GL, UL, and GUL CPRI MUX in co-MPT scenarios are supported.
- Converging and converged parties must be deployed in the same BBU.
- A BBU3910A provides six CPRI ports, all of which support CPRI MUX.

Network Mode Constraints

The supported network mode varies with the BBU3910A model. [Table 4-10](#) lists the details.

Table 4-10 Network modes supported by each BBU3910A model

BBU Model	Network Mode
BBU3910A1	GU, GL, UL
BBU3910A2	GU, GL, UL
BBU3910A3	GU, GL, UL, GUL

4.5 BBU3910C Boards Supporting CPRI MUX and Constraints

Topology Constraints

The constraints on CPRI MUX used with a BBU3910C are listed as follows:

- Converging and converged parties must be deployed in the same BBU.
- The BBU3910C only supports GU CPRI MUX in co-MPT scenarios. A BBU3910C provides one CPRI port.

4.6 BBU5900A Boards Supporting CPRI MUX and Constraints

Topology Constraints

The constraints on CPRI MUX used with a BBU5900A are listed as follows:

- Converging and converged parties must be deployed in the same BBU.
- In non-baseband extension mode, GSM can serve as the converged party, and GSM RF modules are connected to the UBBP_U/L/UL. This eliminates the need for adding a GSM baseband processing unit.
- If a BBU is configured with a multimode co-BBP board, and network modes involved in CPRI MUX are the same as or included in the network modes that work in co-BBP mode, it is recommended that the CPRI port connecting to RF modules be provided by this multimode co-BBP board. For example, if a BBU is configured with a UBBP_GL, it is recommended that the GL CPRI port connecting to RF modules be provided by this UBBP_GL instead of a GO or LO BBP/BRI.

Boards and Their Slot Assignment

[Table 4-11](#) describes BBU5900A boards that can serve as the converging board of CPRI MUX and their slot assignment.

Table 4-11 BBU5900A boards that can serve as the converging board of CPRI MUX and their slot assignment

Network Mode Served by the CPRI Port ⁽¹⁾	Converging Board	Slot of the Converging Board
GU	UBBP_G/U/L/GU/GL/UL/GUL	Slot 0, 2, or 4
GL		
UL		
GUL		
LN	UBBP_L/N/LN	Slot 0, 2, or 4
ULN	UBBP_U/L/N/UL/LN	Slot 0, 2, or 4
Note: In this table, UBBP_G/U/L/GU/GL/UL/GUL/LN/ULN indicates the working mode of the UBBP. For example, UBBP_UL indicates that this UBBP works in UL mode, and there is a possibility that it supports GSM, UMTS, and LTE. (1): When a UBBPh series board serves as the converging board and is used with DSFP optical modules, GSM is supported only over ports 0 to 5.		

4.7 BookBBU5901 Boards Supporting CPRI MUX and Constraints

Topology Constraints

The constraints on CPRI MUX used with a BookBBU5901 are listed as follows:

- Only GU, GL, UL, GUL, and LN CPRI MUX in co-MPT scenarios are supported.
- CPRI MUX is only supported in one BBU. Inter-BBU CPRI MUX is not supported.
- All the three CPRI ports on the BookBBU5901 support CPRI MUX.

4.8 BookBBU5901a Boards Supporting CPRI MUX and Constraints

Topology Constraints

The constraints on CPRI MUX used with a BookBBU5901a are listed as follows:

- Only GU, GL, UL, GUL, and LN CPRI MUX in co-MPT scenarios are supported.
- CPRI MUX is only supported in one BBU. Inter-BBU CPRI MUX is not supported.
- All the three CPRI ports on the BookBBU5901a support CPRI MUX.

4.9 BBU5900C Boards Supporting CPRI MUX and Constraints

Topology Constraints

The constraints on CPRI MUX used with a BBU5900C are listed as follows:

- Only GU, GL, UL, and GUL CPRI MUX in co-MPT scenarios are supported.
- CPRI MUX is only supported in one BBU. Inter-BBU CPRI MUX is not supported.
- All the three CPRI ports on the BBU5900C support CPRI MUX.

5 CPRI MUX Networking Example

5.1 Star Topology with CPRI MUX

This section provides examples of the star topology with CPRI MUX. For the restrictions on BBU board slot assignment, see [4 Boards That Support CPRI MUX and Constraints](#).

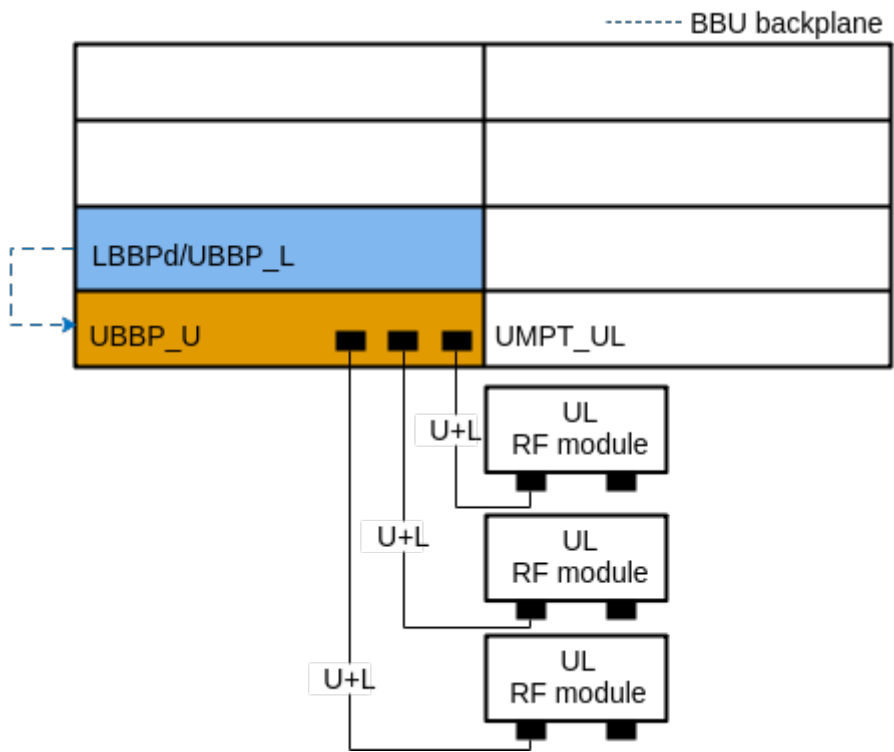
5.1.1 Star Topology with CPRI MUX (BBU3900)

5.1.1.1 Co-MPT Scenarios

Star Topology with UL CPRI MUX

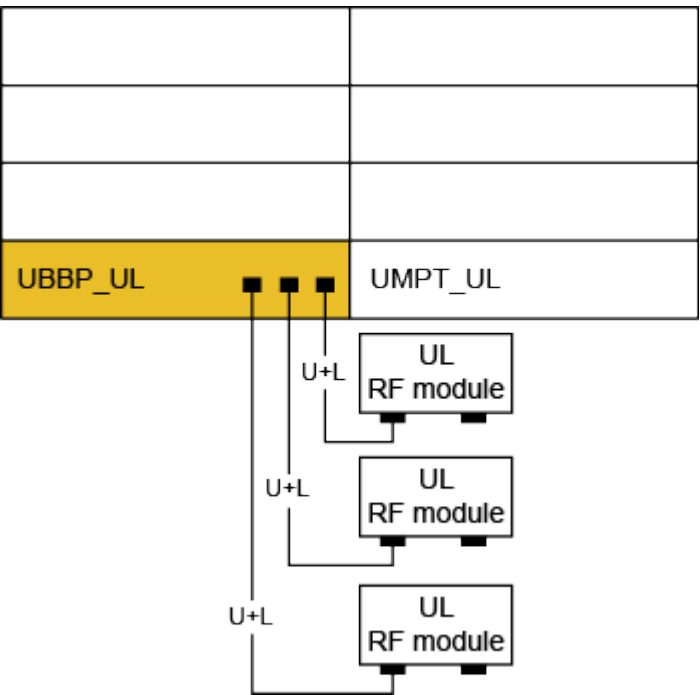
- Scenario 1: The converging board is a single-mode board.
[Figure 5-1](#) shows an example. In this example, UMTS is the converging party and the converging board is a UBBP_U; LTE FDD is the converged party and the converged board is an LBBPd or UBBP_L.

Figure 5-1 UL CPRI MUX in co-MPT scenarios (converging party: UMTS)



- Scenario 2: The converging board is a co-BBP board.
Figure 5-2 shows an example with a UBBP_UL configured in slot 3.

Figure 5-2 UL CPRI MUX with UL co-BBP in co-MPT scenarios



Star Topology with GU CPRI MUX

- Scenario 1: The converging board is a single-mode board.
Figure 5-3 shows an example where GSM uses the non-baseband extension mode. In this example, the converging party is GSM and the converging board is a UBR1b.

Figure 5-3 GU CPRI MUX in co-MPT scenarios (converging board: UBR1b)

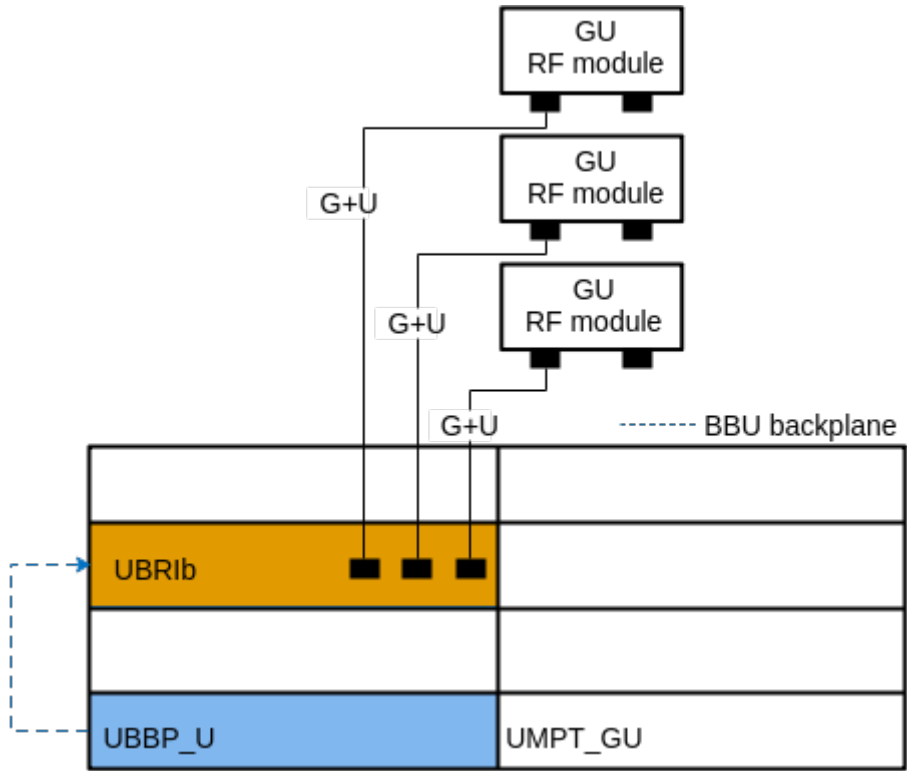
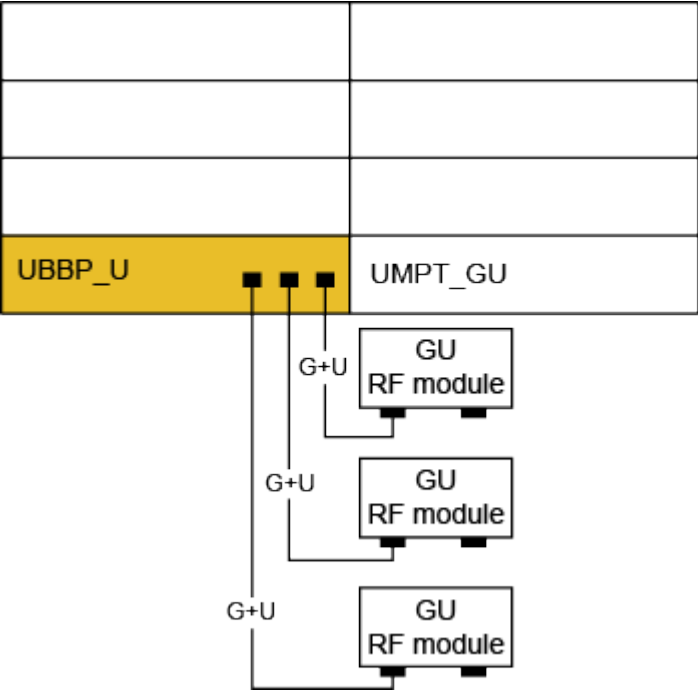


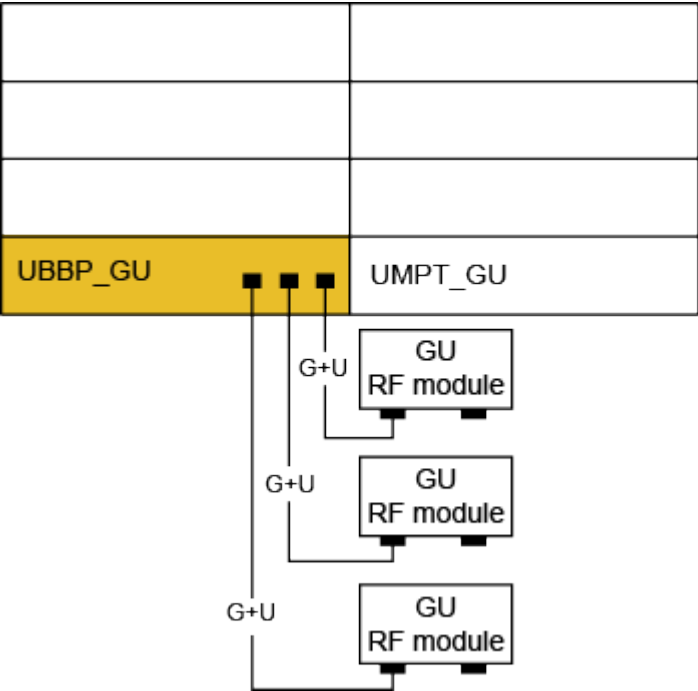
Figure 5-4 shows an example where GSM uses the non-baseband extension mode. In this example, the converging party is UMTS and the converging board is a UBBP_U.

Figure 5-4 GU CPRI MUX in co-MPT scenarios (converging board: UBBP_U)



- Scenario 2: The converging board is a co-BBP board.
[Figure 5-5](#) shows an example with a UBBP_GU configured in slot 3.

Figure 5-5 GU CPRI MUX with GU co-BBP in co-MPT scenarios



Star Topology with GL CPRI MUX

- Scenario 1: The converging board is a single-mode board.

Figure 5-6 shows an example where GSM uses the non-baseband extension mode. In this example, the converging party is GSM and the converging board is a UBR1b.

Figure 5-6 GL CPRI MUX in co-MPT scenarios (converging board: UBR1b)

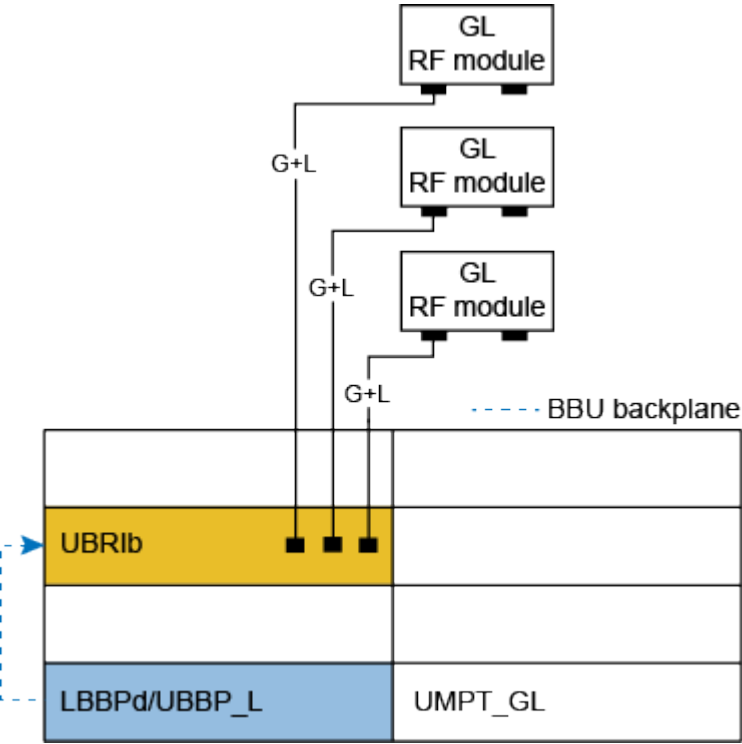
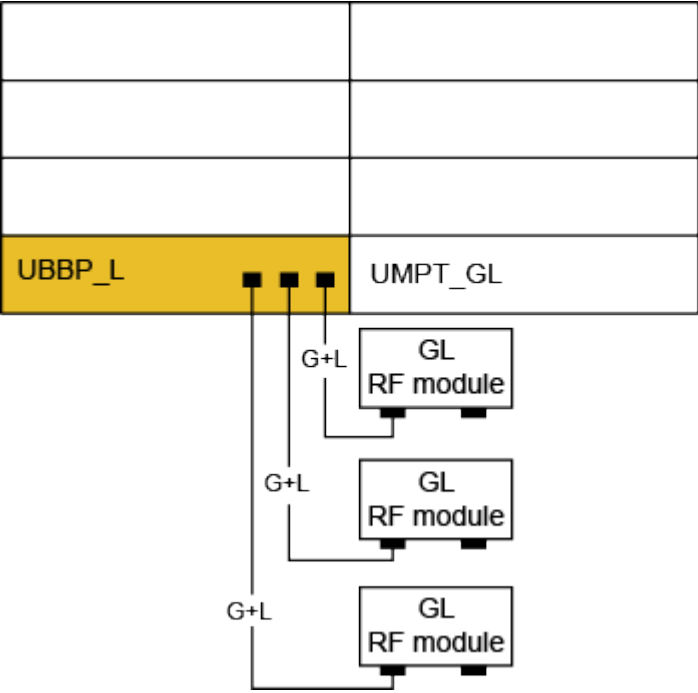


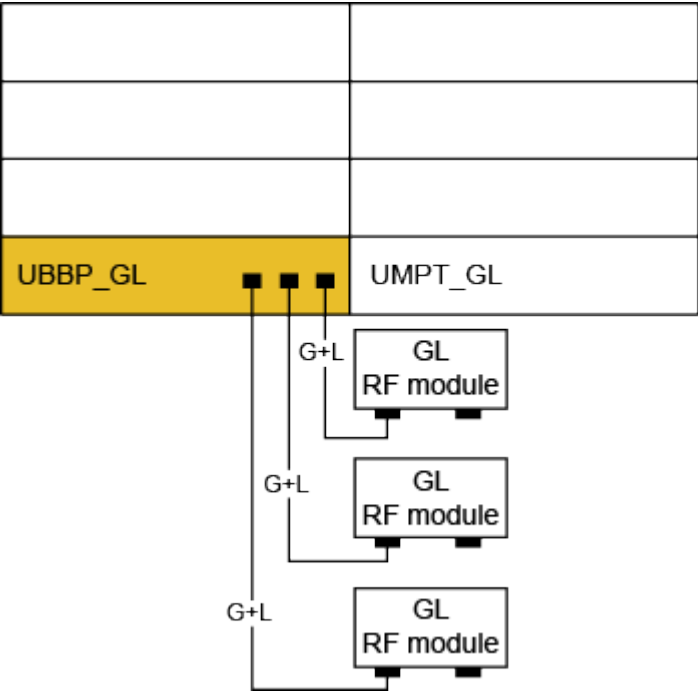
Figure 5-7 shows an example where GSM uses the non-baseband extension mode. In this example, the converging party is LTE and the converging board is a UBBP_L.

Figure 5-7 GL CPRI MUX in co-MPT scenarios (converging board: UBBP_L)



- Scenario 2: The converging board is a co-BBP board.
Figure 5-8 shows an example with a UBBP_GL configured in slot 3.

Figure 5-8 GL CPRI MUX with GL co-BBP in co-MPT scenarios



Star Topology with GUL CPRI MUX

- Scenario 1: The converging board is a single-mode board.

NOTE

- (1) In this scenario, UMTS and LTE cells must be configured on the specified baseband equipment group, and such a baseband equipment group must include only single-mode BBPs.

Figure 5-9 shows an example where GSM uses the non-baseband extension mode. In this example, the converging party is GSM and the converging board is a UBRib.

Figure 5-9 GUL CPRI MUX in co-MPT scenarios (converging board: UBRib)

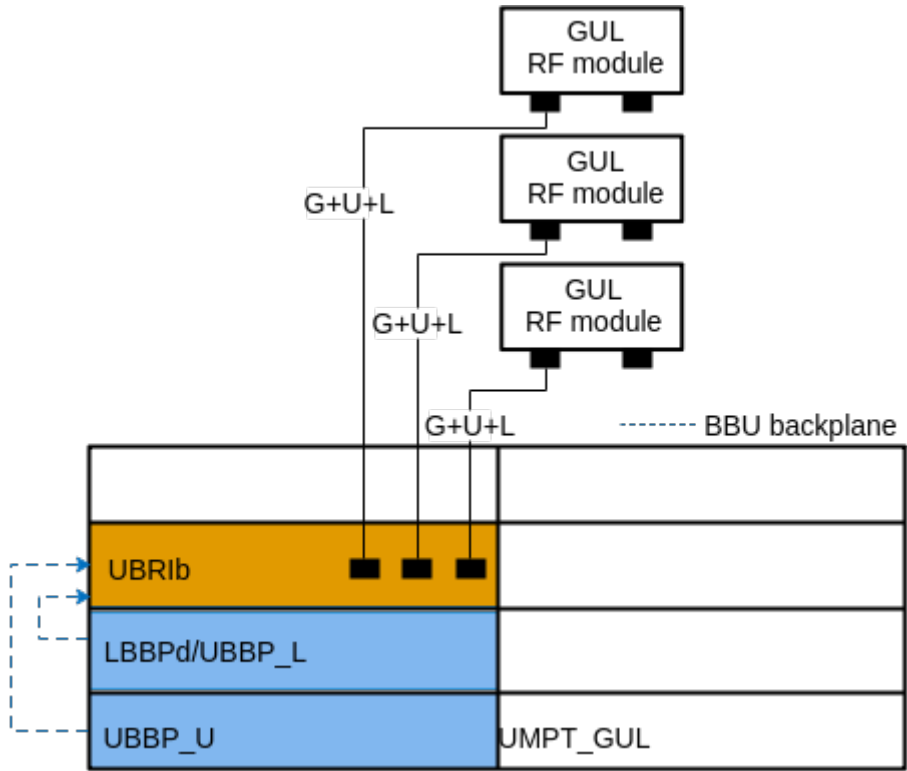
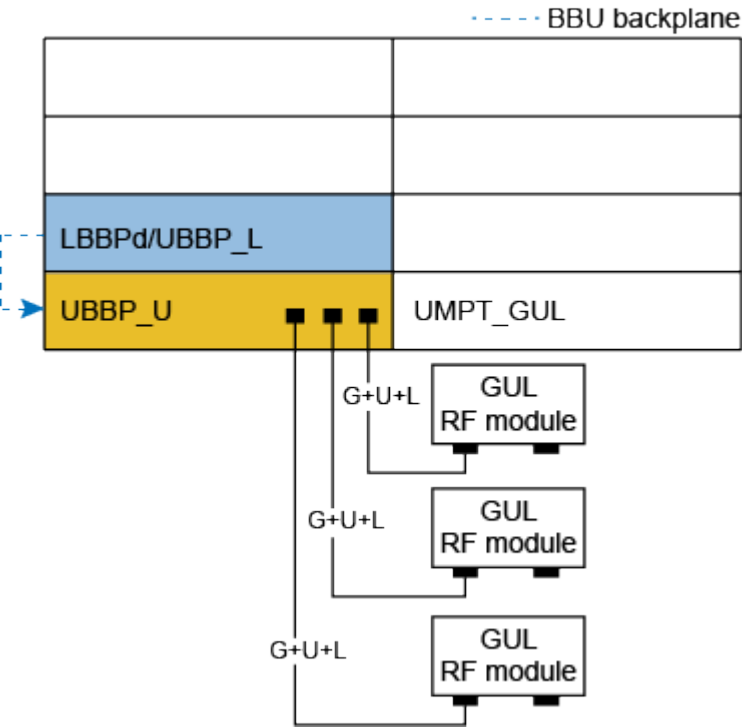


Figure 5-10 shows an example where GSM uses the non-baseband extension mode. In this example, the converging party is UMTS.

Figure 5-10 GUL CPRI MUX in co-MPT scenarios (converging party: UMTS)



- Scenario 2: The converging board is a co-BBP board.
Figure 5-11 shows an example where GSM uses the non-baseband extension mode. In this example, the converging party is UL.

Figure 5-11 GUL CPRI MUX in co-MPT scenarios (converging party: UL)

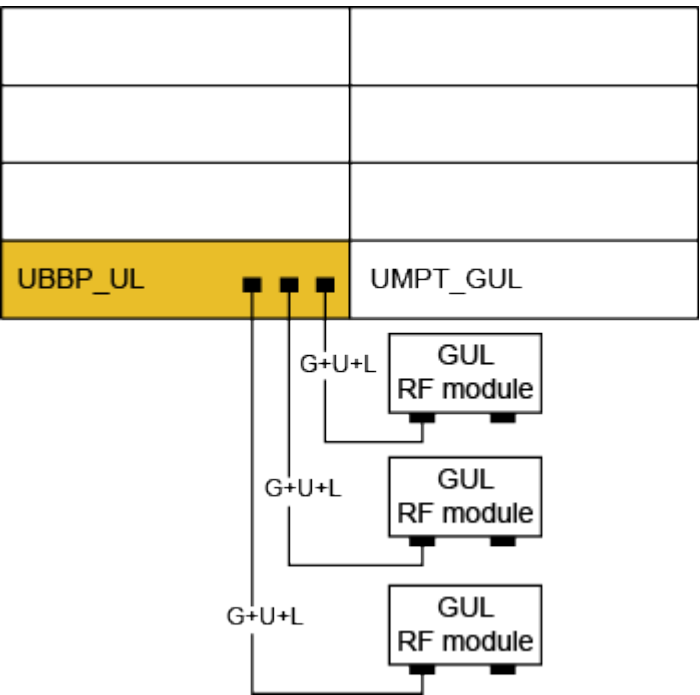
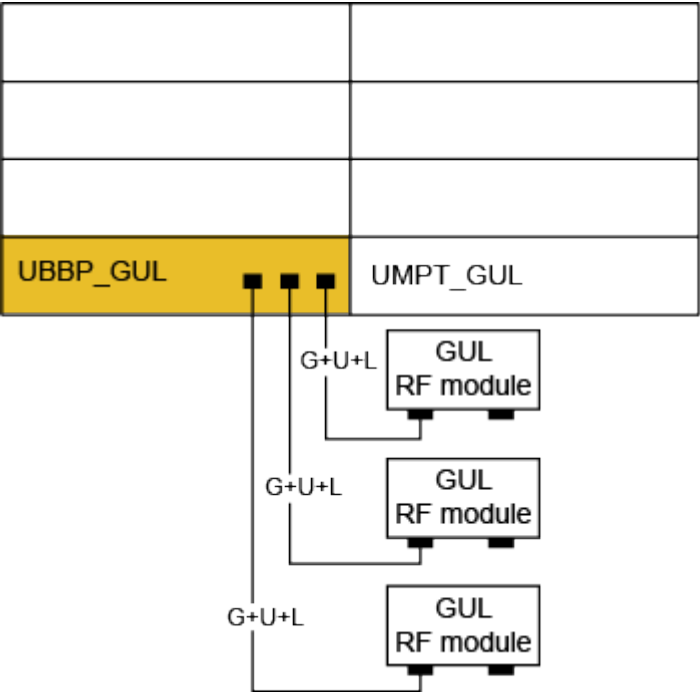


Figure 5-12 shows an example with a **UBBP_GUL** configured in slot 3.

Figure 5-12 GUL CPRI MUX with GUL co-BBP in co-MPT scenarios

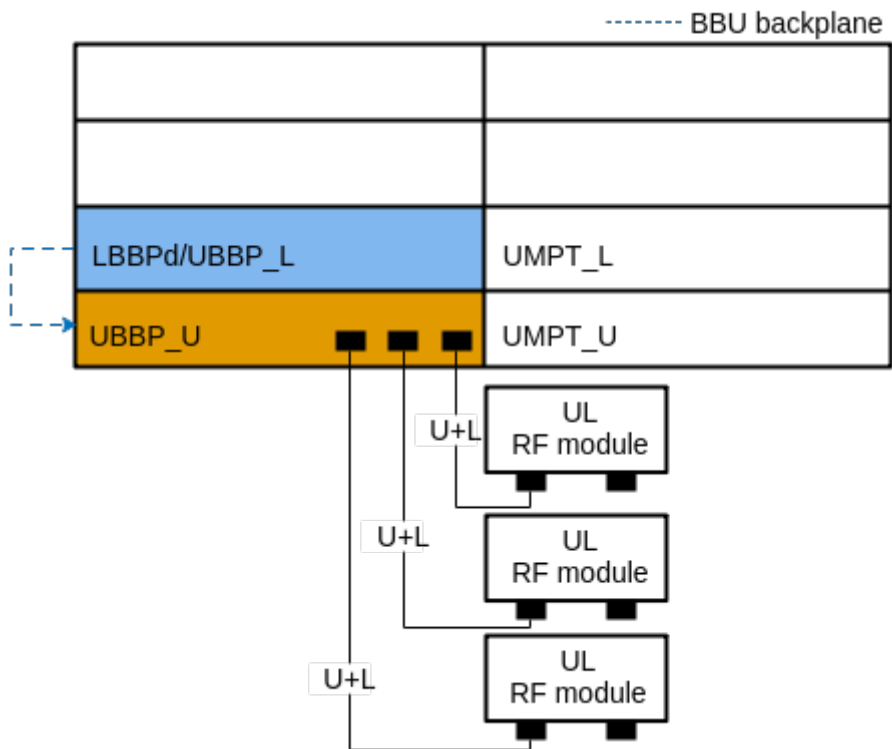


5.1.1.2 Separate-MPT Scenarios

Star Topology with UL CPRI MUX

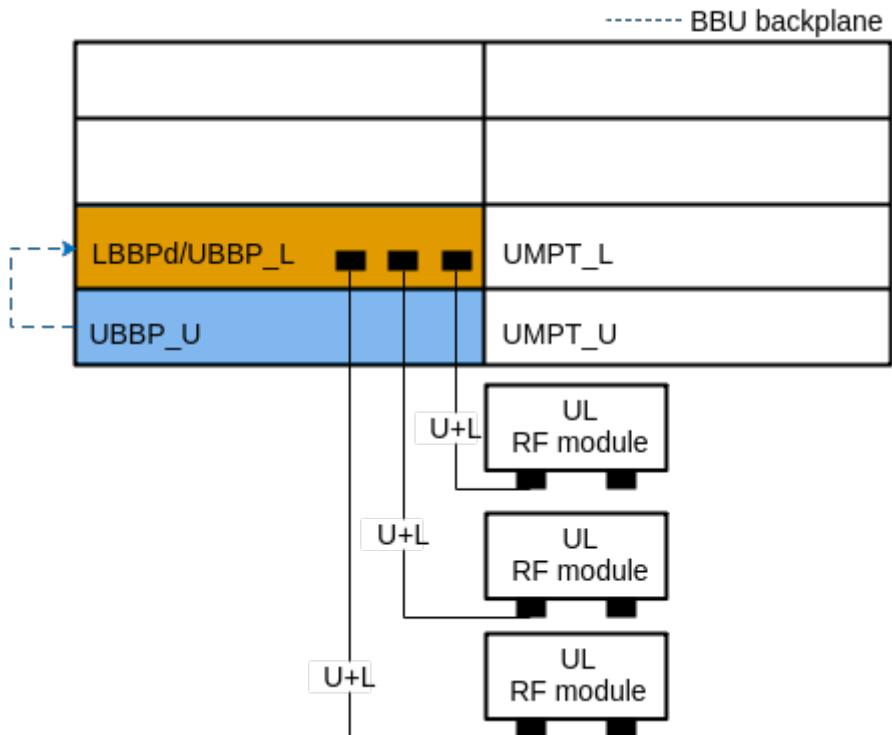
- Figure 5-13 shows an example. In this example, the converging party is UMTS and the converged party is LTE.

Figure 5-13 UL CPRI MUX in separate-MPT scenarios (converging party: UMTS)



- **Figure 5-14** shows an example. In this example, the converging party is LTE and the converged party is UMTS.

Figure 5-14 UL CPRI MUX in separate-MPT scenarios (converging party: LTE)



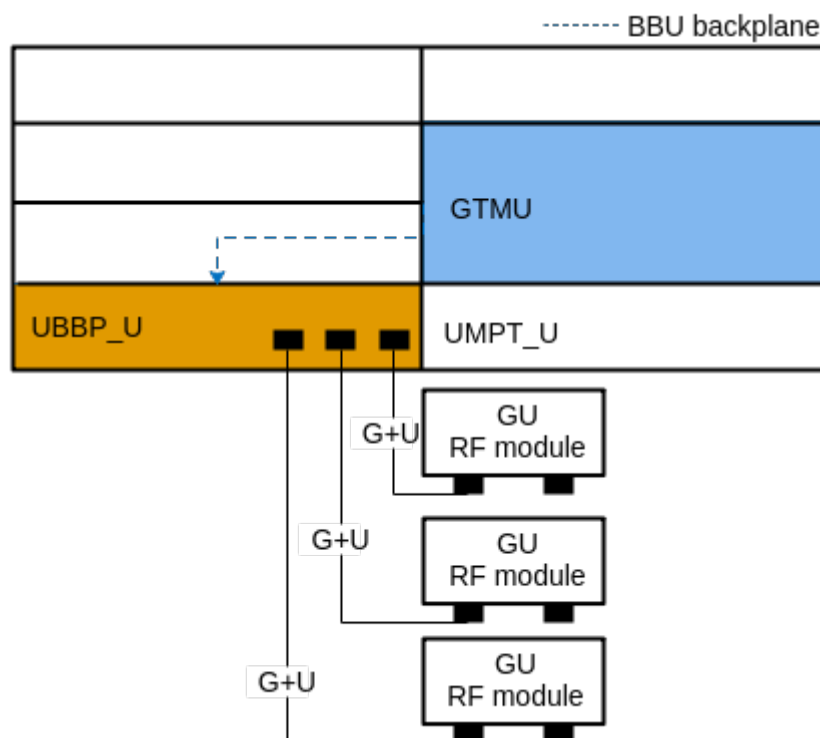
Star Topology with GU CPRI MUX

In GU CPRI MUX, the converging party must be UMTS and the converged party must be GSM. [Figure 5-15](#) shows an example.

NOTICE

In this scenario, data can be converged from slot 6 to either slot 2 or 3, but cannot to both slots.

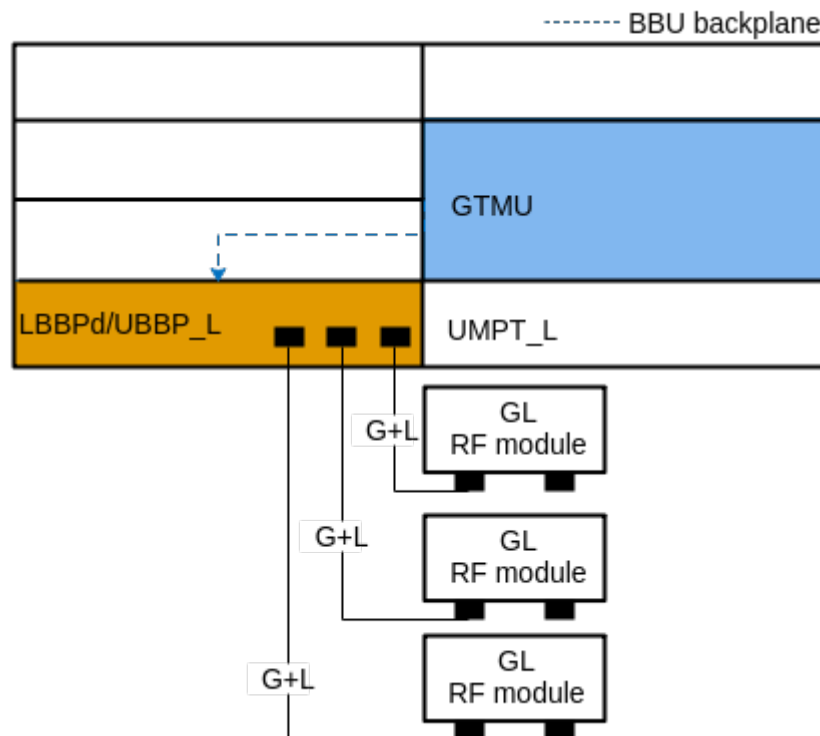
Figure 5-15 GU CPRI MUX in separate-MPT scenarios (converged party: GSM)



Star Topology with GL CPRI MUX

In GL CPRI MUX, the converging party must be LTE and the converged party must be GSM. [Figure 5-16](#) shows an example.

Figure 5-16 GL CPRI MUX in separate-MPT scenarios (converged party: GSM)



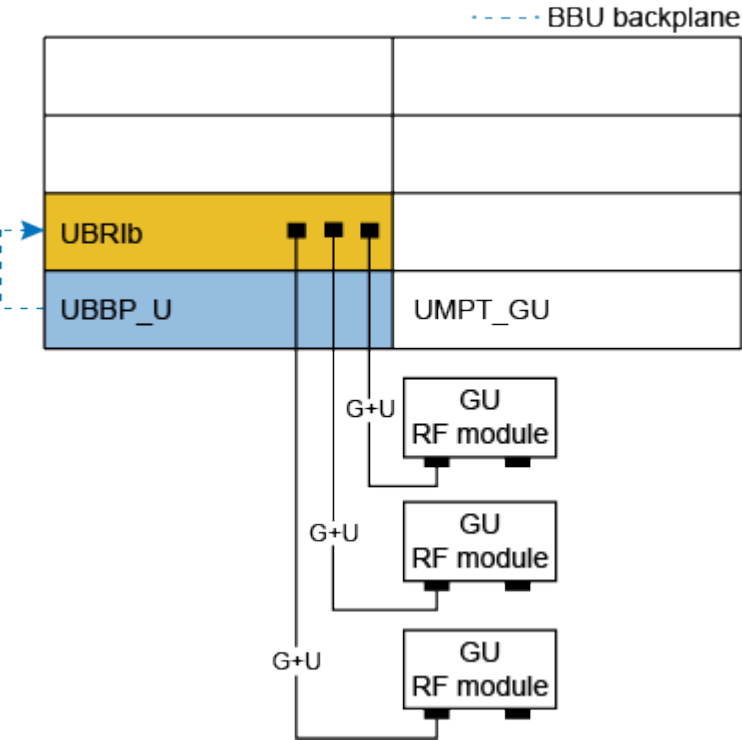
5.1.2 Star Topology with CPRI MUX (BBU3910)

5.1.2.1 Co-MPT Scenarios

This section describes the star topology with CPRI MUX where the converging board is a single-mode BBP, a UBRib, or a multimode BBP. In this section, the star topology with GU CPRI MUX is taken as an example.

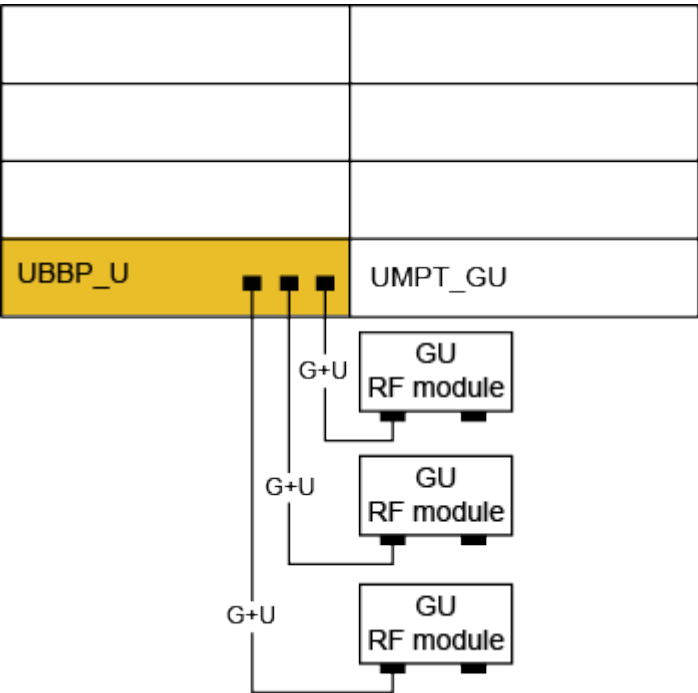
- **Figure 5-17** shows an example when the converging board is a UBRib.

Figure 5-17 GU CPRI MUX in co-MPT scenarios (converging board: UBRib)



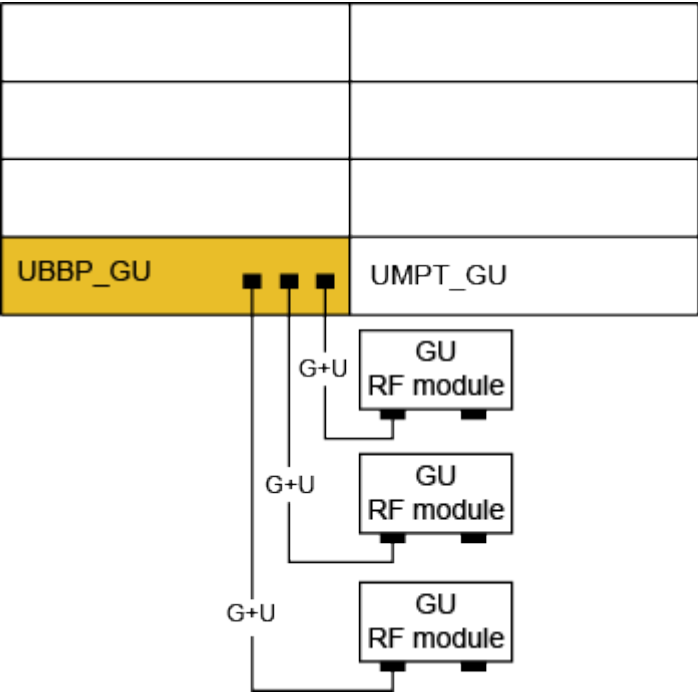
- Figure 5-18 shows an example when the converging board is a UBBP_U.

Figure 5-18 GU CPRI MUX in co-MPT scenarios (converging board: UBBP_U)



- Figure 5-19 shows an example when the converging board is a UBBP_GU.

Figure 5-19 GU CPRI MUX in co-MPT scenarios (converging board: UBBP_GU)

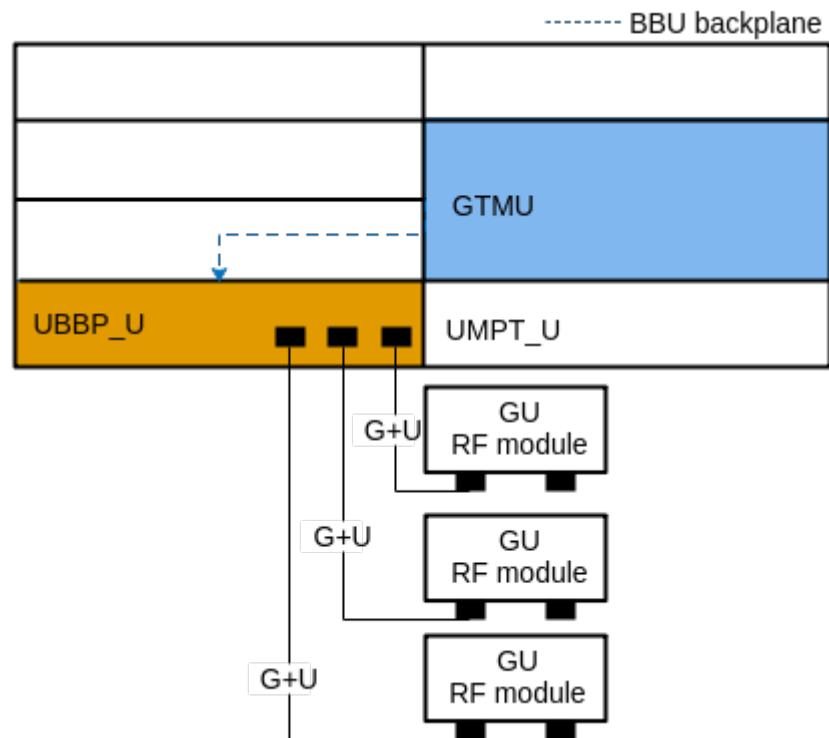


5.1.2.2 Separate-MPT Scenarios

Star Topology with GU or GL CPRI MUX

Figure 5-20 shows an example of GU CPRI MUX. In this example, the converging board is a UBBP_U.

Figure 5-20 GU CPRI MUX in separate-MPT scenarios

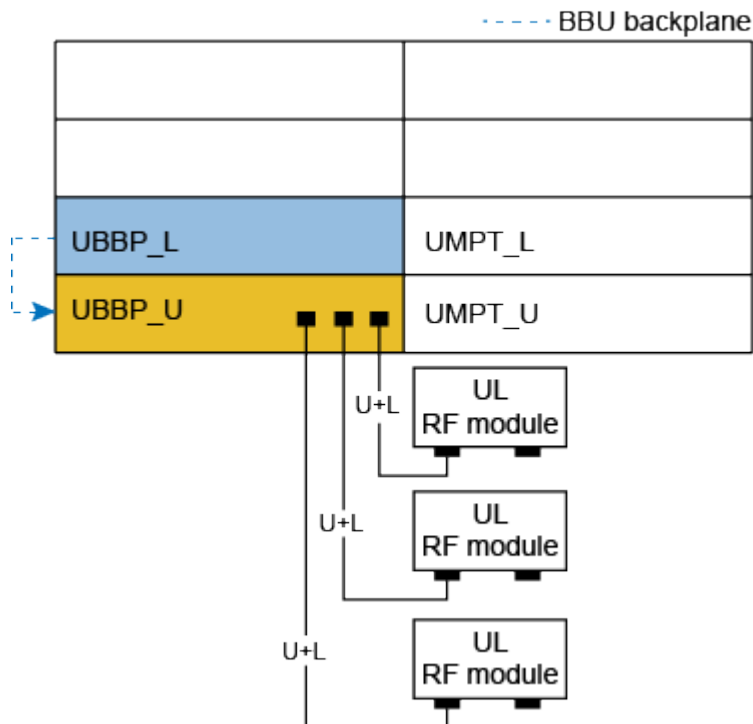


Star Topology with UL or LN CPRI MUX

- One convergence relationship

Figure 5-21 shows an example. In this example, the converging board is a UBBP_U.

Figure 5-21 UL CPRI MUX in separate-MPT scenarios (converging party: UMTS)



- Multiple convergence relationships

In a BBU3910, one converged board corresponds to one or two converging boards. A maximum of three one-to-one convergence relationships or two one-to-two convergence relationships are supported.

Figure 5-22 shows an example of three one-to-one convergence relationships. In this example, the converging party is LTE. In this figure, boards in slots 0 and 1 are converged and converging boards, respectively. These two boards form a group. Similarly, boards in slots 2 and 3 form a group and boards in slots 4 and 5 form a group.

Figure 5-22 UL CPRI MUX in separate-MPT scenarios with three one-to-one convergence relationships

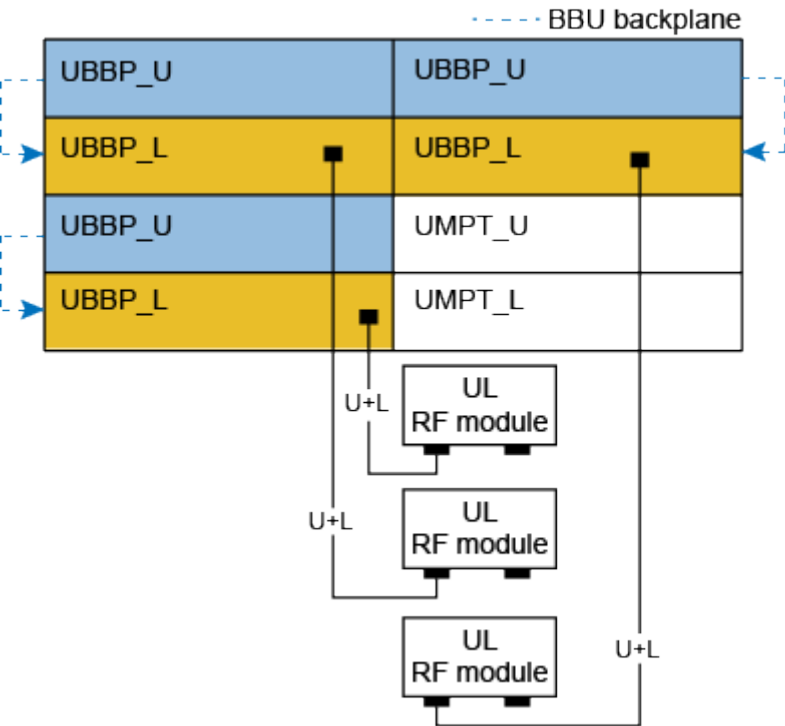
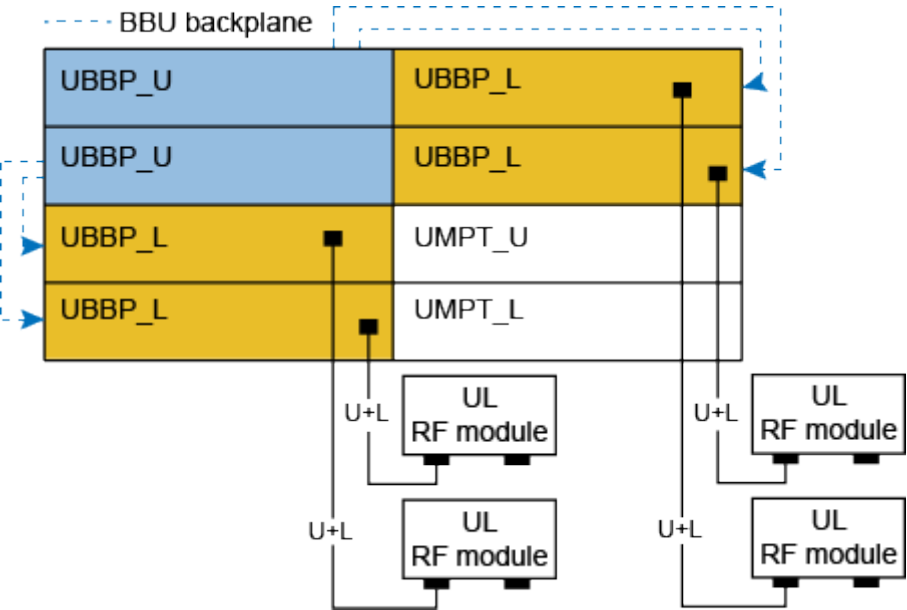


Figure 5-23 shows an example of two one-to-two convergence relationships. In this example, the converging party is LTE. In this figure, the board in slot 0 is a converged board, and boards in slots 4 and 5 are converging boards. These boards form a group. Similarly, boards in slot 1, 2, and 3 form a group.

Figure 5-23 UL CPRI MUX in separate-MPT scenarios with two one-to-two convergence relationships



5.1.2.3 Hybrid-MPT Scenarios

Star Topology with ULN CPRI MUX

Figure 5-24 shows an example in a $U\&[L*N]$ hybrid-MPT scenario when the converging board is the UBBP_N. Data convergence from UBBP_L to UBBP_N and that from UBBP_U to UBBP_N apply to CPRI MUX in co-MPT scenarios and CPRI MUX in separate-MPT scenarios, respectively.

Figure 5-24 ULN CPRI MUX in hybrid-MPT scenarios (converging board: UBBP_N)

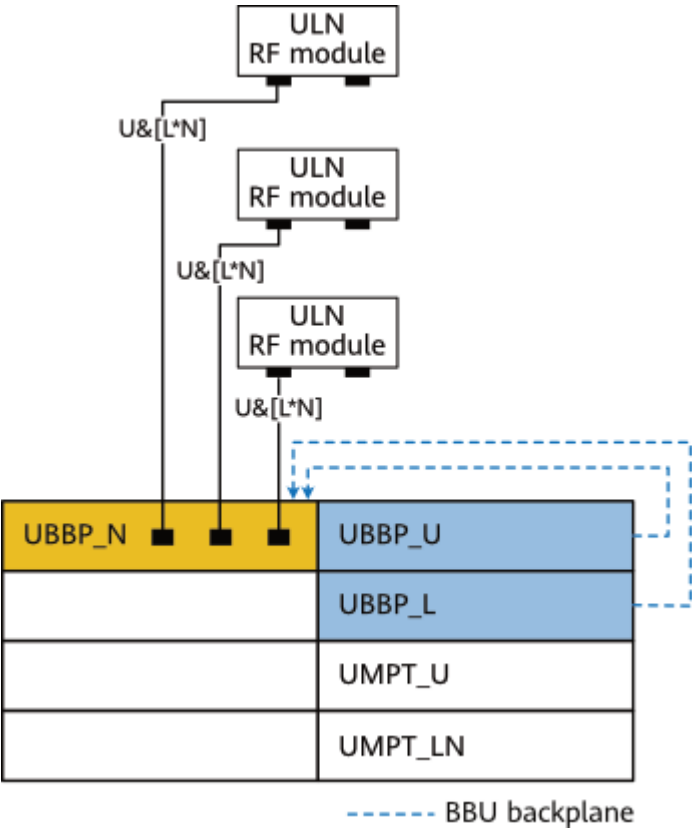
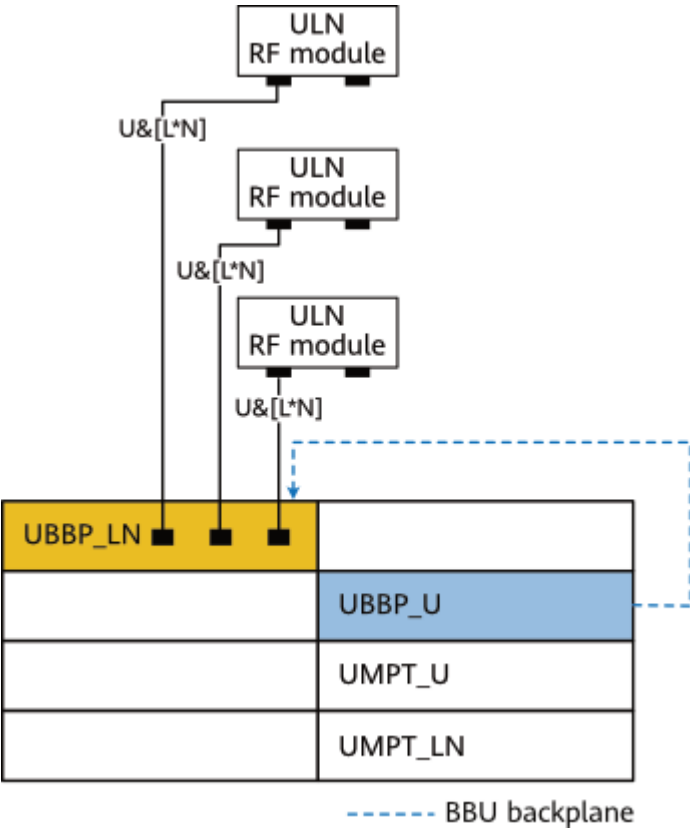


Figure 5-25 shows an example in a $U\&[L*N]$ hybrid-MPT scenario when the converging board is the UBBP_LN.

Figure 5-25 ULN CPRI MUX in hybrid-MPT scenarios (converging board: UBBP_LN)



Star Topology with GULN CPRI MUX

Figure 5-26 shows an example in a $[G*U]&[L*N]$ hybrid-MPT scenario when the converging board is the UBBP_N.

Figure 5-26 GULN CPRI MUX in hybrid-MPT scenarios (converging board: UBBP_N)

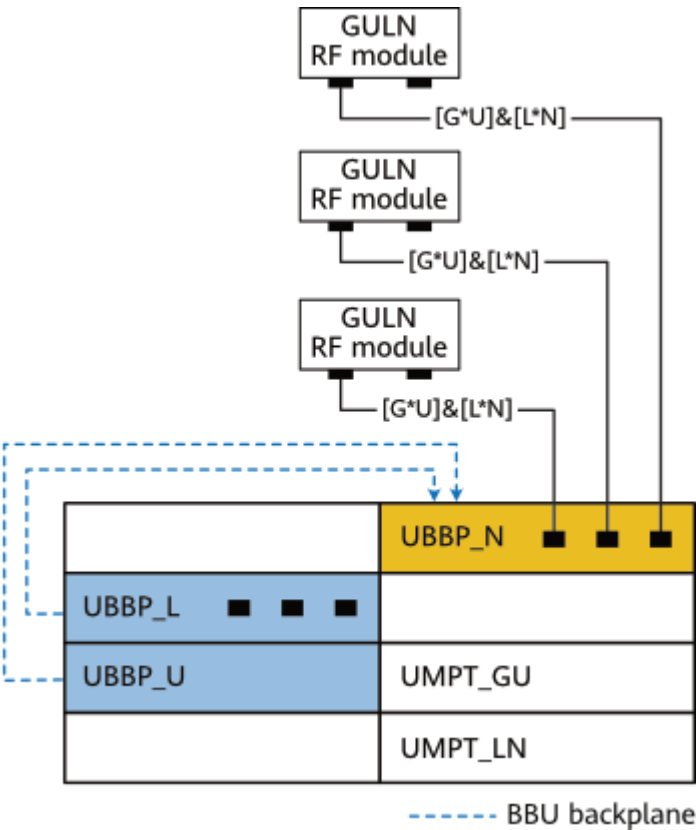
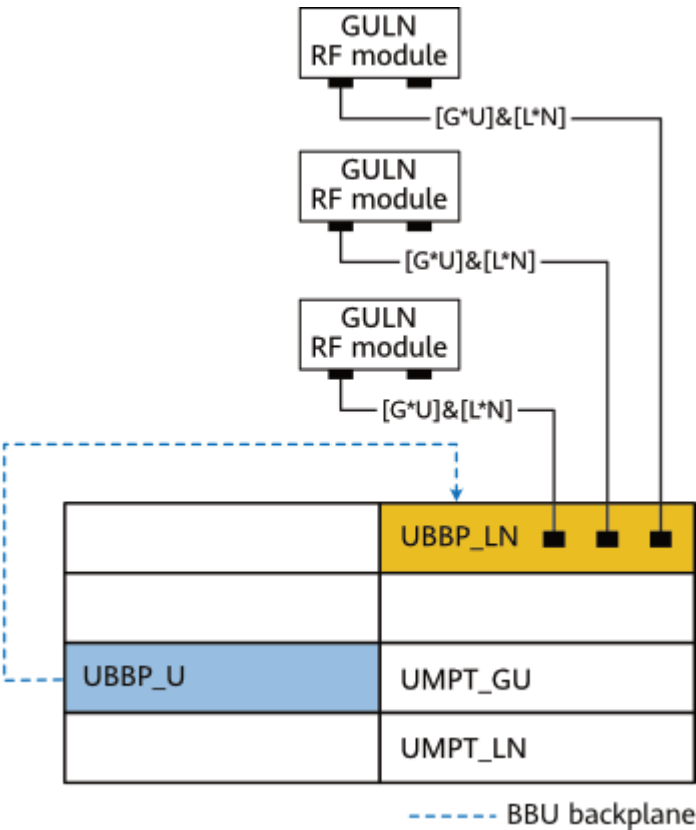


Figure 5-27 shows an example in a [G*U]&[L*N] hybrid-MPT scenario when the converging board is the UBBP_LN.

Figure 5-27 GULN CPRI MUX in hybrid-MPT scenarios (converging board: UBBP_LN)



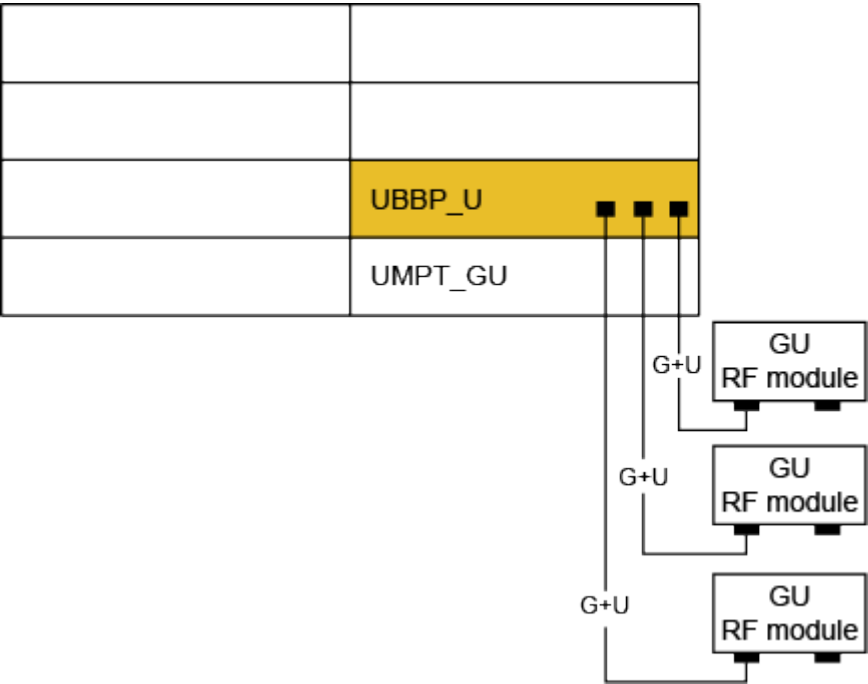
5.1.3 Star Topology with CPRI MUX (BBU5900/BBU5910)

5.1.3.1 Co-MPT Scenarios

Star Topology with GU CPRI MUX

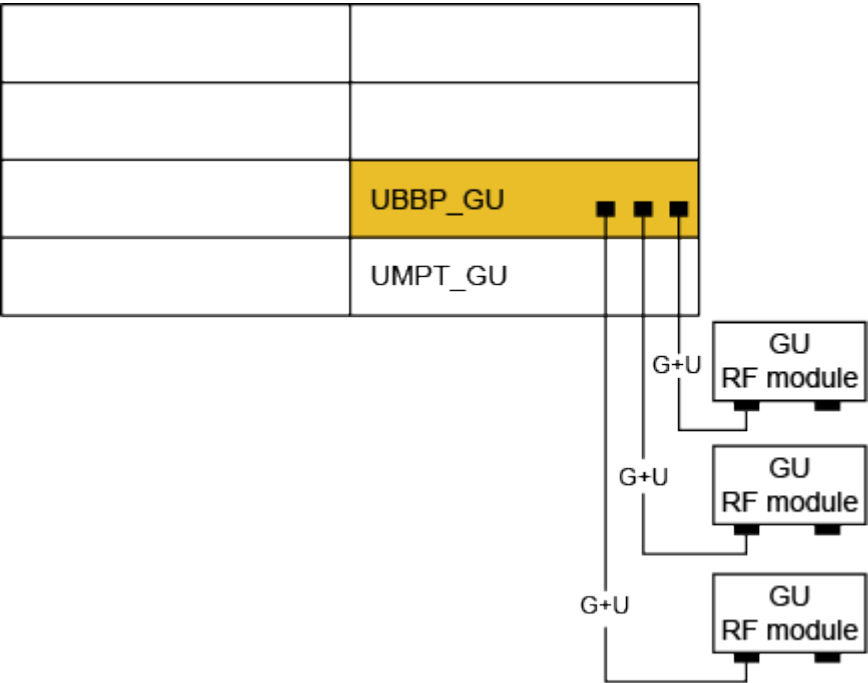
- **Figure 5-28** shows an example. In this example, the converging board is a UBBP_U.

Figure 5-28 GU CPRI MUX in co-MPT scenarios (converging board: UBBP_U)



- **Figure 5-29** shows an example. In this example, the converging board is a UBBP_GU.

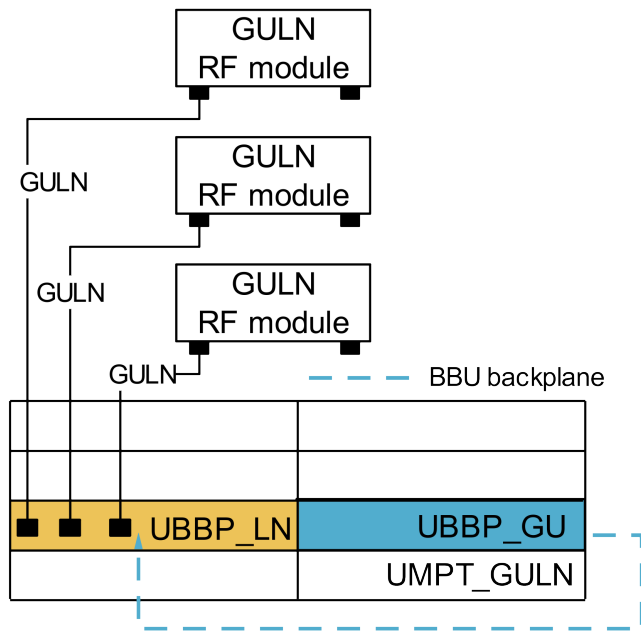
Figure 5-29 GU CPRI MUX in co-MPT scenarios (converging board: UBBP_GU)



Star Topology with GULN CPRI MUX

- **Figure 5-30** shows an example. In this example, the converging board is a UBBP_LN.

Figure 5-30 GULN CPRI MUX in co-MPT scenarios (converging board: UBBP_LN)

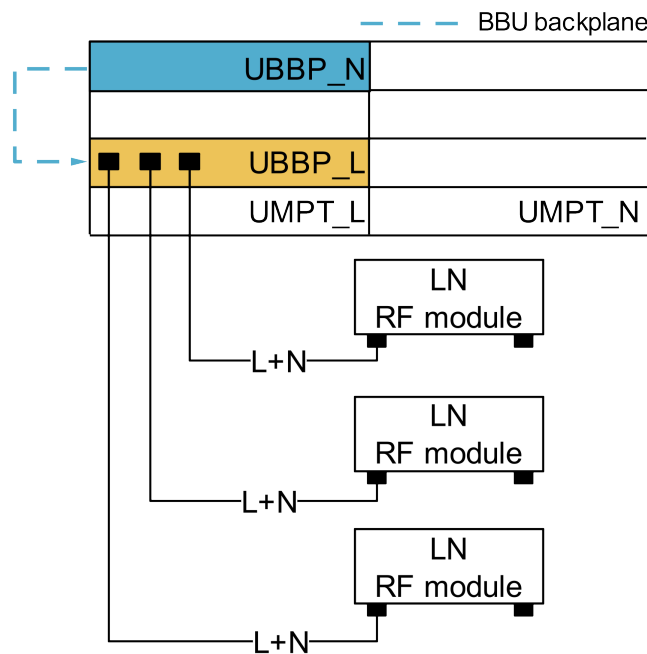


5.1.3.2 Separate-MPT Scenarios

Star Topology with UL or LN CPRI MUX

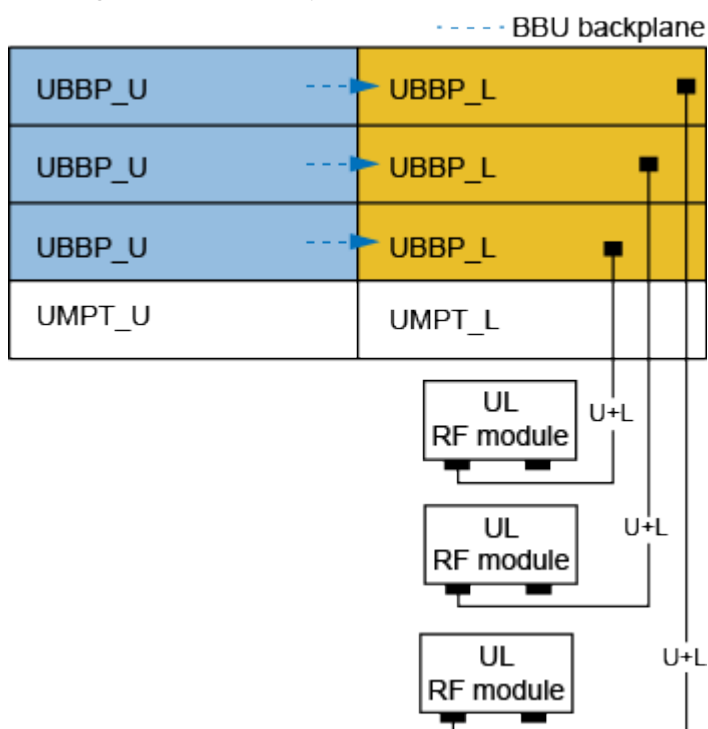
- One convergence relationship
Figure 5-31 shows an example. In this example, the converging board is UBBP_L and the networking is LN CPRI MUX.

Figure 5-31 LN CPRI MUX in separate-MPT scenarios (converging board: UBBP_L)



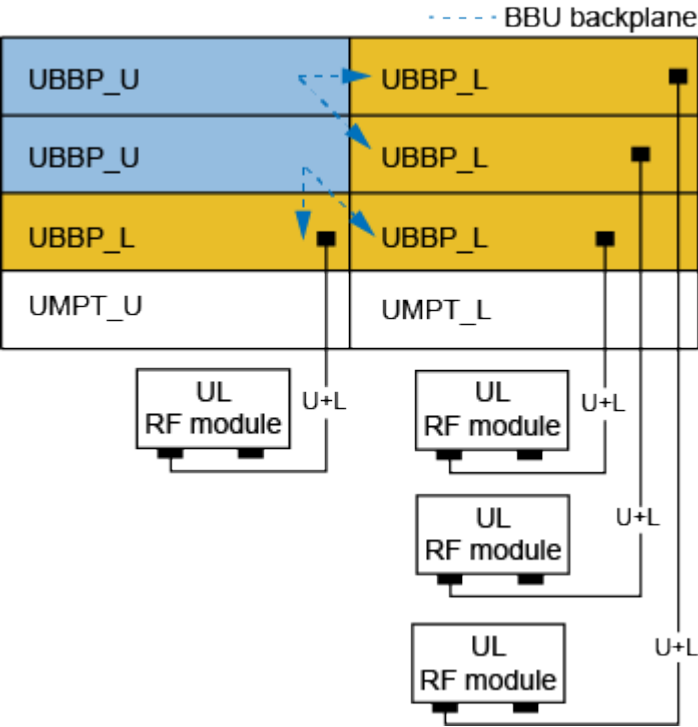
- Multiple convergence relationships
In a BBU5900/BBU5910, one converged board can correspond to one or two converging boards. A maximum of three one-to-one convergence relationships or two one-to-two convergence relationships are supported.
Scenario 1: [Figure 5-32](#) shows an example of three one-to-one convergence relationships. In this figure, boards in slots 0 and 1 are converged and converging boards, respectively. These two boards form a group. Similarly, boards in slots 2 and 3 form a group and boards in slots 4 and 5 form a group.

Figure 5-32 UL CPRI MUX in separate-MPT scenarios with three one-to-one convergence relationships



Scenario 2: [Figure 5-33](#) shows an example of two one-to-two convergence relationships. In this figure, the board in slot 0 is a converged board, and boards in slots 1 and 3 are converging boards. These three boards form a group. Similarly, boards in slots 2, 4, and 5 form a group.

Figure 5-33 UL CPRI MUX in separate-MPT scenarios with two one-to-two convergence relationships



5.1.3.3 Hybrid-MPT Scenarios

Star Topology with ULN CPRI MUX

Figure 5-34 shows an example in a U&[L*N] hybrid-MPT scenario when the converging board is the UBBP_N. Data convergence from UBBP_L to UBBP_N and that from UBBP_U to UBBP_N apply to CPRI MUX in co-MPT scenarios and CPRI MUX in separate-MPT scenarios, respectively.

Figure 5-34 ULN CPRI MUX in hybrid-MPT scenarios (converging board: UBBP_N)

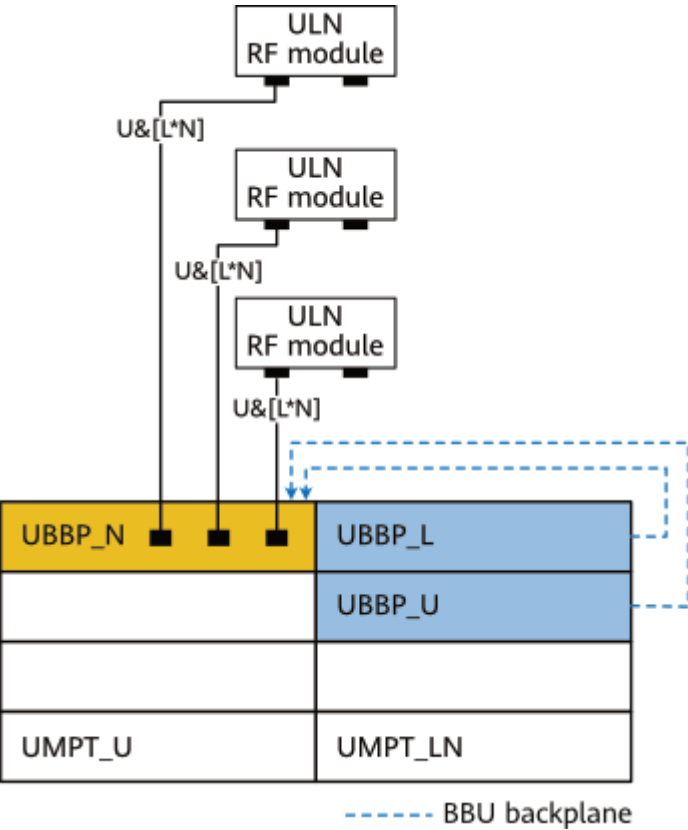
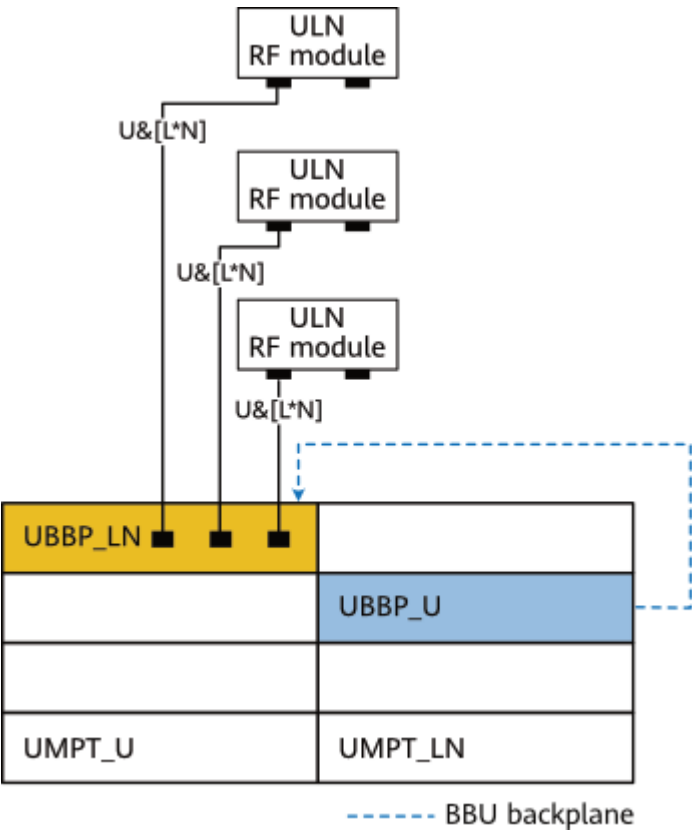


Figure 5-35 shows an example in a U&[L*N] hybrid-MPT scenario when the converging board is the UBBP_LN.

Figure 5-35 ULN CPRI MUX in hybrid-MPT scenarios (converging board: UBBP_LN)



Star Topology with GULN CPRI MUX

Figure 5-36 shows an example in a [G*U]&[L*N] hybrid-MPT scenario when the converging board is the UBBP_N.

Figure 5-36 GULN CPRI MUX in hybrid-MPT scenarios (converging board: UBBP_N)

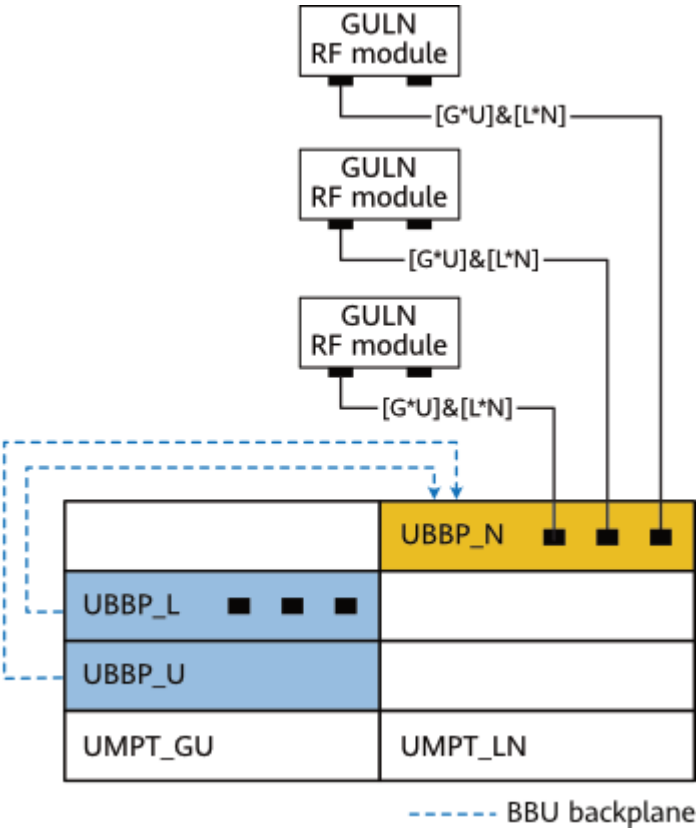
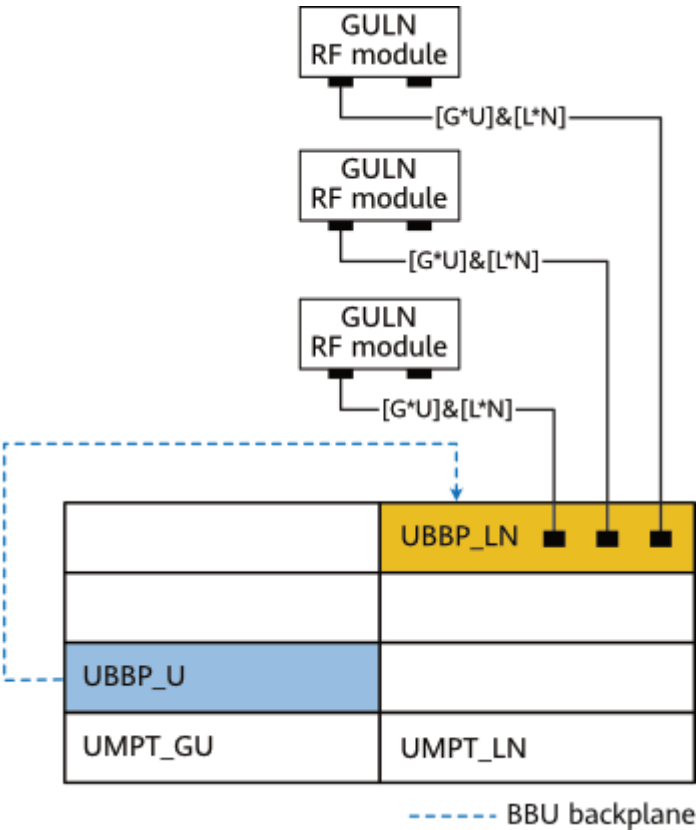


Figure 5-37 shows an example in a [G*U]&[L*N] hybrid-MPT scenario when the converging board is the UBBP_LN.

Figure 5-37 GULN CPRI MUX in hybrid-MPT scenarios (converging board: UBBP_LN)

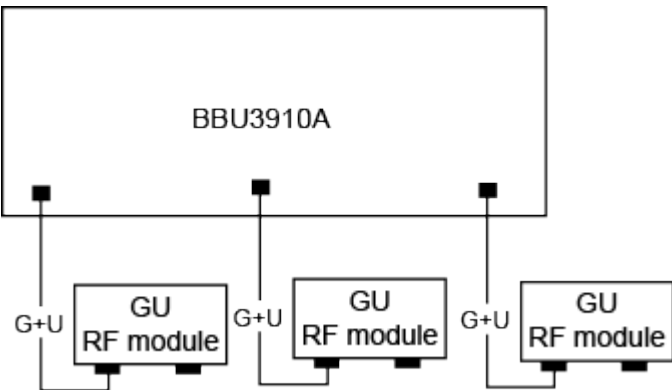


5.1.4 Star Topology with CPRI MUX (BBU3910A)

Star Topology with GU CPRI MUX

Figure 5-38 shows an example of GU CPRI MUX.

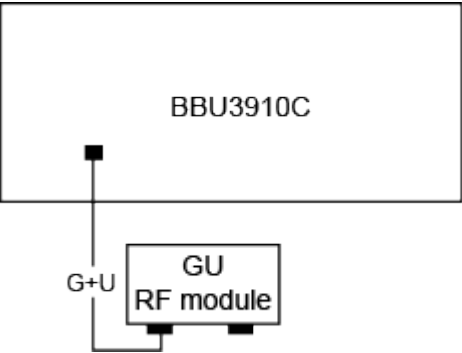
Figure 5-38 GU CPRI MUX used with a BBU3910A in co-MPT scenarios



5.1.5 Star Topology with CPRI MUX (BBU3910C)

Figure 5-39 uses GU CPRI MUX as an example.

Figure 5-39 GU CPRI MUX used with a BBU3910C in co-MPT scenarios

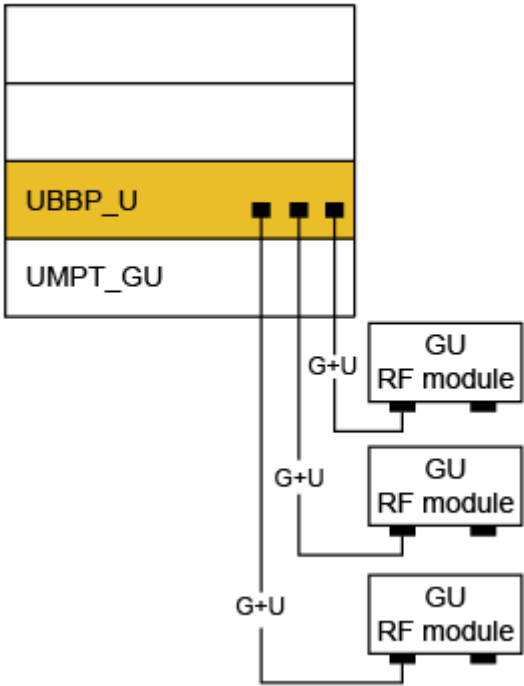


5.1.6 Star Topology with CPRI MUX (BBU5900A)

Star Topology with GU CPRI MUX

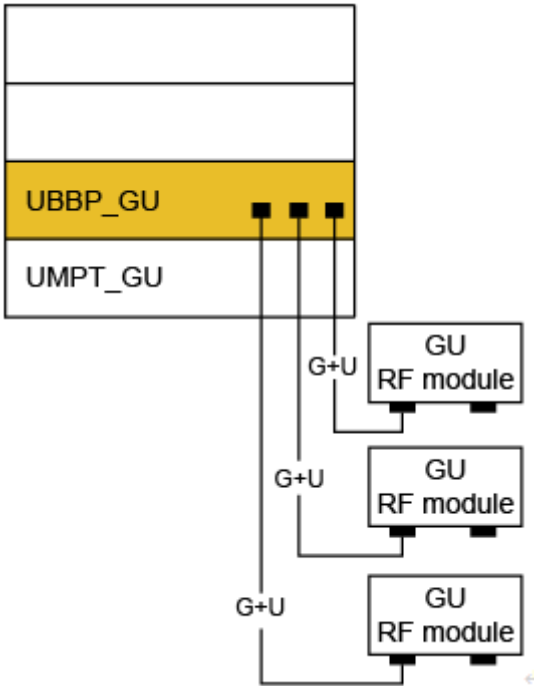
- The following figure shows an example. In this example, the converging board is a UBBP_U.

Figure 5-40 GU CPRI MUX in co-MPT scenarios (converging board: UBBP_U)



- The following figure shows an example. In this example, the converging board is a UBBP_GU.

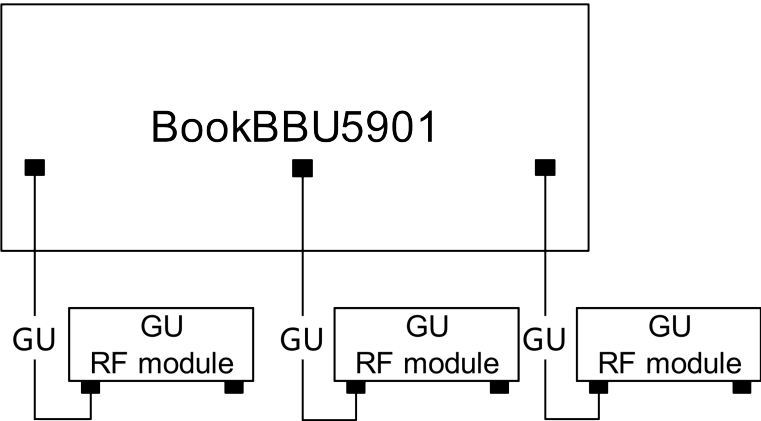
Figure 5-41 GU CPRI MUX in co-MPT scenarios (converging board: UBBP_GU)



5.1.7 Star Topology with CPRI MUX (BookBBU5901)

Figure 5-42 shows an example of GU CPRI MUX.

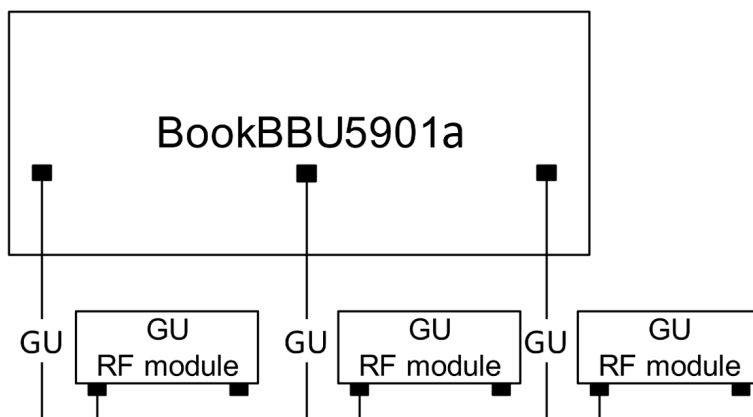
Figure 5-42 GU CPRI MUX used with a BookBBU5901 in co-MPT scenarios



5.1.8 Star Topology with CPRI MUX (BookBBU5901a)

Figure 5-43 shows an example of GU CPRI MUX.

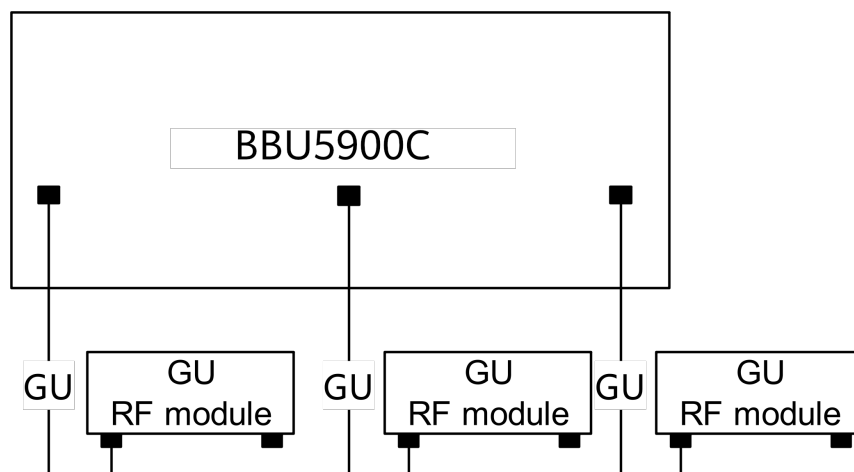
Figure 5-43 GU CPRI MUX used with a BookBBU5901a in co-MPT scenarios



5.1.9 Star Topology with CPRI MUX (BBU5900C)

Figure 5-44 shows an example of GU CPRI MUX.

Figure 5-44 GU CPRI MUX used with a BBU5900C in co-MPT scenarios



5.2 Chain Topology with CPRI MUX

The chain topology with CPRI MUX adopts the same BBU configurations as the star topology. The difference is that in a chain topology, multimode RF modules operating in the same or different network modes can be cascaded.

NOTE

- This section describes RF module cascading solutions and constraints. For BBU configurations, see [4 Boards That Support CPRI MUX and Constraints](#) and [5.1 Star Topology with CPRI MUX](#).
- If a chain contains RF modules whose working mode includes GSM, first activate cells served by such RF modules that are configured at the upper levels, that is, closer to the BBP.
- All RF modules on a chain must work at the same rate.

Cascading Specifications

Table 5-1 lists the maximum number of cascading levels in a chain topology with CPRI MUX.

Table 5-1 Maximum number of cascading levels in a chain topology with CPRI MUX

Network Mode	Maximum Number of Cascading Levels
GU	6
GL	4
UL	4
GUL	4
LN	4
ULN	4

NOTE

- In specific scenarios, the maximum number of cascading levels also depends on factors such as the cell specifications of each network mode and the CPRI MUX access capabilities of the converging party. For details on the CPRI MUX access capabilities, see [6 Specifications of CPRI MUX Capabilities](#).
- If configurations of a base station exceed CPRI MUX access capabilities, the status of certain cells served by this base station may be abnormal.
- For the maximum level of cascading and the maximum distance between an RF module and a BBU, see the description of CPRI port specifications in *RRU Technical Specifications* or *AAU Technical Specifications* for the RF module in *3900 & 5900 Series Base Station Product Documentation*.

5.2.1 Intra-Mode Cascading

Definition and Constraints

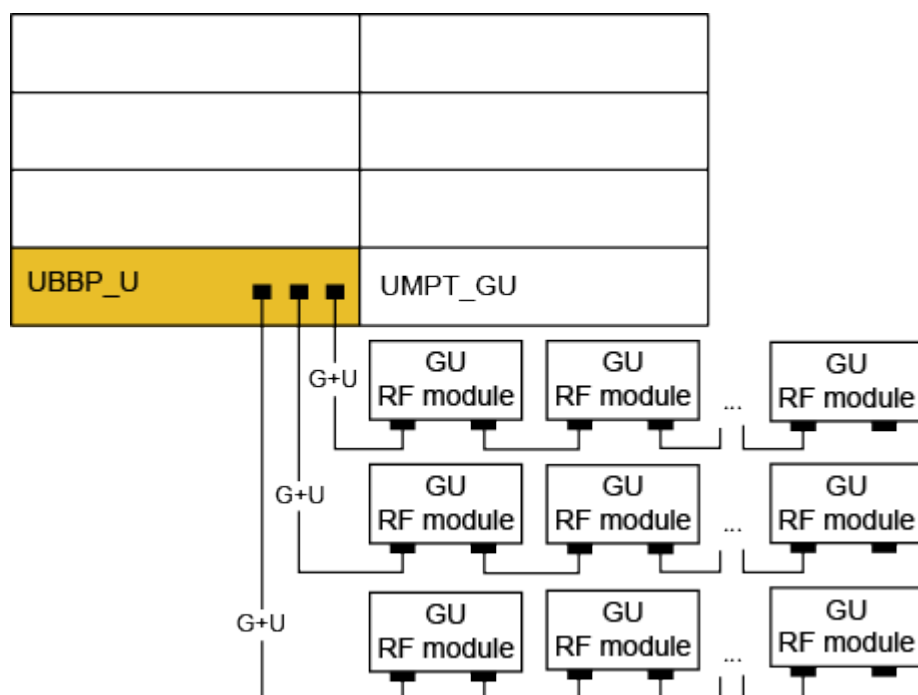
In intra-mode cascading scenarios, all cascaded multimode RF units are working in the same mode.

The intra-mode cascading solution applies to both co-MPT and separate-MPT scenarios. This recommended solution features few limitations in terms of application and is implemented in a broad range of scenarios.

Example

Figure 5-45 shows an example of GU CPRI MUX where a BBU3900 is used and GSM uses the non-baseband extension mode. In this example, the converging party is UMTS and the converging board is a UBBP_U. Multimode RF modules on the chains are all configured to work in GU mode, which is the case on the live network.

Figure 5-45 Intra-mode cascading chain with GU CPRI MUX in co-MPT scenarios where a BBU3900 is used and GSM uses the non-baseband extension mode (converging party: UMTS; converging board: UBBP_U)



5.2.2 Inter-Mode Cascading

Introduction

In inter-mode cascading scenarios, cascaded multimode RF modules work in different modes. The inter-mode cascading solution is provided for use in specific scenarios. Contact Huawei engineers for analysis based on the topology.

NOTE

On an inter-mode cascading chain, if an RF module whose working mode includes GSM is configured with GSM TRXs, it must be configured at the upper level, that is, closer to a BBP.

5.2.2.1 Co-MPT Scenarios

Constraints

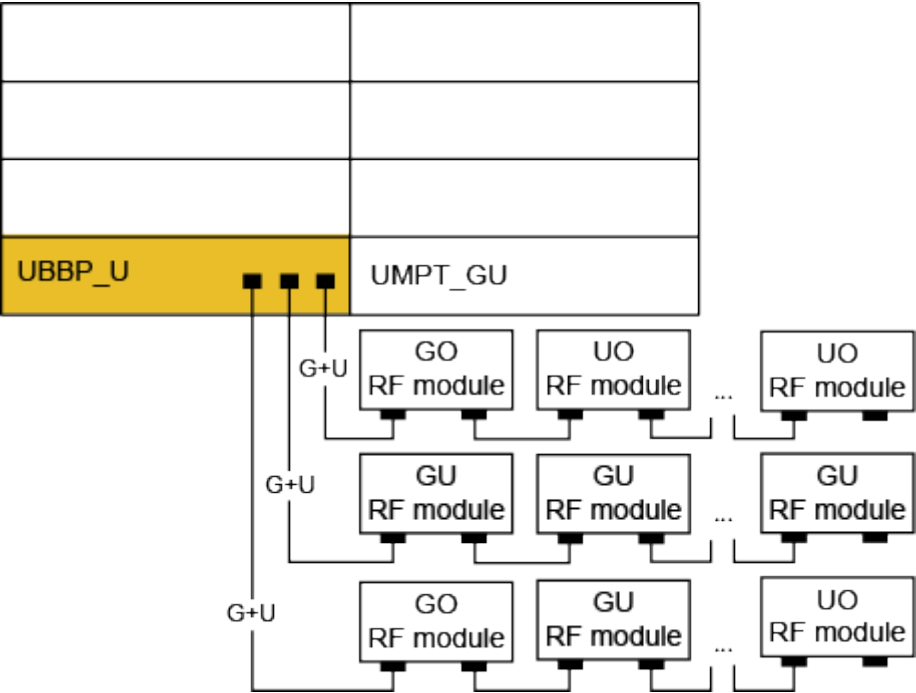
1. If GSM is included in the working mode of any multimode RF module on an inter-mode cascading chain, all multimode RF modules on this chain must support GSM. In other words, if any multimode RF module on such a chain works in GO, GU, GL, GUL, GLN, or GULN mode, then this chain can only contain GU, GL, GUL, GLN, and GULN RF modules. [Figure 5-46](#) shows an example.
2. If GSM is excluded from the working modes of all multimode RF modules on an inter-mode cascading chain, this chain can contain any type of multimode RF modules. In other words, if all multimode RF modules on such a chain

work in UO, LO, NO, UL, UN, LN, or ULN mode, then this chain can contain any type of RF modules. [Figure 5-47](#) shows an example.

Example

- The following figure takes GU CPRI MUX in co-MPT scenarios as an example of constraint 1. In this example, a BBU3900 is used, GSM uses the non-baseband extension mode, the converging party is UMTS, and the converging board is a UBBP_U.

Figure 5-46 Inter-mode cascading chain with GU CPRI MUX in co-MPT scenarios where a BBU3900 is used and GSM uses the non-baseband extension mode (converging party: UMTS; converging board: UBBP_U)

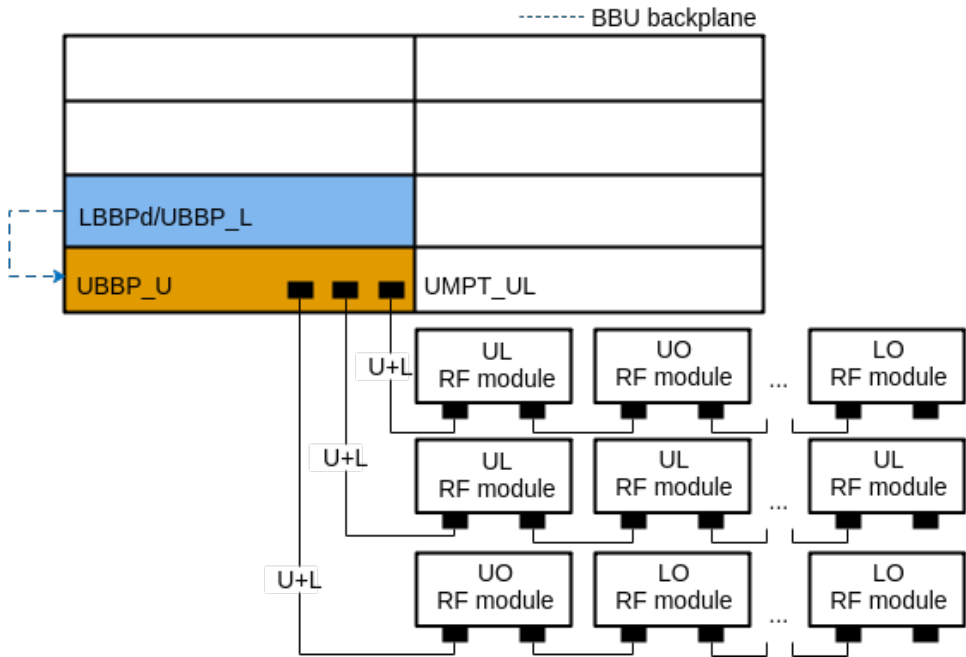


NOTE

In this figure, UO RF modules are multimode RF modules that work in UO mode and also support GSM. In other words, they are GU or GUL RF modules.

- The following figure takes UL CPRI MUX in co-MPT scenarios as an example of constraint 2. In this example, a BBU3900 is used and the converging party is UMTS.

Figure 5-47 Inter-mode cascading chain with UL CPRI MUX in co-MPT scenarios where a BBU3900 is used (converging party: UMTS)



5.2.2.2 Separate-MPT Scenarios

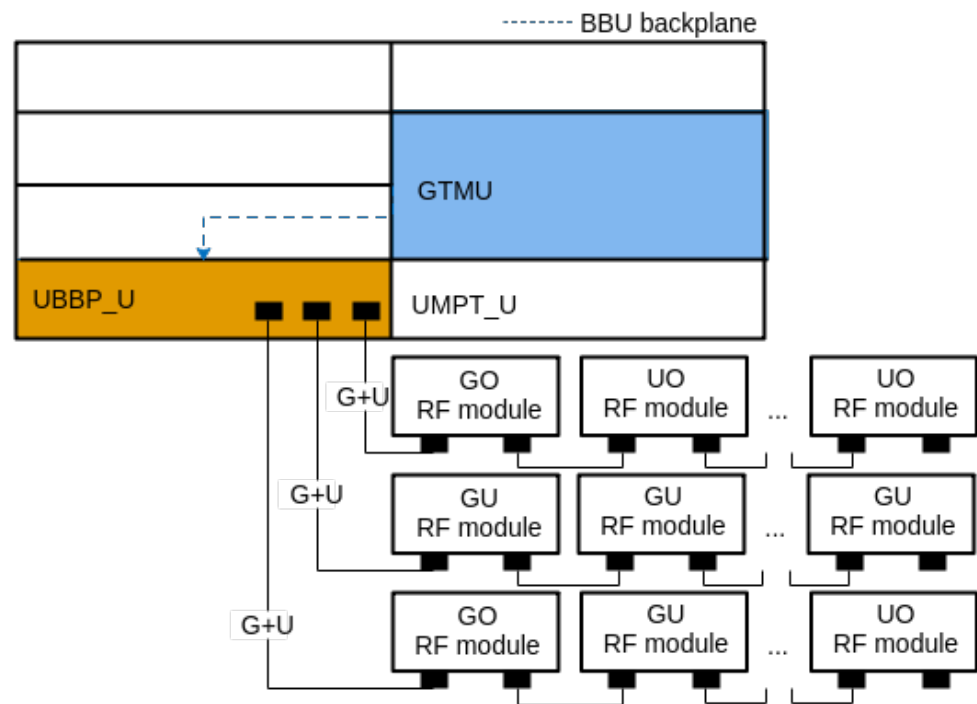
Constraints

In separate-MPT scenarios, an inter-mode cascading chain can contain multimode RF modules that work in a single network mode or dual network modes. These RF modules must support both converging and converged parties.

Example

Figure 5-48 takes GU CPRI MUX in separate-MPT scenarios as an example. In this example, multimode RF modules on the inter-mode cascading chains must be GU or GUL RF modules that work in GO, UO, or GU mode.

Figure 5-48 Inter-mode cascading chain with GU CPRI MUX in separate-MPT scenarios where a BBU3900 is used (converging party: UMTS)



5.3 Load Sharing Topology with CPRI MUX

This topology is one of multimode load sharing topologies. In the load sharing topology with CPRI MUX, a multimode RF module has two CPRI links. These CPRI links together carry services of two or more network modes, and one of them carry services of multiple network modes on its own.

Currently, the load sharing topology with CPRI MUX is supported in co-MPT scenarios using multimode RF modules.

Figure 5-49 shows an example. The two CPRI links carry GU+L services. Specifically, one carries GU services and the other carries LTE services.

Figure 5-49 GU+L multimode load sharing topology

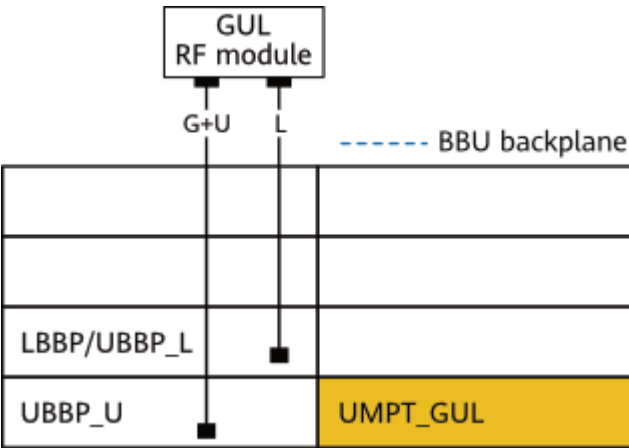
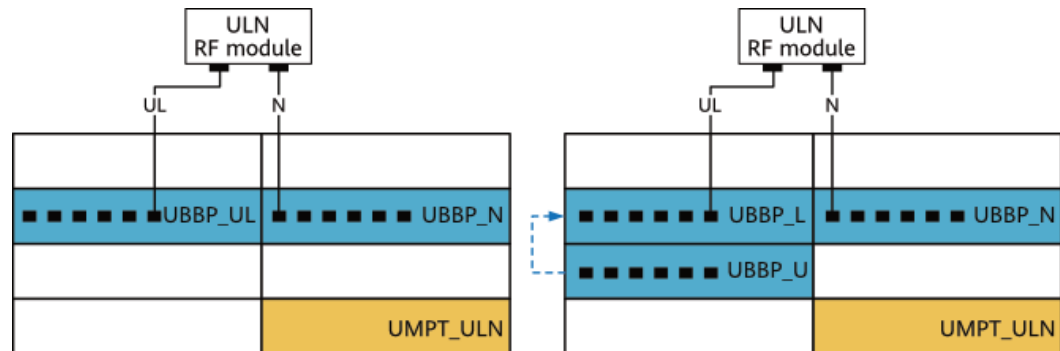


Figure 5-50 shows the examples. The two CPRI links carry UL+N services. Specifically, one carries UL services and the other carries NR services. In inter-board load sharing scenarios, if cells served by an RF module are configured on three or more BBPs, the optical fibers must connect to the two BBPs with the highest CPRI bandwidth requirements. Otherwise, cells may be unavailable or the service capability of cells may degrade due to insufficient backplane bandwidth between BBPs that provide optical fiber connections and the other BBPs.

Figure 5-50 UL+N multimode load sharing topology



NOTICE

- In the multimode load sharing topology, the two CPRI links of an RF module must be connected to two BBPs/BRIs that work in different network modes. Only the BBU3900/BBU3910/BBU5900/BBU5910 supports the multimode load sharing topology.
- In the multimode load sharing topology, the number of LTE cells cannot exceed the limit supported by each CPRI link. The maximum number of LTE cells varies depending on the CPRI port rate and the number of cells supported by the BBU. For details about the number of LTE cells supported by CPRI ports with different data rates, see the following:
For 3900 series BBUs, see section "UBBP" or "LBBP" in *3900 & 5900 Series Base Station Product Documentation > BBU3900 and BBU3910 Hardware Description > "BBU3900 and BBU3910 Boards"*.
For the BBU5900, see section "UBBP" in *3900 & 5900 Series Base Station Product Documentation > BBU5900 Hardware Description > "BBU5900 Boards"*.
For the BBU5910, see section "UBBP" in *3900 & 5900 Series Base Station Product Documentation > BBU5910 Hardware Description > "BBU5910 Boards"*.
- If a UBBP is used in the multimode load sharing topology, the UBBP must have been configured with baseband resources of its working mode. Otherwise, carriers cannot be successfully allocated to the CPRI ports on this UBBP.
- In the multimode load sharing topology, if the CPRI port or the board hosting the CPRI port is faulty, services of certain network modes may fail to be established. After the fault is rectified, services of all network modes will automatically recover.
- In the multimode load sharing topology, if the network modes served by the CPRI ports include GSM and a UBBPh series board is used with DSFP optical modules, only ports 0 to 5 on this board can provide optical fiber connections.

5.4 Dual-Star Topology with CPRI MUX

The dual-star topology with CPRI MUX applies to triple-mode base stations and quadruple-mode base stations. In a triple-mode dual-star topology, two network modes use CPRI MUX and form a dual-star topology with the third one in separate-MPT scenarios. In a quadruple-mode dual-star topology, four network modes are divided into two groups, and two modes in each group use CPRI MUX and form a dual-star topology between groups in separate-MPT scenarios. Only the BBU3910, BBU3900, BBU5910, and BBU5900 support this topology.

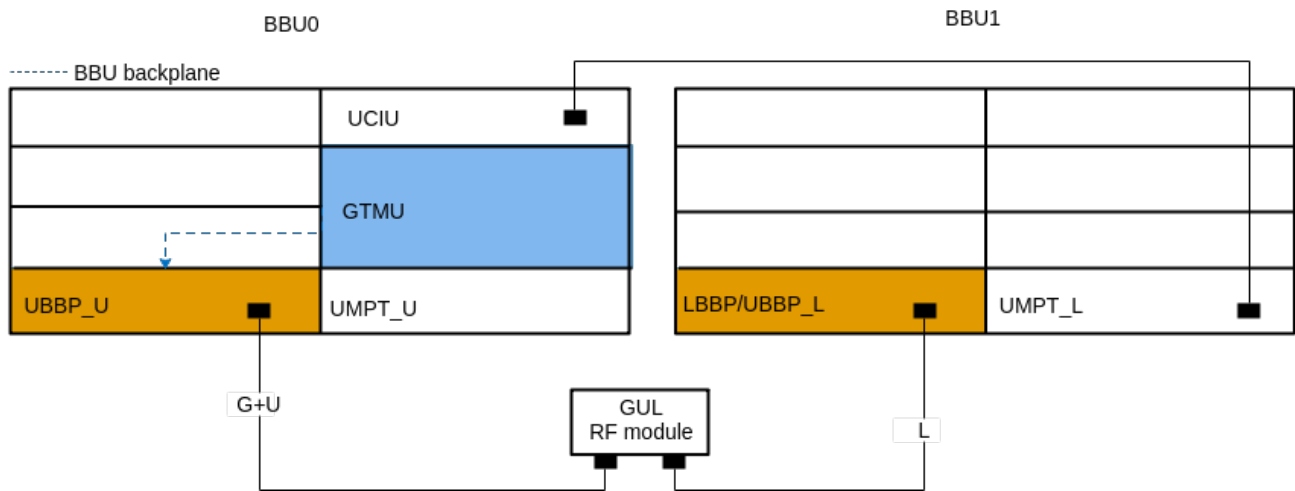
NOTE

The converging party and the converged party must share one BBU and cannot be deployed in different BBUs.

GU+L Dual-Star Topology

As shown in [Figure 5-51](#), BBU0 uses GU CPRI MUX in separate-MPT scenarios. In BBU0, a UBBP_U is connected to a multimode RF module through a CPRI link which carries GU data. BBU1 is deployed only with LTE. In BBU1, a UBBP_L/LBBP is connected to the multimode RF module through another CPRI link. This way, a GU +L dual-star topology is formed.

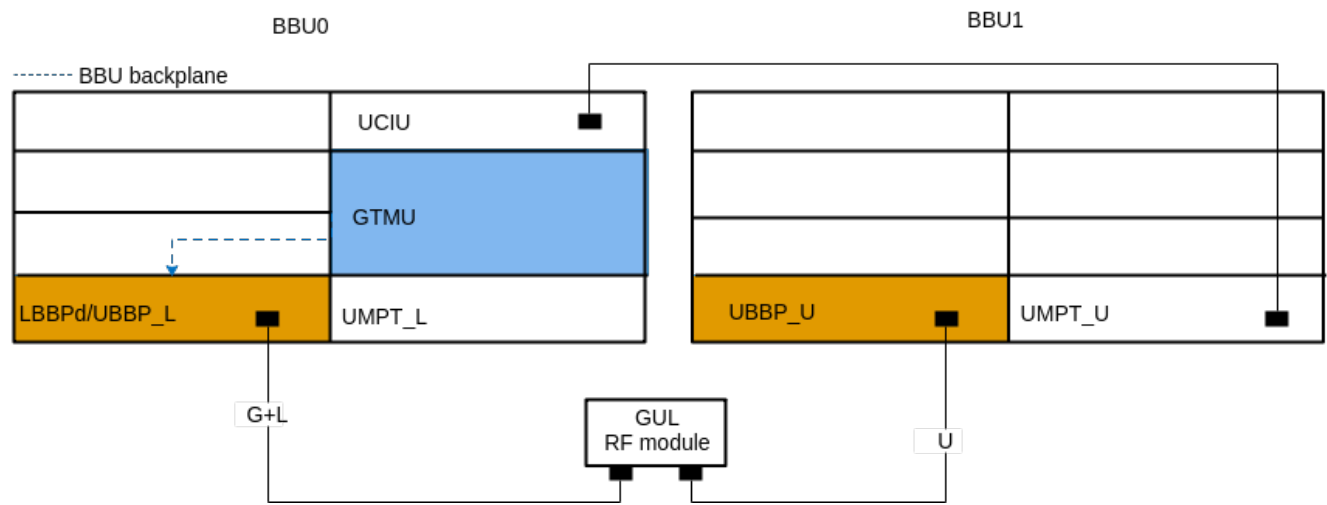
Figure 5-51 GU+L triple-mode dual-star topology



GL+U Dual-Star Topology

As shown in [Figure 5-52](#), BBU0 uses GL CPRI MUX in separate-MPT scenarios. In BBU0, a UBBP_L/LBBP is connected to a multimode RF module through a CPRI link which carries GL data. BBU1 is deployed only with UMTS. In BBU1, a UBBP_U is connected to the multimode RF module through another CPRI link. This way, a GL+U dual-star topology is formed.

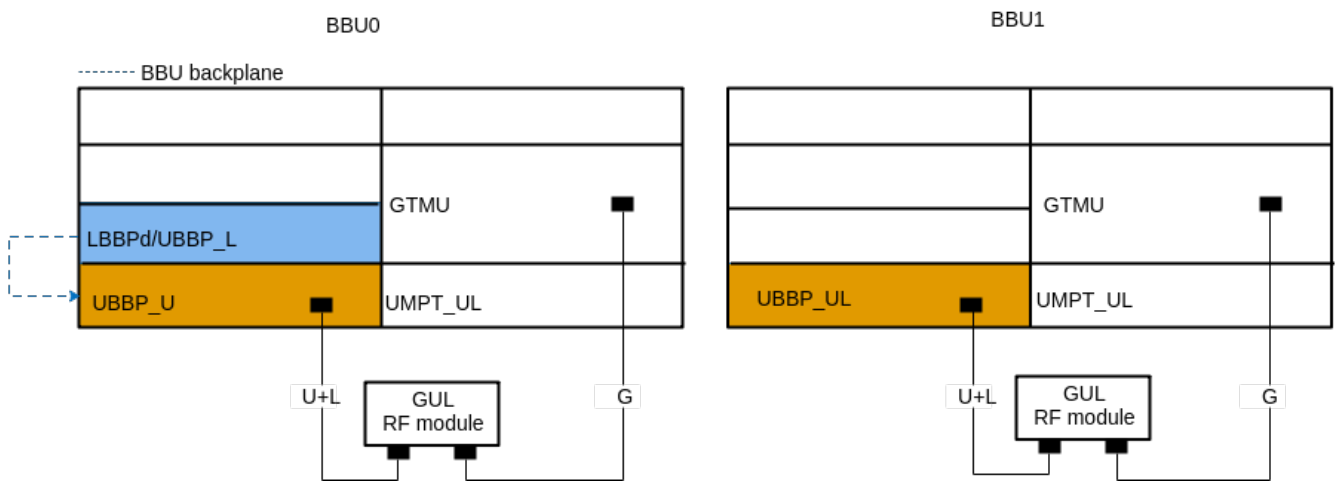
Figure 5-52 GL+U triple-mode dual-star topology



G[U*L] Dual-Star Topology

UMTS and LTE share one main control board, and a UBBP_U, LBBPd/UBBP_L, or UBBP_UL is connected to a multimode RF module through a CPRI link which carries UL data. GSM uses another main control board GTMU which is connected to the multimode RF module through another CPRI link. This way, a G[U*L] dual-star topology is formed. [Figure 5-53](#) shows an example of the topology where a UBBP_U provides the CPRI port for UL CPRI MUX.

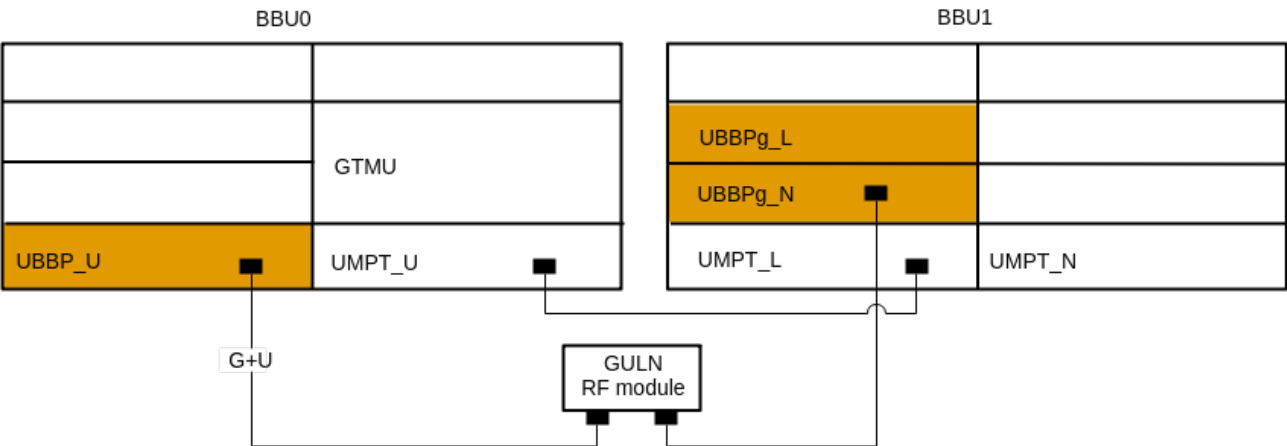
Figure 5-53 G[U*L] triple-mode dual-star topology



GU+LN Dual-Star Topology

In [Figure 5-54](#), BBU0 uses separate-MPT GU CPRI MUX. The UBBP_U provides a CPRI link to connect to the multimode RF module. BBU1 adopts LN separate-MPT, and the UBBP_N provides a CPRI link to connect to the multimode RF module. In this way, a quadruple-mode inter-BBU dual-star topology is formed.

Figure 5-54 GU+LN dual-star topology



NOTE

BBU0 can be a BBU3900/BBU3910, and BBU1 can be a BBU5900/BBU5910.

6 Specifications of CPRI MUX Capabilities

In a topology with CPRI MUX, the CPRI MUX access capability depends on the CPRI line rate and board capabilities of the converging party.

6.1 CPRI MUX Access Capabilities

Table 6-1 lists the access capabilities of the converging party in multimode scenarios where CPRI MUX is used with a BBU3900/BBU3910/BBU5900/BBU5910/BBU5900A. **Table 6-3** lists the access capability in multimode scenarios where CPRI MUX is used with a BBU3910A. **Table 6-4** lists the access capability in multimode scenarios where CPRI MUX is used with a BBU3910C. **Table 6-5** lists the access capability in multimode scenarios where CPRI MUX is used with a BookBBU5901. **Table 6-6** lists the access capability in multimode scenarios where CPRI MUX is used with a BookBBU5901a. **Table 6-7** lists the access capability in multimode scenarios where CPRI MUX is used with a BBU5900C.

 NOTE

- The CPRI line rate, access capabilities of a CPRI link, and access capabilities of a board all refer to those of the converging party.
- Access capabilities mentioned in the tables apply to common cells. For the impact of this feature working in different RATs on cell capacity, see the corresponding feature parameter description.
- If GSM uses the baseband extension mode, GSM-related data in the tables refers to capacity specifications when the antenna TX/RX mode is 1T2R.
- UMTS-related data in the tables refers to capacity specifications when the antenna TX/RX mode is 1T2R.
- The LTE-related data in the table refers to the capacity specifications when CPRI compression is not performed in 2T2R scenarios.
- Unless otherwise specified, the NR-related data in the table refers to the capacity specifications when CPRI compression is not performed in 2T2R scenarios or 2:1 CPRI compression is performed in 4T4R scenarios.
- The capabilities of a BBU3910A or a converging board in a BBU3900/BBU3910/BBU5900/BBU5910/BBU5900A/BBU5900C cannot exceed the cell or carrier specifications of a site. For the cell and carrier specifications, see *BBU Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*.
- To manually obtain the access capabilities of a CPRI chain/ring, the calculation of the cell CPRI bandwidth is required. For details about how to calculate the cell CPRI bandwidth, see *CPRI Compression in eRAN Feature Documentation* and *5G RAN Feature Documentation*. It is recommended that 10% of the CPRI bandwidth be reserved in the case of a CPRI line rate of 10.1 Gbit/s or lower, and 5% of the CPRI bandwidth be reserved in the case of a CPRI line rate of 24.3 Gbit/s. For details about how to calculate the reserved bandwidth, contact Huawei engineers.

Access Capabilities When CPRI MUX Is Used with a BBU3900/BBU3910/BBU5900/BBU5910/BBU5900A

Table 6-1 Access capabilities in GUL scenarios where CPRI MUX is used with a BBU3900/BBU3910/BBU5900/BBU5910/BBU5900A

CPRI Line Rate	2.5 Gbit/s	4.9 Gbit/s	9.8 Gbit/s
Access Capabilities of a CPRI Link (GU)	16 GSM TRXs + 4 UMTS cells	32 GSM TRXs ⁽¹⁾ + 8 UMTS cells	32 GSM TRXs ⁽¹⁾ + 16 UMTS cells
Access Capabilities of a Board (GU)	96 GSM TRXs ⁽²⁾ + 24 UMTS cells	126 GSM TRXs ⁽²⁾ + 24 UMTS cells	126 GSM TRXs ⁽²⁾ + 24 UMTS cells
Access Capabilities of a CPRI Link (GL)	16 GSM TRXs + LTE 1x10 MHz	32 GSM TRXs ⁽¹⁾ + LTE 1x20 MHz	32 GSM TRXs ⁽¹⁾ + LTE 2x20 MHz

CPRI Line Rate	2.5 Gbit/s	4.9 Gbit/s	9.8 Gbit/s
Access Capabilities of a Board (GL)	96 GSM TRXs ⁽²⁾ + LTE 6x10 MHz	126 GSM TRXs ⁽²⁾ + LTE ⁽⁴⁾ 6x20 MHz	126 GSM TRXs ⁽²⁾⁽³⁾ + LTE ⁽⁴⁾ 8x20 MHz
Access Capabilities of a CPRI Link (UL)	4 UMTS cells + LTE 1x10 MHz	8 UMTS cells + LTE 1x20 MHz	8 UMTS cells + LTE 2x20 MHz or 3 UMTS cells + LTE 3x20 MHz
Access Capabilities of a Board (UL)	24 UMTS cells + LTE 6x10 MHz	24 UMTS cells + LTE ⁽⁴⁾ 6x20 MHz	24 UMTS cells + LTE ⁽⁴⁾ 8x20 MHz
Access Capabilities of a CPRI Link (GUL)	6 GSM TRXs + 2 UMTS cells + LTE 1x10 MHz	12 GSM TRXs + 4 UMTS cells + LTE 1x20 MHz	24 GSM TRXs + 8 UMTS cells + LTE 2x20 MHz ⁽⁵⁾
Access Capabilities of a Board (GUL)	36 GSM TRXs + 12 UMTS cells + LTE 6x10 MHz	72 GSM TRXs + 24 UMTS cells + LTE ⁽⁴⁾ 6x20 MHz	126 GSM TRXs ⁽²⁾ + 24 UMTS cells + LTE ⁽⁴⁾ 8x20 MHz

 NOTE

- (1): For GU/GL CPRI MUX used with a BBU3910 or with a UBBPg1a serving as the converging board in separate-MPT scenarios, if the number of ports converging data is less than or equal to five, a single CPRI link can carry 32 GSM TRXs. However, if the number of ports converging data is six, a single CPRI link can only carry 24 GSM TRXs.
- If GSM uses the non-baseband extension mode, GSM-related data in this table refers to capacity specifications when the maximum distance between the BBU and RF modules is 20 km in time division multiplexing (TDM) transmission mode.

In IP over E1 or IP over FE transmission mode, specifications in this table take precedence if these specifications do not exceed the site capacity specifications. Otherwise, site capacity specifications take precedence.

- (2): The capacity specifications refer to those when GSM uses the non-baseband extension mode. If GSM uses the baseband extension mode, a board serves a maximum of 72 GSM TRXs.
- (3): In separate-MPT GL CPRI MUX where a BBU3900 is used, if the converging board is an LBBPd and the CPRI line data rate is 9.8 Gbit/s, the board provides a maximum of three converging ports, which must be ports 0, 1, and 2.

The converging board can provide the following port combinations for data convergence if the CPRI line rates of the ports include both 9.8 Gbit/s and 4.9 Gbit/s:

- Three 9.8 Gbit/s ports and one 4.9 Gbit/s port
- Two 9.8 Gbit/s ports and two 4.9 Gbit/s ports
- One 9.8 Gbit/s port and four 4.9 Gbit/s ports

A 9.8 Gbit/s port converging data must be port 0, 1, or 2.

- For LTE access capabilities in this table:

An LTE FDD cell with a bandwidth of 10 MHz is equivalent to four LTE NB-IoT cells, each with a bandwidth of 200 kHz.

An LTE FDD cell with a bandwidth of 20 MHz is equivalent to eight LTE NB-IoT cells, each with a bandwidth of 200 kHz.

- (4): If LTE cells are served by a board in slot 0, 1, 4, or 5 of a BBU3900, the board serves a maximum of four 20 MHz 2T2R cells.
- (5): Only the UBBP supports GUL CPRI MUX with a CPRI line rate of 9.8 Gbit/s.

NOTICE

The minimum CPRI line rate of the converging board is 2.5 Gbit/s.

Table 6-2 Access capabilities in NR-included multimode scenarios where CPRI MUX is used with a BBU3910/BBU5900/BBU5910/BBU5900A

CPRI Line Rate	9.8 Gbit/s	10.1 Gbit/s	24.3 Gbit/s
Access Capabilities of a CPRI Link (UN)	3 UMTS cells + NR 3x20 MHz	3 UMTS cells + NR 4x20 MHz	8 UMTS cells + NR 10x20 MHz
Access Capabilities of a Board (UN)	18 UMTS cells + NR 12x20 MHz	18 UMTS cells + NR 12x20 MHz	24 UMTS cells + NR 24x20 MHz

CPRI Line Rate	9.8 Gbit/s	10.1 Gbit/s	24.3 Gbit/s
Access Capabilities of a CPRI Link (LN)	LTE 1x20 MHz + NR 2x20 MHz	LTE 2x20 MHz + NR 2x20 MHz	LTE 1x20 MHz + NR 10x20 MHz
	LTE 2x20 MHz + NR 1x20 MHz		LTE 10x20 MHz + NR 1x20 MHz
Access Capabilities of a Board (LN)	LTE 6x20 MHz + NR 12x20 MHz	LTE 9x20 MHz + NR 12x20 MHz	LTE 6x20 MHz + NR 24x20 MHz
	LTE 12x20 MHz + NR 6x20 MHz	LTE 12x20 MHz + NR 9x20 MHz	LTE 24x20 MHz + NR 6x20 MHz
Access Capabilities of a CPRI Link (GLN)	32 GSM TRXs + LTE 1x20 MHz + NR 1x20 MHz	32 GSM TRXs + LTE 1x20 MHz + NR 2x20 MHz	32 GSM TRXs + LTE 1x20 MHz + NR 9x20 MHz
			32 GSM TRXs + LTE 9x20 MHz + NR 1x20 MHz
Access Capabilities of a Board (GLN)	126 GSM TRXs + LTE 6x20 MHz + NR 6x20 MHz	126 GSM TRXs + LTE 6x20 MHz + NR 12x20 MHz	126 GSM TRXs + LTE 6x20 MHz + NR 24x20 MHz
			126 GSM TRXs + LTE 24x20 MHz + NR 6x20 MHz
Access Capabilities of a CPRI Link (ULN)	3 UMTS cells + LTE 1x20 MHz + NR 2x20 MHz	3 UMTS cells + LTE 2x20 MHz + NR 2x20 MHz	8 UMTS cells + LTE 1x20 MHz + NR 9x20 MHz
	3 UMTS cells + LTE 2x20 MHz + NR 1x20 MHz		8 UMTS cells + LTE 9x20 MHz + NR 1x20 MHz

CPRI Line Rate	9.8 Gbit/s	10.1 Gbit/s	24.3 Gbit/s
Access Capabilities of a Board (ULN)	18 UMTS cells + LTE 6x20 MHz + NR 12x20 MHz	18 UMTS cells + LTE 9x20 MHz + NR 12x20 MHz	24 UMTS cells + LTE 6x20 MHz + NR 24x20 MHz
	18 UMTS cells + LTE 12x20 MHz + NR 6x20 MHz		24 UMTS cells + LTE 24x20 MHz + NR 6x20 MHz
Access Capabilities of a CPRI Link (GULN)	24 GSM TRXs + 8 UMTS cells + LTE 1x20 MHz + NR 1x20 MHz	24 GSM TRXs + 8 UMTS cells + LTE 1x20 MHz + NR 2x20 MHz	24 GSM TRXs + 8 UMTS cells + LTE 1x20 MHz + NR 8x20 MHz
			24 GSM TRXs + 8 UMTS cells + LTE 8x20 MHz + NR 1x20 MHz
Access Capabilities of a Board (GULN)	126 GSM TRXs + 24 UMTS cells + LTE 6x20 MHz + NR 6x20 MHz	126 GSM TRXs + 24 UMTS cells + LTE 6x20 MHz + NR 12x20 MHz	126 GSM TRXs + 24 UMTS cells + LTE 6x20 MHz + NR 24x20 MHz
			126 GSM TRXs + 24 UMTS cells + LTE 24x20 MHz + NR 6x20 MHz

Access Capabilities When CPRI MUX Is Used with a BBU3910A

Table 6-3 Access capabilities in multimode scenarios where CPRI MUX is used with a BBU3910A

CPRI Line Rate	2.5 Gbit/s	4.9 Gbit/s	9.8 Gbit/s
Access Capabilities of a CPRI Link (GU)	16 GSM TRXs + 4 UMTS cells	32 GSM TRXs + 8 UMTS cells	32 GSM TRXs + 12 UMTS cells
Access Capabilities of a BBU (GU)	48 GSM TRXs + 12 UMTS cells	48 GSM TRXs + 12 UMTS cells	48 GSM TRXs + 12 UMTS cells
Access Capabilities of a CPRI Link (GL)	16 GSM TRXs + LTE 1x10 MHz	32 GSM TRXs + LTE 1x20 MHz	32 GSM TRXs + LTE 2x20 MHz
Access Capabilities of a BBU (GL)	48 GSM TRXs + LTE 6x10 MHz	48 GSM TRXs + LTE 6x20 MHz	48 GSM TRXs + LTE 6x20 MHz
Access Capabilities of a CPRI Link (UL)	4 UMTS cells + LTE 1x10 MHz	6 UMTS cells + LTE 1x20 MHz	6 UMTS cells + LTE 2x20 MHz or 3 UMTS cells + LTE 3x20 MHz
Access Capabilities of a BBU (UL)	6 UMTS cells + LTE 3x10 MHz	6 UMTS cells + LTE 3x20 MHz	6 UMTS cells + LTE 3x20 MHz
Access Capabilities of a CPRI Link (GUL)	6 GSM TRXs + 2 UMTS cells + LTE 1x10 MHz	12 GSM TRXs + 4 UMTS cells + LTE 1x20 MHz	12 GSM TRXs + 6 UMTS cells + LTE 2x20 MHz
Access Capabilities of a BBU (GUL)	36 GSM TRXs + 6 UMTS cells + LTE 3x10 MHz	48 GSM TRXs + 6 UMTS cells + LTE 3x20 MHz	48 GSM TRXs + 6 UMTS cells + LTE 3x20 MHz

NOTE

GSM-related data in this table refers to capacity specifications when GSM uses the non-baseband extension mode, the transmission operates in IP over E1 or IP over FE mode, and the maximum distance between the BBU and RF modules is 20 km.

Access Capabilities When CPRI MUX Is Used with a BBU3910C

Table 6-4 Access capabilities in multimode scenarios where CPRI MUX is used with a BBU3910C

CPRI Line Rate	2.5 Gbit/s	4.9 Gbit/s
Access Capabilities of a CPRI Link (GU)	24 GSM TRXs + 6 UMTS cells	24 GSM TRXs + 6 UMTS cells
Access Capabilities of a BBU (GU)	24 GSM TRXs + 6 UMTS cells	24 GSM TRXs + 6 UMTS cells

NOTE

The BBU3910C does not support the baseband extension mode. GSM-related data in this table refers to capacity specifications when GSM uses the non-baseband extension mode, the transmission operates in IP over FE mode, and the maximum distance between the BBU3910C and RF modules is reached.

Access Capabilities When CPRI MUX Is Used with a BookBBU5901

Table 6-5 Access capabilities in multimode scenarios where CPRI MUX is used with a BookBBU5901

CPRI Line Rate	2.5 Gbit/s	4.9 Gbit/s	9.8 Gbit/s	10.1 Gbit/s	24.3 Gbit/s
Access Capabilities of a CPRI Link (GU)	16 GSM TRXs + 4 UMTS cells	24 GSM TRXs + 6 UMTS cells	24 GSM TRXs + 6 UMTS cells	24 GSM TRXs + 6 UMTS cells	24 GSM TRXs + 6 UMTS cells
Access Capabilities of a BBU (GU)	24 GSM TRXs + 6 UMTS cells	24 GSM TRXs + 6 UMTS cells	24 GSM TRXs + 6 UMTS cells	24 GSM TRXs + 6 UMTS cells	24 GSM TRXs + 6 UMTS cells
Access Capabilities of a CPRI Link (GL)	16 GSM TRXs + LTE 1x10 MHz	24 GSM TRXs + LTE 1x20 MHz	24 GSM TRXs + LTE 2x20 MHz	24 GSM TRXs + LTE 3x20 MHz	24 GSM TRXs + LTE 6x20 MHz
Access Capabilities of a BBU (GL)	24 GSM TRXs + LTE 3x10 MHz	24 GSM TRXs + LTE 3x20 MHz	24 GSM TRXs + LTE 6x20 MHz	24 GSM TRXs + LTE 6x20 MHz	24 GSM TRXs + LTE 6x20 MHz

CPRI Line Rate	2.5 Gbit/s	4.9 Gbit/s	9.8 Gbit/s	10.1 Gbit/s	24.3 Gbit/s
Access Capabilities of a CPRI Link (UL)	4 UMTS cells + LTE 1x10 MHz	6 UMTS cells + LTE 1x20 MHz	6 UMTS cells + LTE 2x20 MHz or 3 UMTS cells + LTE 3x20 MHz	6 UMTS cells + LTE 3x20 MHz	6 UMTS cells + LTE 6x20 MHz
Access Capabilities of a BBU (UL)	6 UMTS cells + LTE 3x10 MHz	6 UMTS cells + LTE 3x20 MHz	6 UMTS cells + LTE 6x20 MHz	6 UMTS cells + LTE 6x20 MHz	6 UMTS cells + LTE 6x20 MHz
Access Capabilities of a CPRI Link (GUL)	6 GSM TRXs + 2 UMTS cells + LTE 1x10 MHz	12 GSM TRXs + 4 UMTS cells + LTE 1x20 MHz	12 GSM TRXs + 6 UMTS cells + LTE 2x20 MHz	12 GSM TRXs + 6 UMTS cells + LTE 3x20 MHz	24 GSM TRXs + 6 UMTS cells + LTE 6x20 MHz
Access Capabilities of a BBU (GUL)	24 GSM TRXs + 6 UMTS cells + LTE 3x10 MHz	24 GSM TRXs + 6 UMTS cells + LTE 3x20 MHz	24 GSM TRXs + 6 UMTS cells + LTE 6x20 MHz	24 GSM TRXs + 6 UMTS cells + LTE 6x20 MHz	24 GSM TRXs + 6 UMTS cells + LTE 6x20 MHz
Access Capabilities of a CPRI Link (LN)	Not supported	LTE 1x10 MHz + NR 1x40 MHz	LTE 1x20 MHz + NR 1x100 MHz	LTE 2x20 MHz + NR 1x100 MHz	LTE 4x20 MHz + NR 2x100 MHz
Access Capabilities of a BBU (LN)	Not supported	LTE 4x10 MHz + NR 2x40 MHz	LTE 4x20 MHz + NR 2x100 MHz	LTE 4x20 MHz + NR 2x100 MHz	LTE 4x20 MHz + NR 2x100 MHz

NOTE

The BookBBU5901 does not support the baseband extension mode. GSM-related data in this table refers to capacity specifications when GSM uses the non-baseband extension mode, the transmission operates in IP over FE mode, and the maximum distance between the BookBBU5901 and RF modules is reached.

Access Capabilities When CPRI MUX Is Used with a BookBBU5901a**Table 6-6** Access capabilities in multimode scenarios where CPRI MUX is used with a BookBBU5901a

CPRI Line Rate	2.5 Gbit/s	4.9 Gbit/s	9.8 Gbit/s	10.1 Gbit/s	24.3 Gbit/s
Access Capabilities of a CPRI Link (GU)	16 GSM TRXs + 4 UMTS cells	24 GSM TRXs + 6 UMTS cells	24 GSM TRXs + 6 UMTS cells	24 GSM TRXs + 6 UMTS cells	24 GSM TRXs + 6 UMTS cells
Access Capabilities of a BBU (GU)	24 GSM TRXs + 6 UMTS cells	24 GSM TRXs + 6 UMTS cells	24 GSM TRXs + 6 UMTS cells	24 GSM TRXs + 6 UMTS cells	24 GSM TRXs + 6 UMTS cells
Access Capabilities of a CPRI Link (GL)	16 GSM TRXs + LTE 1x10 MHz	24 GSM TRXs + LTE 1x20 MHz	24 GSM TRXs + LTE 2x20 MHz	24 GSM TRXs + LTE 3x20 MHz	24 GSM TRXs + LTE 6x20 MHz
Access Capabilities of a BBU (GL)	24 GSM TRXs + LTE 3x10 MHz	24 GSM TRXs + LTE 3x20 MHz	24 GSM TRXs + LTE 6x20 MHz	24 GSM TRXs + LTE 6x20 MHz	24 GSM TRXs + LTE 6x20 MHz
Access Capabilities of a CPRI Link (UL)	4 UMTS cells + LTE 1x10 MHz	6 UMTS cells + LTE 1x20 MHz	6 UMTS cells + LTE 2x20 MHz or 3 UMTS cells + LTE 3x20 MHz	6 UMTS cells + LTE 3x20 MHz	6 UMTS cells + LTE 6x20 MHz
Access Capabilities of a BBU (UL)	6 UMTS cells + LTE 3x10 MHz	6 UMTS cells + LTE 3x20 MHz	6 UMTS cells + LTE 6x20 MHz	6 UMTS cells + LTE 6x20 MHz	6 UMTS cells + LTE 6x20 MHz
Access Capabilities of a CPRI Link (GUL)	6 GSM TRXs + 2 UMTS cells + LTE 1x10 MHz	12 GSM TRXs + 4 UMTS cells + LTE 1x20 MHz	12 GSM TRXs + 6 UMTS cells + LTE 2x20 MHz	12 GSM TRXs + 6 UMTS cells + LTE 3x20 MHz	24 GSM TRXs + 6 UMTS cells + LTE 6x20 MHz

CPRI Line Rate	2.5 Gbit/s	4.9 Gbit/s	9.8 Gbit/s	10.1 Gbit/s	24.3 Gbit/s
Access Capabilities of a BBU (GUL)	24 GSM TRXs + 6 UMTS cells + LTE 3x10 MHz	24 GSM TRXs + 6 UMTS cells + LTE 3x20 MHz	24 GSM TRXs + 6 UMTS cells + LTE 6x20 MHz	24 GSM TRXs + 6 UMTS cells + LTE 6x20 MHz	24 GSM TRXs + 6 UMTS cells + LTE 6x20 MHz
Access Capabilities of a CPRI Link (LN)	Not supported	LTE 1x10 MHz + NR 1x40 MHz	LTE 1x20 MHz + NR 1x100 MHz	LTE 2x20 MHz + NR 1x100 MHz	LTE 4x20 MHz + NR 2x100 MHz
Access Capabilities of a BBU (LN)	Not supported	LTE 4x10 MHz + NR 2x40 MHz	LTE 4x20 MHz + NR 2x100 MHz	LTE 4x20 MHz + NR 2x100 MHz	LTE 4x20 MHz + NR 2x100 MHz

NOTE

The BookBBU5901a does not support the baseband extension mode. GSM-related data in this table refers to capacity specifications when GSM uses the non-baseband extension mode, the transmission operates in IP over FE mode, and the maximum distance between the BookBBU5901a and an RF module is reached.

Access Capabilities When CPRI MUX Is Used with a BBU5900C

Table 6-7 Access capabilities in multimode scenarios where CPRI MUX is used with a BBU5900C

CPRI Line Rate	2.5 Gbit/s	4.9 Gbit/s	9.8 Gbit/s	10.1 Gbit/s
Access Capabilities of a CPRI Link (GU)	16 GSM TRXs + 4 UMTS cells	24 GSM TRXs + 6 UMTS cells	24 GSM TRXs + 6 UMTS cells	24 GSM TRXs + 6 UMTS cells
Access Capabilities of a BBU (GU)	24 GSM TRXs + 6 UMTS cells	24 GSM TRXs + 6 UMTS cells	24 GSM TRXs + 6 UMTS cells	24 GSM TRXs + 6 UMTS cells
Access Capabilities of a CPRI Link (GL)	16 GSM TRXs + LTE 1x10 MHz	24 GSM TRXs + LTE 1x20 MHz	24 GSM TRXs + LTE 2x20 MHz	24 GSM TRXs + LTE 3x20 MHz

CPRI Line Rate	2.5 Gbit/s	4.9 Gbit/s	9.8 Gbit/s	10.1 Gbit/s
Access Capabilities of a BBU (GL)	24 GSM TRXs + LTE 3x10 MHz	24 GSM TRXs + LTE 3x20 MHz	24 GSM TRXs + LTE 3x20 MHz	24 GSM TRXs + LTE 3x20 MHz
Access Capabilities of a CPRI Link (UL)	4 UMTS cells + LTE 1x10 MHz	6 UMTS cells + LTE 1x20 MHz	6 UMTS cells + LTE 2x20 MHz or 3 UMTS cells + LTE 3x20 MHz	6 UMTS cells + LTE 3x20 MHz
Access Capabilities of a BBU (UL)	6 UMTS cells + LTE 3x10 MHz	6 UMTS cells + LTE 3x20 MHz	6 UMTS cells + LTE 3x20 MHz	6 UMTS cells + LTE 3x20 MHz
Access Capabilities of a CPRI Link (GUL)	6 GSM TRXs + 2 UMTS cells + LTE 1x10 MHz	12 GSM TRXs + 4 UMTS cells + LTE 1x20 MHz	12 GSM TRXs + 6 UMTS cells + LTE 2x20 MHz	12 GSM TRXs + 6 UMTS cells + LTE 3x20 MHz
Access Capabilities of a BBU (GUL)	24 GSM TRXs + 6 UMTS cells + LTE 3x10 MHz	24 GSM TRXs + 6 UMTS cells + LTE 3x20 MHz	24 GSM TRXs + 6 UMTS cells + LTE 3x20 MHz	24 GSM TRXs + 6 UMTS cells + LTE 3x20 MHz

NOTE

- Access capabilities mentioned in the tables apply to common cells. For the impact of this feature working in different RATs on cell capacity, see the corresponding feature parameter description.
- UMTS-related data in the tables refers to capacity specifications when the antenna TX/RX mode is 1T2R.
- The LTE-related data in the table refers to the capacity specifications when CPRI compression is not performed in 2T2R scenarios.
- The BBU5900C does not support the baseband extension mode. GSM-related data in this table refers to capacity specifications when GSM uses the non-baseband extension mode, the transmission operates in IP over FE mode, and the maximum distance between the BBU5900C and RF modules is reached.

6.2 LNR Backplane Switching Specifications

Table 6-8 BBU3900 LTE backplane switching specifications

Converging Board	Converged Board	Backplane Switching Specifications
UBBPd/UBBPe/ UBBPg1/UBBPg1a	UBBPd/UBBPe/ UBBPg1/UBBPg1a	20 x 20 MHz 2T2R ⁽¹⁾
UBRlb	UBBPd/UBBPe/ UBBPg1/UBBPg1a	10 x 20 MHz 2T2R ⁽²⁾

Table 6-9 BBU3910 LNR backplane switching specifications

Converging Board	Converged Board	Backplane Switching Specifications
UBBPd/UBBPe/UBBPei/ UBBPg1/UBBPg1a/ UBBPg2	UBBPd/UBBPe/UBBPei/ UBBPg1/UBBPg1a/ UBBPg2	20 x 20 MHz 2T2R
UBBPd/UBBPe/UBBPei/ UBBPg1/UBBPg1a/ UBBPg2	UBBPg2a	20 x 20 MHz 2T2R
UBBPg2a	UBBPd/UBBPe/UBBPei/ UBBPg1/UBBPg1a/ UBBPg2	20 x 20 MHz 2T2R
UBBPg2a	UBBPg2a	50 x 20 MHz 2T2R

Table 6-10 BBU5900/BBU5910/BBU5900A LNR backplane switching specifications

Converging Board	Converged Board	Backplane Switching Specifications
UBBPd/UBBPe/UBBPei/ UBBPg1/UBBPg1a/ UBBPg2	UBBPd/UBBPe/UBBPei/ UBBPg1/UBBPg1a/ UBBPg2	20 x 20 MHz 2T2R
UBBPd/UBBPe/UBBPei/ UBBPg1/UBBPg1a/ UBBPg2	UBBPg2a/UBBPg3/ UBBPg3b/UBBPh1/ UBBPh1u/UBBPh2/ UBBPh2a/UBBPh3/ UBBPh3a/UBBPh3b	20 x 20 MHz 2T2R

Converging Board	Converged Board	Backplane Switching Specifications
UBBPg2a/UBBPg3/ UBBPg3b/UBBPh1/ UBBPh1u/UBBPh2/ UBBPh2a	UBBPd/UBBPe/UBBPei/ UBBPg1/UBBPg1a/ UBBPg2	20 x 20 MHz 2T2R
UBBPg2a/UBBPg3/ UBBPg3b/UBBPh1/ UBBPh1u/UBBPh2/ UBBPh2a	UBBPg2a/UBBPg3/ UBBPg3b/UBBPh1/ UBBPh1u/UBBPh2/ UBBPh2a/UBBPh3/ UBBPh3a/UBBPh3b	50 x 20 MHz 2T2R
UBBPh3/UBBPh3a/ UBBPh3b	UBBPd/UBBPe/UBBPei/ UBBPg1/UBBPg1a/ UBBPg2	20 x 20 MHz 2T2R
UBBPh3/UBBPh3a/ UBBPh3b	UBBPg2a/UBBPg3/ UBBPg3b/UBBPh1/ UBBPh1u/UBBPh2/ UBBPh2a	50 x 20 MHz 2T2R
UBBPh3/UBBPh3a/ UBBPh3b	UBBPh3/UBBPh3a/ UBBPh3b	100 x 20 MHz 2T2R

 NOTE

- (1) in [Table 6-8](#): In a BBU3900, if the converging and converged boards are installed in slots 2 and 3, slots 3 and 2, or slots 3 and 1, respectively, the backplane switching specifications are 20 x 20 MHz 2T2R. If only the converging board is installed in slot 2 or 3 but not the converged board (excluding the scenarios where the converging and converged boards are installed in slots 3 and 1), the backplane switching specifications are only 10 x 20 MHz 2T2R.
- (2) in [Table 6-8](#): In a BBU3900, if the converging and converged boards are installed in slots 1 and 0 or slots 1 and 3, respectively, the backplane switching specifications are 10 x 20 MHz 2T2R. If the converging and converged boards are installed in slots 1 and 4 or slots 1 and 5, respectively, the backplane switching specifications are only 5 x 20 MHz 2T2R.
- For LTE FDD and NR FDD cells enabled with LTE FDD and NR spectrum sharing, when the LTE and NR co-carrier co-CPRI data function is configured, if the spectrum-sharing LTE FDD and NR FDD cells are served by the same baseband processing unit and require backplane bandwidth resources, the two cells use the same backplane bandwidth resources.
- The preceding cell specifications apply when 2:1 CPRI compression is used for LTE/NR.
- If other CPRI compression modes are used, the backplane switching specifications must be calculated based on the configured CPRI compression ratios. For details about how to calculate CPRI compression ratios for LTE and NR, see *CPRI Compression* in *eRAN Feature Documentation* and *5G RAN Feature Documentation*.
- The UBBPd/UBBPe indicates the series of boards that support the backplane switching specifications.

6.3 Capacity Specifications

The following are methods for calculating capacity specifications for each network mode with different configurations:

- GSM
 - In GSM non-baseband extension mode, the number of GSM TRXs are reduced by half if the following condition is met:
 $20\text{ km} < \text{Distance between the BBU and RF modules} \leq 40\text{ km}.$
- UMTS
 - The number of cells supported when the antenna TX/RX mode is 1T2R is the same as that supported when the antenna TX/RX mode is 2T2R.
 - The antenna TX/RX mode is in inversely proportion to the supported number of cells. For example, the number of cells supported when the antenna TX/RX mode is 1T2R is twice that supported when the antenna TX/RX mode is 2T4R.
- LTE
 - With the antenna TX/RX mode and CPRI line rate fixed, the number of cells supported by a 3 MHz, 5 MHz, 10 MHz, or 20 MHz cell is in inverse proportion to the bandwidth of LTE cells. For example, the supported number of 3 MHz cells is twice that of 5 MHz cells, and the supported number of 5 MHz cells is twice that of 10 MHz cells.
 - With the antenna TX/RX mode and CPRI line rate fixed, the supported number of 1.4 MHz cells equals that of 3 MHz, and the supported number of 15 MHz cells equals that of 20 MHz cells.
 - With the cell bandwidth fixed, the antenna TX/RX mode of 1T1R, 2T2R, 8T8R, 32T32R, and 4T4R is in inverse proportion to the number of cells. For example, if the cell bandwidth is 20 MHz, the number of cells supported when the antenna TX/RX mode is 2T2R is twice that supported when the antenna TX/RX mode is 4T4R. If the cell bandwidth is 20 MHz, the number of cells supported when the antenna TX/RX mode is 8T8R is four times that supported when the antenna TX/RX mode is 32T32R.
 - With the cell bandwidth fixed, the number of cells supported when the antenna TX/RX mode is 1T2R equals that supported when the antenna TX/RX mode is 2T2R, and the number of cells supported when the antenna TX/RX mode is 2T4R equals that supported when the antenna TX/RX mode is 4T4R.
- NR
 - When antenna TX/RX capabilities and CPRI line rates are the same, the number of supported cells is in inverse proportion to the cell bandwidth. For example, the supported number of 10 MHz cells is twice that of 20 MHz cells, and the supported number of 20 MHz cells is twice that of 40 MHz cells.
 - With the cell bandwidth and CPRI line rate fixed, the antenna TX/RX mode of 1T1R, 2T2R, 8T8R, 32T32R, and 4T4R is in inverse proportion to the number of cells. For example, if the cell bandwidth is 20 MHz, the number of cells supported when the antenna TX/RX mode is 2T2R is twice that supported when the antenna TX/RX mode is 4T4R. If the cell

- bandwidth is 20 MHz, the number of cells supported when the antenna TX/RX mode is 8T8R is four times that supported when the antenna TX/RX mode is 32T32R.
- When the cell bandwidth and CPRI line rate are the same, the number of cells supported by CPRI compression is in direct proportion to the CPRI compression ratio. For example, when the cell bandwidth is 20 MHz, the number of cells supported by 2:1 CPRI compression is twice that supported by non-CPRI compression.

6.4 Examples

Scenario 1: BBU3900, GL 1800 MHz, GSM S2/2/2, LTE 3x20 MHz 2T2R

This scenario requires that:

- Each CPRI link serves two GSM TRXs and one 20 MHz 2T2R LTE cell.
- Each board serves six GSM TRXs and three 20 MHz 2T2R LTE cells.

According to [Table 6-1](#), each CPRI link can serve one 20 MHz 2T2R LTE cell if the CPRI line rate is 4.9 Gbit/s. In this case, each converging board can serve a maximum of 126 GSM TRXs and four 20 MHz 2T2R LTE cells.

The capacity requirements are met and this scenario is supported.

Scenario 2: BBU3900, GL 1800 MHz, GSM S2/2/2, LTE 3x10 MHz 4T4R

This scenario requires that:

- Each CPRI link serves two GSM TRXs and one 10 MHz 4T4R LTE cell.
- Each board serves six GSM TRXs and three 10 MHz 4T4R LTE cells.

One 10 MHz 4T4R cell is equivalent to two 10 MHz 2T2R cells, and then equivalent to one 20 MHz 2T2R cell.

According to [Table 6-1](#), each CPRI link can serve one 10 MHz 4T4R cell if the CPRI line rate is 4.9 Gbit/s. In this case, each converging board can serve a maximum of 126 GSM TRXs and four 20 MHz 2T2R LTE cells.

Therefore, the capacity requirements can be met and this scenario is supported.

Scenario 3: BBU3900, GL 1800 MHz, GSM S2/2/2/2/2/2, LTE 6x20 MHz 2T2R

This scenario requires that:

- Each CPRI link serves two GSM TRXs and one 20 MHz 2T2R LTE cell.
- Each board serves twelve GSM TRXs and six 20 MHz 2T2R LTE cells.

According to [Table 6-1](#), each CPRI link can serve one 20 MHz 2T2R LTE cell if the CPRI line rate is 4.9 Gbit/s. In this case, each converging board can serve a maximum of 126 GSM TRXs and four 20 MHz 2T2R LTE cells.

Therefore, the capacity requirements cannot be met and this scenario is not supported. Contact Huawei engineers for technical support.

7 Network Analysis

7.1 Benefits

CPRI MUX provides benefits in the following scenarios:

- New base station deployment
CPRI MUX enables multiple network modes to use one CPRI link for data transmission, which considerably decreases the number of required optical modules and optical fibers. This helps reduce cable purchase, installation, and maintenance costs.
- Refarming
If only one CPRI cable is used in a base station before refarming, new network modes can share the existing CPRI cable and RF module after refarming without the need for adjusting or adding CPRI cables. This reduces both engineering expenses and service interruption duration.
- Multimode coverage for highways or railways
CPRI MUX enables multimode RF modules to be cascaded and provide multimode coverage along highways or railways.

7.2 Impacts

7.2.1 Services

CPRI MUX in Separate-MPT/Hybrid-MPT Scenarios

- Operations or exceptions of the converging party
Data and voice services of the converged party will be interrupted if any of the following occurs on the converging party. Most interruptions last less than 3 minutes and the longest does not exceed 3.5 minutes.
 - The software of the converging board or the main control board is reset, the converging board or the main control board is reset through a power cycle, or the converging board or the main control board is removed and then reinserted.

- The converging board is blocked.
- CPRI ports on the converging board are being commissioned.
- A CPRI port of the converging board becomes faulty.
- Operations or exceptions of the converged party
 - If the software of the main control board of the converged party is reset, the rate of data services of the converging party may be decreased for less than 3s in most cases and the longest does not exceed 10s. However, voice services will not be affected.
 - If the main control board of the converged party is reset through a power cycle or if it is removed and then reinserted, data and voice services of the converging party will be interrupted for less than 1 minute in most cases and the longest does not exceed 3 minutes.

CPRI MUX in Co-MPT Scenarios

- Data and voice services of the converged party will be interrupted if any of the following occurs on the converging party. Most interruptions last less than 3 minutes and the longest does not exceed 3.5 minutes.
 - The software of the converging board is reset, the converging board is reset through a power cycle, the converging board is blocked, or the converging board is removed and then reinserted.
 - CPRI ports on the converging board are being commissioned.
 - A CPRI port of the converging board becomes faulty.
- If an LBBPd serves as the converging board and an LTE application is deleted or reset, cells of the converged party will become unavailable for no more than 5 minutes.
- If a UBRlb serves as the converging board, removing or resetting a GSM application does not affect the converged party.
- If a UBBP serves as the converging board, removing or resetting the applications of the network modes configured on this board does not affect the converged party.
- Adding baseband resources to one network mode configured on a UBBP interrupts services of other network modes configured on this board. However, this does not affect services of the converged party that communicates with this board through the BBU backplane.

NOTE

Service interruption duration does not include the duration of manual operations, such as blocking a board or removing and then reinserting a board.

7.2.2 Features

- GBFD-510104 Multi-site Cell
For hardware capability and cell capacity specifications when both CPRI MUX and Multi-site Cell are used, see *Multi-site Cell Feature Parameter Description* in *GBSS Feature Documentation*.

7.2.3 System Capacity

None

7.2.4 Network Performance

None

8 Requirements

8.1 Licenses

None

8.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.2.1 Prerequisite Functions

None

8.2.2 Mutually Exclusive Functions

Feature ID	Feature Name	Description
GBFD-113701	Frequency Hopping (RF frequency hopping and baseband frequency hopping)	In separate-MPT scenarios, GU/GL CPRI MUX does not support baseband frequency hopping, RF frequency hopping, or antenna frequency hopping of the following types: Between an RF module connecting to a GTMUC and an RF module connecting to a UBRI/UBRIB/UBBP_G
GBFD-113703	Antenna Frequency Hopping	Between an RF module connecting to a converging board and an RF module connecting to a UBRI/UBRIB/UBBP_G If a UBRIB in a BBU3900 is the converging board in co-MPT scenarios, CPRI MUX does not support baseband frequency hopping, RF frequency hopping, or antenna frequency hopping between an RF module connecting to a GTMUC that works in evolution mode and an RF module connecting to a UBRIB.

8.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations)

Boards

For boards that support CPRI MUX and constraints, see [4 Boards That Support CPRI MUX and Constraints](#).

RF Modules

For RF modules that support CPRI MUX and constraints, see [4 Boards That Support CPRI MUX and Constraints](#).

8.4 Networking

Network planning must be performed according to the topologies and principles described in [5.1 Star Topology with CPRI MUX](#), [5.2 Chain Topology with CPRI MUX](#), [5.3 Load Sharing Topology with CPRI MUX](#), and [5.4 Dual-Star Topology with CPRI MUX](#).

8.5 Others

Perform hardware planning by referring to [4 Boards That Support CPRI MUX and Constraints](#). The CPRI rates of BBPs/BRIs, optical modules, and RF modules must correspond and meet the minimum requirements outlined in [Specifications of CPRI MUX Capabilities](#).

9 Operation and Maintenance

For scenarios where CPRI MUX is used with the RMU solution, see the corresponding "Operation and Maintenance" section in *RMU*.

9.1 When to Use

CPRI MUX for multimode base stations is recommended in the following scenarios:

- A single-mode base station is to be reconstructed into a multimode base station with the existing CPRI topology unchanged to reduce engineering costs.
- A co-MPT dual-mode or multimode base station is to be deployed with lower costs of CPRI cables.
- Multimode RF modules are to be cascaded, for example, to provide coverage for stadiums, subways, and high-speed railways.

9.2 Precautions

The following information must be collected before feature deployment:

- Names of baseband processing units
- Names of RF modules
- Information of optical modules
- Planned site capacity

RF modules have been installed. Cables between RF modules and BBUs have been connected as required by the CPRI MUX feature. For details, see *Base Station Installation Guide* or *3900 & 5900 Series Base Station Maintenance Guide* in *3900 & 5900 Series Base Station Product Documentation*.

NOTE

If CPRI MUX is to be deployed in a site where a single-mode base station is to be reconstructed into a multimode base station, the hardware must meet requirements for the CPRI MUX feature. Otherwise, it is necessary to replace the hardware.

9.3 Data Configuration

9.3.1 Data Preparation

Separate-MPT/Hybrid-MPT Scenarios

A multimode RF module that operates in multiple network modes must follow the following configuration rules:

- The RF module is configured and monitored by all network modes.
- The RF module must have identical configurations in all network modes. Otherwise, ALM-26274 Inter-System Board Object Configuration Conflict will be reported.

Parameters in [Table 9-1](#) must be configured to the same values for the RF module in all network modes.

Table 9-1 Common parameters

Parameter Name on the GSM (GBTS) Side	Parameter ID on the GSM (GBTS) Side	Parameter Name on the GSM (eGBTS)/ UMTS/LTE/NR Side	Parameter ID on the GSM (eGBTS)/ UMTS/LTE/NR Side
Topo Type	<i>TT</i>	Topo Type	<i>TT (LTE eNodeB, 5G gNodeB)</i>
Cabinet No.	<i>CN</i>	Cabinet No.	<i>CN (LTE eNodeB, 5G gNodeB)</i>
Subrack No.	<i>SRN</i>	Subrack No.	<i>SRN (LTE eNodeB, 5G gNodeB)</i>
Slot No.	<i>SN</i>	Slot No.	<i>SN (LTE eNodeB, 5G gNodeB)</i>
Board Type	<i>BT</i>	RRU type	<i>RT (LTE eNodeB, 5G gNodeB)</i>
Working Standard	<i>WORKINGSTANDARD</i>	RF Unit Working Mode	<i>RS (LTE eNodeB, 5G gNodeB)</i>

In separate-MPT/hybrid-MPT scenarios, the RRU configuration of the converged party takes effect only after the RRU configuration of the converging party is complete.

There are no specific requirements for the configurations of the converging party. For the configurations, see section "Data Preparation" in "Data Configuration" of chapter "Topologies" > "Operation and Maintenance" for a specific base station type (GBTS/eGBTS/NodeB/eNodeB/gNodeB/multimode base station) in *RF Unit and Topology Management*.

CPRI MUX requires that **Access Type** be set to **PEERPORT** for the RRU chain or ring of the converged party. [Table 9-2](#) provides key parameter settings.

- If the converged party is GSM (GBTS), set parameters in a **BTSRXUCHAIN** MO.
- If the converged party is GSM (eGBTS)/UMTS/LTE, set parameters in an **RRUCHAIN** MO.

Table 9-2 Key parameters in a **BTSRXUCHAIN/RRUCHAIN** MO

Parameter Name	Parameter ID	Setting Notes
Access Type	<i>AT (LTE eNodeB, 5G gNodeB)</i>	Set this parameter to PEERPORT .
Head Cabinet No.	<i>HCN (LTE eNodeB, 5G gNodeB)</i>	Set these parameters to the cabinet number, subrack number, slot number, and optical port number of the CPRI port on the BBP/BRI of the peer end, respectively.
Head Subrack No.	<i>HSRN (LTE eNodeB, 5G gNodeB)</i>	
Head Slot No.	<i>HSN (LTE eNodeB, 5G gNodeB)</i>	
Head Port No.	<i>HPN (LTE eNodeB, 5G gNodeB)</i>	

Parameter Name	Parameter ID	Setting Notes
Local Slot No.	<i>LSN (LTE eNodeB, 5G gNodeB)</i>	• BBU3900: This parameter is set to 255 by default and the slot number does not need to be configured for the converged party. If GSM (GBTS) is the converged party, this parameter cannot be specified. • BBU3910: – In GU/GL CPRI MUX (GBTS on the GSM side) in separate-MPT scenarios, set this parameter to 255, and do not specify the slot number for the converged party. – In GU/GL CPRI MUX (eGBTS on the GSM side) or UL CPRI MUX in separate-MPT scenarios, set the slot number for the converged party based on site conditions. • BBU5900/BBU5910: For a BBU5900 used in a separate-MPT UL or LN base station using CPRI MUX, set this parameter to the slot number of the converged board.
Bit Rate Set	<i>BITRATESET</i>	This parameter specifies the bit rate set for rate negotiation. This parameter takes effect only when the user-defined rate negotiation switch is turned on.
Rate Change Period	<i>RATECHGPERIOD</i>	This parameter specifies the duration of a port working in each negotiated rate. This parameter takes effect only when the user-defined rate negotiation switch is turned on.

 NOTE

When configuring a **BTSRXUCHAIN** MO, you do not need to set the **Local Slot No.**, **User-Defined Rate Negotiation Switch**, **Bit Rate Set**, and **Rate Change Period** parameters.

Co-MPT Scenarios

If the topology meets the requirements for co-MPT scenarios specified in [4 Boards That Support CPRI MUX and Constraints](#), CPRI MUX is supported by default and software configuration is not required.

- The data configurations for the star or chain topology with CPRI MUX in a co-MPT base station are identical to those for the star or chain topology in a

single-mode base station. For details about how to configure a star or chain topology, see section "Data Configuration" > "Data Preparation" > "Scenario-specific Data" > "Scenario 1: Star topology" and "Scenario 2: Chain topology" of chapter "Topologies" > "Operation and Maintenance" for a specific base station type (GBTS/eGBTS/NodeB/eNodeB/gNodeB/multimode base station) in *RF Unit and Topology Management*.

- Data configurations for the load sharing topology with CPRI MUX in a co-MPT base station are identical to those for the load sharing topology in an LTE-only base station. For details, see section "Scenario-specific Data Preparation" > "Scenario 4: load sharing topology" of chapter "Topologies" > "Operation and Maintenance (eNodeB)" > "Data Configuration" > "Data Preparation" and "Topologies" > "Operation and Maintenance (gNodeB)" > "Data Configuration" > "Data Preparation" in *RF Unit and Topology Management*.

Dual-Star Topology

- GU+L triple-mode dual-star topology
For details, see [Figure 5-51](#).
- GL+U triple-mode dual-star topology
For details, see [Figure 5-52](#).
- G[U*L] triple-mode dual-star topology
For details, see [Figure 5-53](#).
- GU+LN dual-star topology
For details, see [Figure 5-54](#).

For these topologies, see [Separate-MPT/Hybrid-MPT Scenarios](#) for data preparation of separate-MPT GU/GL/LN CPRI MUX and [Co-MPT Scenarios](#) for data preparation of co-MPT U*L CPRI MUX. For data preparation for GSM/UMTS/LTE/NR, see section "Scenario-specific Data" > "Scenario 1: Dual-star topology" of chapter "Topologies" > "Operation and Maintenance (Multimode Base Station)" > "Data Configuration" > "Data Preparation" in *RF Unit and Topology Management*.

9.3.2 Using MML Commands

Before using MML commands, refer to [8 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

In co-MPT scenarios, CPRI MUX is supported by default and software configuration is not required when the requirements outlined in this document are met. For data configurations of RRUs and corresponding chains or rings, see operation and maintenance sections in *RF Unit and Topology Management*. This section describes the procedure for activating CPRI MUX in separate-MPT scenarios.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Using MML Commands

- Step 1** On the converging party, run the **ADD RRUCHAIN** command to add an RRU/RFU chain or ring.
- Step 2** On the converging party, run the **ADD RRU** command to add an RRU or RFU to the RRU/RFU chain or ring.
- Step 3** On the converged party, run the following commands to add an RRU/RFU chain or ring. For key parameter settings, see [Table 9-2](#).
 - ADD BTSRXUCHAIN** if the converged party is GSM (GBTS)
 - ADD RRUCHAIN** if the converged party is GSM (eGBTS), UMTS, or LTE
- Step 4** On the converged party, run the following command to add an RRU or RFU to the RRU/RFU chain or ring:
 - ADD BTSRXUBRD** if the converged party is GSM (GBTS).
 - ADD RRU** if the converged party is GSM (eGBTS), UMTS, or LTE

----End

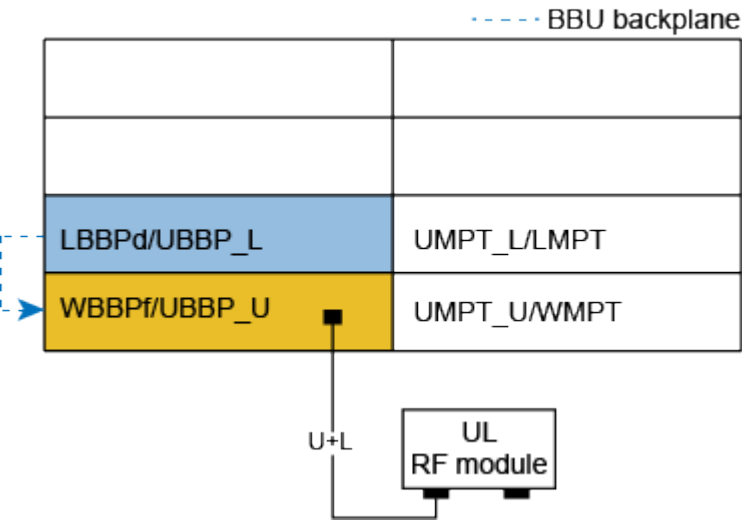
NOTE

Data configurations must be identical for RF modules on the converging party and converged party sides. For the parameters, see [Table 9-1](#).

MML Command Examples

- BBU3900**
[Figure 9-1](#) shows the topology used in the following command example.

Figure 9-1 UL CPRI MUX used with a BBU3900 in separate-MPT scenarios (converging party: UMTS)

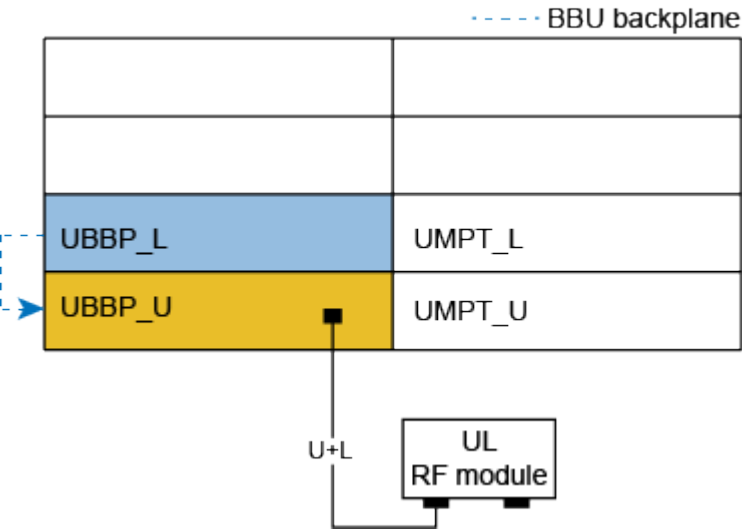


```
//Adding an RRU chain or ring on the converging party
ADD RRUCHAIN: RCN=0, TT=CHAIN, BM=COLD, AT=LOCALPORT, HCN=0, HSRN=0, HSN=3, HPN=0;
//Adding an RRU or RFU to the RRU/RFU chain or ring on the converging party
```

```
ADD RRU: CN=0, SRN=60, SN=0, TP=TRUNK, RCN=0, PS=0, RT=MRRU, RS=UL, RXNUM=2, TXNUM=2;  
//Adding an RRU chain or ring on the converged party  
ADD RRUCHAIN: RCN=0, TT=CHAIN, BM=COLD, AT=PEERPORT, HCN=0, HSRN=0, HSN=3, HPN=0;  
//Adding an RRU or RFU to the RRU/RFU chain or ring on the converged party  
ADD RRU: CN=0, SRN=60, SN=0, TP=TRUNK, RCN=0, PS=0, RT=MRRU, RS=UL, RXNUM=2, TXNUM=2;
```

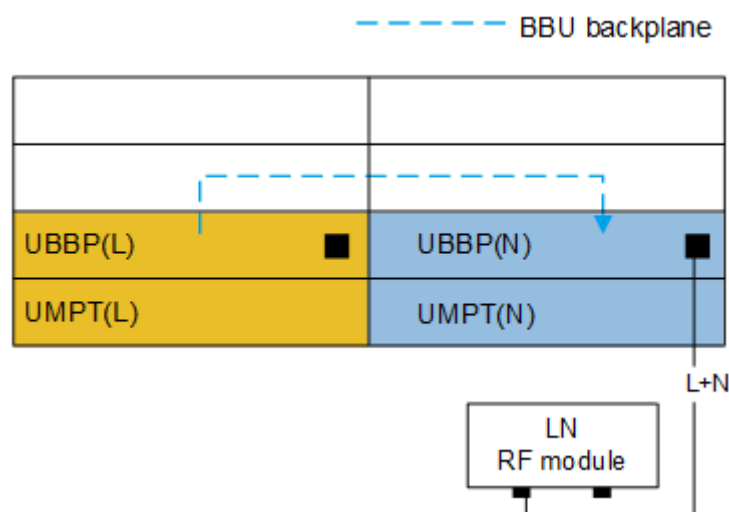
- **BBU3910**
Figure 9-2 shows the topology used in the following command example.

Figure 9-2 UL CPRI MUX used with a BBU3910 in separate-MPT scenarios (converging party: UMTS)



```
//Adding an RRU chain or ring on the converging party  
ADD RRUCHAIN: RCN=0, TT=CHAIN, BM=COLD, AT=LOCALPORT, HCN=0, HSRN=0, HSN=3, HPN=0;  
//Adding an RRU or RFU to the RRU/RFU chain or ring on the converging party  
ADD RRU: CN=0, SRN=60, SN=0, TP=TRUNK, RCN=0, PS=0, RT=MRRU, RS=UL, RXNUM=2, TXNUM=2;  
//Adding an RRU chain or ring on the converged party  
ADD RRUCHAIN: RCN=0, TT=CHAIN, BM=COLD, AT=PEERPORT, HCN=0, HSRN=0, HSN=3, HPN=0, LSN=2;  
//Adding an RRU or RFU to the RRU/RFU chain or ring on the converged party  
ADD RRU: CN=0, SRN=60, SN=0, TP=TRUNK, RCN=0, PS=0, RT=MRRU, RS=UL, RXNUM=2, TXNUM=2;
```

- **BBU5900/BBU5910:**
The following figure shows the topology used in the command example.

Figure 9-3 LN CPRI MUX used with a BBU5900/BBU5910 in separate-MPT scenarios (converging party: NR)

//Adding an RRU chain or ring on the converging party

```
ADD RRUCHAIN: RCN=0, TT=CHAIN, BM=COLD, AT=LOCALPORT, HCN=0, HSRN=0, HSN=5, HPN=0;
```

//Adding an RRU or RFU to the RRU/RFU chain or ring on the converging party

```
ADD RRU: CN=0, SRN=60, SN=0, TP=TRUNK, RCN=0, PS=0, RT=MRRU, RS=LN, RXNUM=2, TXNUM=2;
```

//Adding an RRU chain or ring on the converged party

```
ADD RRUCHAIN: RCN=0, TT=CHAIN, BM=COLD, AT=PEERPORT, HCN=0, HSRN=0, HSN=5, HPN=0, LSN=4;
```

//Adding an RRU or RFU to the RRU/RFU chain or ring on the converged party

```
ADD RRU: CN=0, SRN=60, SN=0, TP=TRUNK, RCN=0, PS=0, RT=MRRU, RS=LN, RXNUM=2, TXNUM=2;
```

9.3.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

9.4 Activation Verification

Step 1 Verify that RRUs do not report related alarms.

Step 2 Run the following command to check whether the RRUs operate properly.

- Separate-MPT scenarios
 - **DSP BTSBRD** for GSM (GBTS)
 - **DSP BRD** for GSM (eGBTS), UMTS, or LTE
- Co-MPT scenarios
 - **DSP BRD**

- Step 3** Run the following command to check whether the CPRI line rate of the converging port is consistent with the plan:
- **DSP CPRIPORT** on the converging party side in separate-MPT scenarios
 - **DSP CPRIPORT** in co-MPT scenarios
- End

9.5 Network Monitoring

Converging and converged parties have separate O&M platforms in separate-MPT scenarios, and share one O&M platform in co-MPT scenarios. The following table lists CPRI MUX-related MML commands, which can be executed on the O&M platform.

Table 9-3 MML commands for maintenance

MML Command	Application Scenario	Support by the Converging Party	Support by the Converged Party (Only in Separate-MPT Scenarios)
DSP RRUCHAIN	To query the number of RRUs on a chain or ring connecting to the optical port of the converging party and the number of cascading levels.	Fully supported	Fully supported
DSP RRUCHAINPHYTO PO	To query the connections and physical information of RRUs connecting to the optical port of the converging party.	Fully supported	Fully supported
DSP CPRIPORT	To query the real-time information including the data rate over the optical port of the converging party. To query whether the optical port converges data.	Fully supported	Partially supported: Use this command to query data rates on the CPRI port and the port aggregation status.

MML Command	Application Scenario	Support by the Converging Party	Support by the Converged Party (Only in Separate-MPT Scenarios)
DSP CPRILBR	To query the current CPRI line rate of the specified chain or ring, the maximum CPRI line rate of the BBP to which the chain or ring is configured, and the maximum CPRI line rate of each RRU or RFU on the chain or ring.	Fully supported	Partially supported: Use this command to query the CPRI line rate of the specified chain/ring and the maximum CPRI line rate of each RRU or RFU on a chain or ring.
STR CPRILBRNEG	To manually start a negotiation on the CPRI line rate of the optical port. You can run this command if the RRU chain or ring cannot be set up or cannot operate at the maximum CPRI line rate due to changes in parameter settings of the RRU chain or ring. Executing this command causes link disconnection, interrupting services.	Fully supported	Not supported
MOD RRUCHAIN	To set breakpoints on an RRU chain or ring.	Fully supported	Not supported
DSP CELL for LTE DSP ULOCCELL for UMTS DSP GTRXSTAT for GSM (GBTS) DSP GTRX for GSM (eGBTS)	To query the cell status. You can run this command if cell setup fails or cell capability degrades due to insufficient CPRI bandwidth.	Fully supported	Fully supported

9.6 Reconfiguration

You can add an RRU/RFU to or remove an RRU/RFU from a cascading chain in the star or chain topology with CPRI MUX.

For guidance on how to add RF modules to or remove them from a chain or ring on the MAE-Deployment, see section "Adding or Deleting an RF Module" of "Device Data Reconfiguration" in *eRAN Reconfiguration Guide in 3900 & 5900 Series Base Station Product Documentation*.

Separate-MPT Scenarios

- Adding an RRU/RFU to a cascading chain
To add an RRU/RFU to a cascading chain without changing the original topology, you need to add the information of this RRU/RFU on both the converging and converged parties.
 - a. On the converging party, run the **MOD RRUCHAIN** command to set a breaking point.
 - b. On both the converging and converged parties, run the **ADD RRU** command for GSM (eGBTS)/UMTS/LTE or **ADD BTSRXUBRD** for GSM (GBTS) to add an RRU/RFU.
 - c. Adjust the cable connections and add an RRU/RFU at the specified position.
 - d. On the converging party, run the **MOD RRUCHAIN** command to remove the breaking point.
- Removing an RRU/RFU from a cascading chain
To remove an RRU/RFU from a cascading chain without changing the original topology, you need to remove the information of this RRU/RFU on both the converging and converged parties.
 - a. On the converging party, run the **MOD RRUCHAIN** command to set a breaking point.
 - b. On both the converging and converged parties, run the **RMV RRU** command for GSM (eGBTS)/UMTS/LTE or **RMV BTSRXUBRD** for GSM (GBTS) to remove an RRU/RFU.
 - c. Adjust the cable connections and remove the RRU/RFU at the specified position.
 - d. On the converging party, run the **MOD RRUCHAIN** command to remove the breaking point.

Co-MPT Scenarios

- Adding an RRU/RFU to a cascading chain
To add an RRU/RFU to a cascading chain without changing the original topology, perform the following steps:
 - a. Run the **MOD RRUCHAIN** command to set a breaking point.
 - b. Run the **ADD RRU** command to add an RRU/RFU.
 - c. Adjust the cable connections and add an RRU/RFU at the specified position.
 - d. Run the **MOD RRUCHAIN** command to remove the breaking point.
- Removing an RRU/RFU from a cascading chain
To remove an RRU/RFU from a cascading chain without changing the original topology, perform the following steps:

- a. Run the **MOD RRUCHAIN** command to set a breaking point.
- b. Run the **RMV RRU** command to remove an RRU/RFU.
- c. Adjust the cable connections and remove the RRU/RFU at the specified position.
- d. Run the **MOD RRUCHAIN** command to remove the breaking point.

10 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.

----End

11 Terminology

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

12 Reference Documents

1. 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
2. Documents in *3900 & 5900 Series Base Station Product Documentation*:
 - *BBU Technical Specifications*
 - *RRU Technical Specifications*
 - *AAU Technical Specifications*
 - *RFU Technical Specifications*
 - *BBU5900 Hardware Description*
 - *BBU5910 Hardware Description*
 - *BBU5900A Hardware Description*
 - *BBU5900C Hardware Description*
 - *BBU3900 and BBU3910 Hardware Description*
 - *3900 & 5900 Series Base Station Maintenance Guide*
 - *eRAN Reconfiguration Guide*
 - *RMU*
3. Feature parameter description documents in *eRAN Feature Documentation* and *5G RAN Feature Documentation*:
 - *LTE FDD and NR SUL Dynamic Spectrum Sharing*
 - *eCPRI*
 - *Cell Management*
 - *RF Unit and Topology Management*
 - *CPRI Compression*
 - *CPRI MUX (LampSite)*
4. Feature parameter description documents in *RAN Feature Documentation*:
 - *NodeB Carrier Management Feature Parameter Description*
5. Feature parameter description documents in *GBSS Feature Documentation*:
 - *GSM Carrier Management*
 - *Multi-site Cell*